

2. General → 2.6 Geometric relations and rule code

2.6 Geometric relations and rule code

The rule codes, in general, use the description of the geometric relation (e.g. "S" for spacing) and also the design layer/definition code (e.g. "M1" for Metal 1). Rule codes are intended for short design rule check error messages.

Design rules that are not checked by DRC are indicated by printing the rule code in red underlined italics. The designer must ensure manually that these rules are adhered to.

'QnA(B)' - Query/ warning based rules are not considered to be DRC failures.

The rule types M_A, N_A, G_A are used to check that predefined devices adhere to the following statement: "The layout of the device is predefined and must not be changed". Please note, that for these devices, it is a requirement that the layout is not to be changed (except for the described conditions). The checks related to the rule codes M_A, N_A, G_A are a supporting tool to achieve this requirement. These rules are available for viewing only via the SpecXplorer website.

Rule text	Description	Rule code
(arbitrary text)	Illegal construct	BnA(B)
A crossing B edge is not allowed	Not allowed for layer A polygon to have portions both inside and outside layer B	BnA
A width	Distance inside_A - inside_A	WnA
A size	(Distance inside_A - inside_A) x (Distance inside_A - inside_A)	WnA
A spacing / notch	Distance outside_A - outside_A	SnA
A spacing to B	Distance outside_A - outside_B	SnAB or SnBA
A enclosure of B	Distance inside_A - outside_B (A contains B)	EnAB or EnBA
A extension beyond B	Distance inside_A - outside_B (A must intersect B)	EnAB or EnBA
A overlap of B	Distance inside_A - inside_B	OnAB or OnBA
A area	Area	AnA
A enclosed area	Area of opening enclosed by A	AnA
ratio of A to B	Ratio	RnAB
(arbitrary text)	Recommend investigation (query or warning)	QnA(B)
A radius / angle	Curvature	CnA
Predefined area must have A	Area restriction	M_A
Predefined area must not have A	Area restriction	N_A
Predefined geometry of A violated	Area restriction	G_A

2. General → 2.6 Geometric relations and rule code

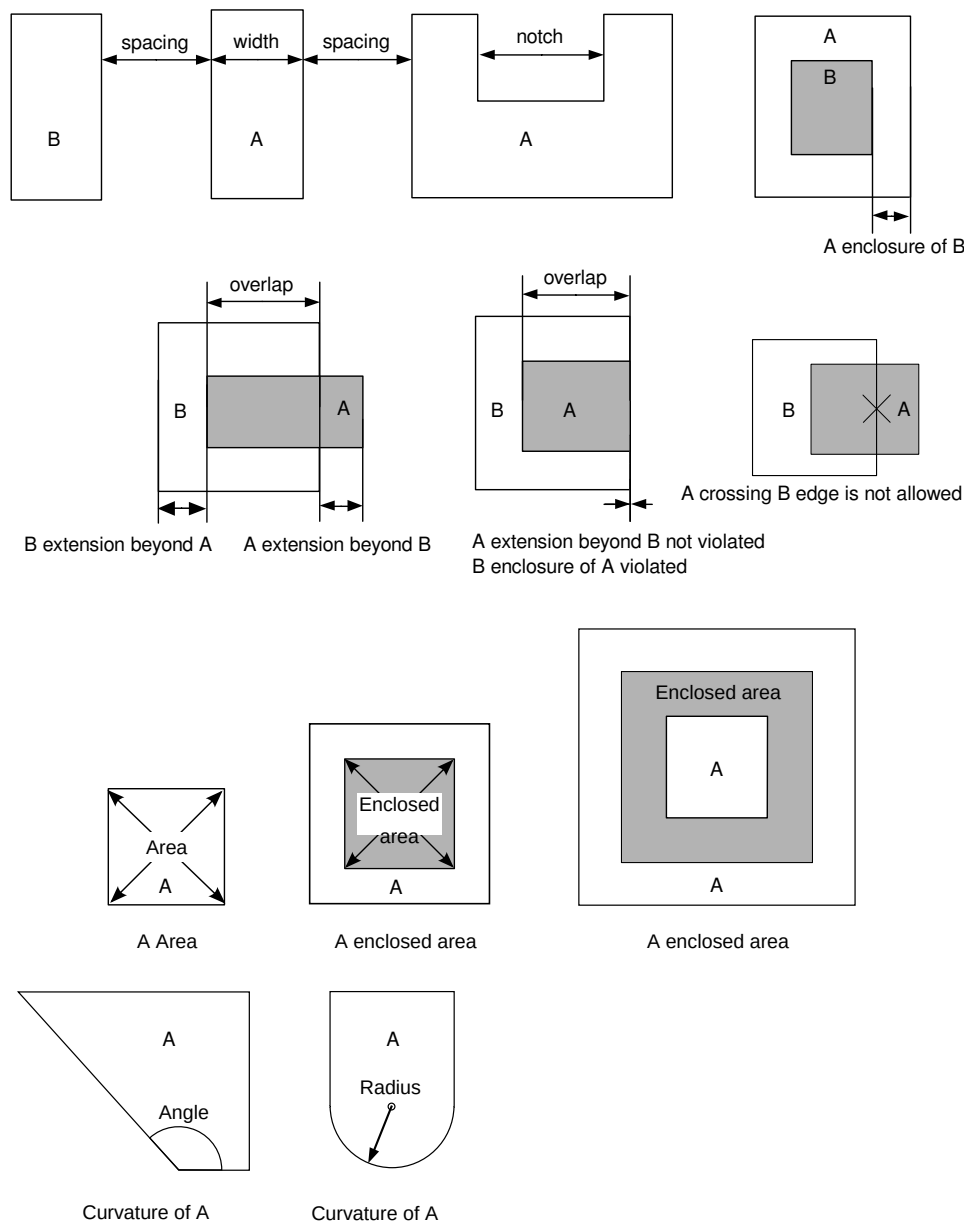


Figure 2.1 Geometric relations

2. General → 2.7 General requirements

2.7 General requirements

Grid	0.005 μm
Recommended working grid	0.01 μm
Corners	90°, 135°
Data extrema including Peripheral Ring and Scribe Lane Note: Data extrema requirements cover demands of the mask generation procedure and the wafer manufacturing process only.	multiple of 1 μm

It is **MANDATORY** that all layout data, including path edges (centre line & width defined), lie on the minimum grid. Designers are recommended to work with a 0.01 μm grid because the data edges can be 0.005 μm when 0.01 μm is used for the centre line of 'paths'.

45 degree data must not be used for:

- minimum width lines;
- minimum spaced lines;
- or for components with a critical tolerance;

since 'snapping to grid' can occur at mask generation.

Self intersecting polygons must be avoided.

Off-grid data points are checked by X-FAB at GDS tape-in. Off-grid elements will be snapped on-grid for mask generation.

3. Layer and Device rules

3. Layer and Device rules

3.1 LPMOS main module

3.1.1 Layer rules

NWELL

This layer is also relevant to the NLEAK/ PLEAK guidelines. It is required to follow the NLEAK/ PLEAK guidelines to ensure functionality of the circuit design.

Name	Description	Value	Unit
W1NW	Minimum NWELL width (except nedia)	0.86	μm
S1NW	Minimum NWELL spacing/notch	0.6	μm
S2NW	Minimum NWELL spacing (different net)	1.4	μm
<u>S3NW</u>	Minimum NWELL spacing (different net) Note: Valid if one NWELL to PSUB voltage is less than 1 V.	2.0	μm
A1NW	Minimum NWELL area	2.25	μm ²

Note: It is not recommended to take 'spurs' of device N-Wells to make resistors.

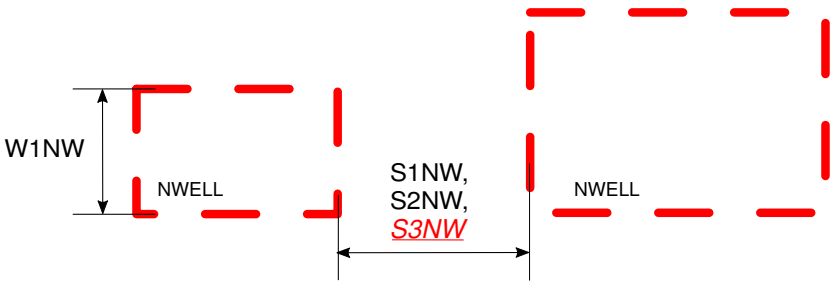


Figure 3.1 NWELL

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → NWELL1, NWELL2, PWELL...

NWELL1, NWELL2, PWELL1, PWELL2

These rules are related to a combination of design layers (refer to section "2.4 Definitions").

Name	Description	Value	Unit
W1W1	Minimum NWELL1 width (except nedia)	0.86	μm
	Note: Not valid between adjacent corners if the virtual line which joins them is orthogonal with respect to the coordinate system axes. The minimum length of this line is 0.56μm.		
W1W2	Minimum PWELL1 width	0.6	μm
	Note: Not valid between adjacent corners if the virtual line which joins them is orthogonal with respect to the coordinate system axes. The minimum length of this line is 0.56μm.		
W1W3	Minimum NWELL2 width	0.86	μm
W1W4	Minimum PWELL2 width	0.6	μm
A1W1	Minimum NWELL1 area	2.25	μm ²
A1W2	Minimum PWELL1 area	2.25	μm ²
A1W3	Minimum NWELL2 area	2.25	μm ²
A1W4	Minimum PWELL2 area	2.25	μm ²

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ NWELL1, NWELL2, PWEL...

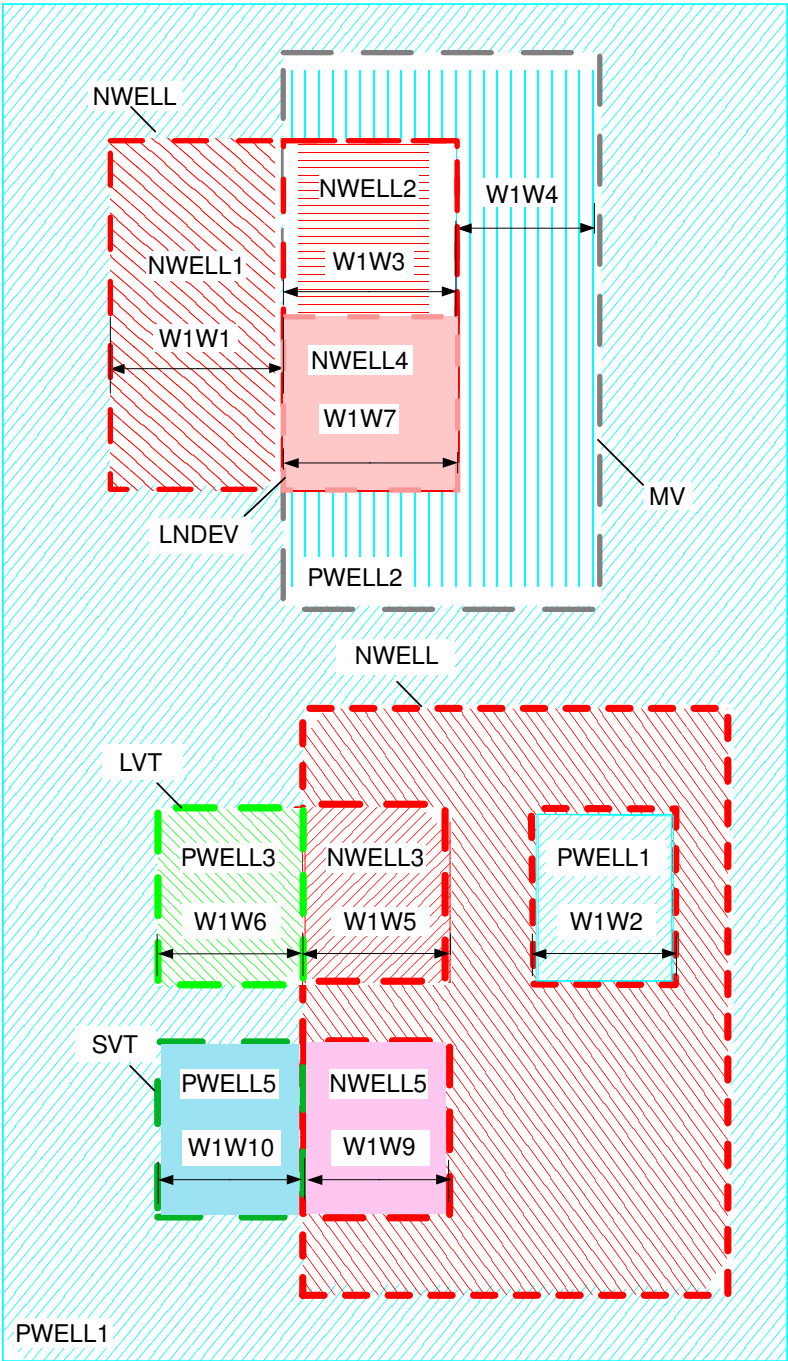


Figure 3.2 NWELL1, PWELL1, NWELL2, PWELL2, NWELL3, PWELL3, NWELL4, NWELL5, PWELL5

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ DIFF

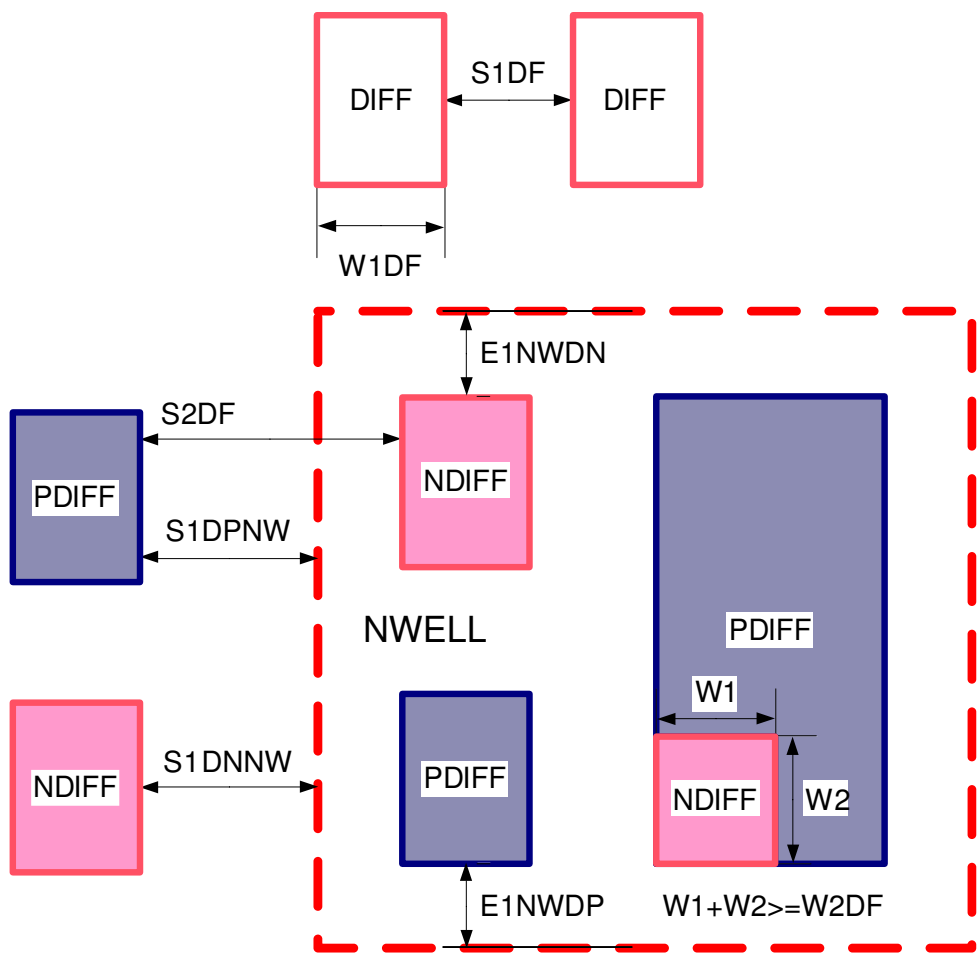


Figure 3.4 DIFF

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → DIFFDUMMY

DIFFDUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1YD	DIFFDUMMY overlap of DIFF is not allowed	-	-
B2YD	DIFFDUMMY overlap of NIMP or PIMP is not allowed	-	-
B3YD	DIFFDUMMY overlap of CONT is not allowed	-	-
B5YD	DIFFDUMMY overlap of SBLK is not allowed	-	-
B6YD	DIFFDUMMY overlap of HVDEPL is not allowed	-	-
B7YD	DIFFDUMMY overlap of PHODEF is not allowed	-	-
B8YD	DIFFDUMMY overlap of EMIR is not allowed	-	-
W1YD	Minimum DIFFDUMMY width	0.4	μm
S1YD	Minimum DIFFDUMMY spacing/notch	0.5	μm
S10YD	Minimum DIFFDUMMY spacing to LOCKED	3.6	μm
S11YD	Minimum DIFFDUMMY spacing to LOCKED1	3.6	μm
S12YD	Minimum DIFFDUMMY spacing to LOCKED2	3.6	μm
S13YD	Minimum DIFFDUMMY spacing to LOCKED3	3.6	μm
S14YD	Minimum DIFFDUMMY spacing to LOCKED4	3.6	μm
S1YDDF	Minimum DIFFDUMMY spacing to DIFF	3.0	μm
S1YDEI	Minimum DIFFDUMMY spacing to EMIR	3.0	μm
S1YDHL	Minimum DIFFDUMMY spacing to HVDEPL	5.0	μm
S1YDHW	Minimum DIFFDUMMY spacing to HNW	10.0	μm
S1YDIN	Minimum DIFFDUMMY spacing to NIMP	0.3	μm
S1YDIP	Minimum DIFFDUMMY spacing to PIMP	0.3	μm
S1YDNW	Minimum DIFFDUMMY spacing to NW4DMY	5.0	μm
	Note: For DNWELLMV design layer valid if DNWELLMV is outside DNWELL.		
S1YDP1	Minimum DIFFDUMMY spacing to POLY1	3.0	μm
S1YDPI	Minimum DIFFDUMMY spacing to PW4DMY	5.0	μm
S1YDSB	Minimum DIFFDUMMY spacing to SBLK	3.0	μm
S1YDWD	Minimum DIFFDUMMY spacing to DNWELL	10.0	μm
E1YDHW	Minimum HNW enclosure of DIFFDUMMY	10.0	μm
E1YDNW	Minimum NW4DMY enclosure of DIFFDUMMY	5.0	μm
	Note: For DNWELLMV design layer valid if DNWELLMV is outside DNWELL.		
E1YDPI	Minimum PW4DMY enclosure of DIFFDUMMY	5.0	μm
E1YDWD	Minimum DNWELL enclosure of DIFFDUMMY	10.0	μm
A1YD	Minimum DIFFDUMMY area	1.2	μm ²

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → POLY1

POLY1

This layer is also relevant to the NLEAK/ PLEAK guidelines. It is required to follow the NLEAK/ PLEAK guidelines to ensure functionality of the circuit design.

Name	Description	Value	Unit
B1GA	90 degree bent GATE is not allowed Note: Only valid for POLY1 over DIFF or for DIFF over POLY1 structures with > 4 vertices.	-	-
B1P1	(POLY1 and SBLK and not (HRES or MRES)) without NIMP or PIMP is not allowed (except dpol, davl#) Note: NIMP touching PIMP under (POLY1 and SBLK and not (HRES or MRES)) is not allowed.	-	-
B2GA	Illegal GATE construct Note: GATE region has been identified as having no association with supported devices.	-	-
BDP1	Not allowed to be used by customers Note: Only valid for POLY1 purpose NOBLK. Reserved layer.	-	-
W1P1	Minimum POLY1 width	0.18	μm
W2P1	Minimum POLY1 width (for 45 degree bent GATE)	0.21	μm
S1P1	Minimum POLY1 spacing/notch	0.25	μm
S3P1	Minimum POLY1 spacing (different net, tag_60v) Note: Valid for spacing of all POLY1 shapes with label tag_60v to all POLY1 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.5	μm
S1P1DF	Minimum POLY1 spacing to DIFF	0.1	μm
S5P1DF	Minimum POLY1 spacing to DIFF (different net, tag_60v) Note: Valid for spacing of all POLY1/DIFF shapes with label tag_60v to all DIFF/POLY1 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.5	μm
E1DFGA	Minimum DIFF extension beyond GATE (except davl#)	0.32	μm
E1P1GA	Minimum POLY1 extension beyond GATE (except davl#)	0.22	μm
A1P1	Minimum POLY1 area	0.118	μm ²
A2P1	Maximum (POLY1 and not SBLK) area	17000.0	μm ²
R1P1	Minimum ratio of POLY1 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	14.0	%
R2P1	Maximum ratio of POLY1 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	200.0	-
Q1P1	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ POLY1

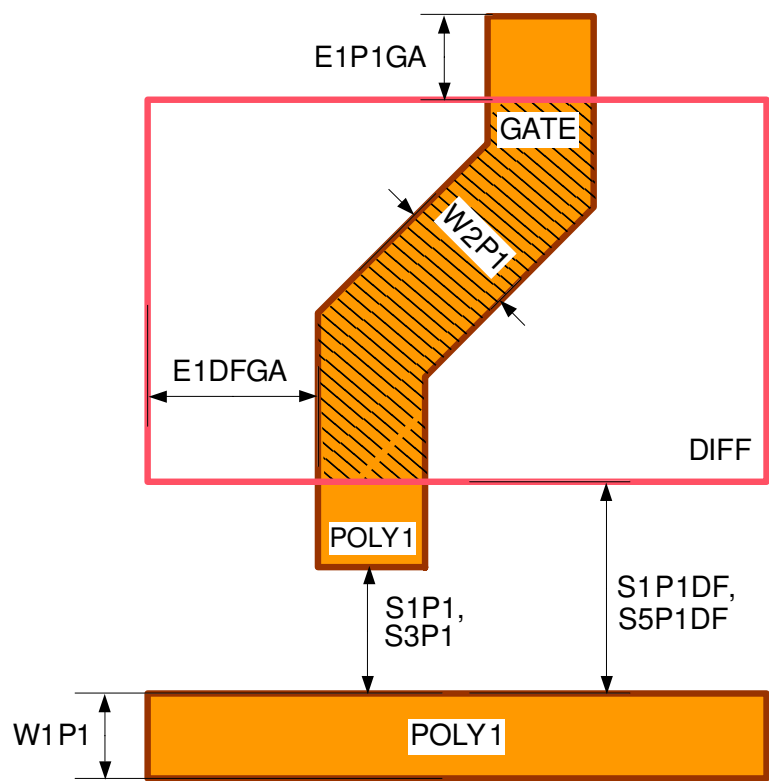
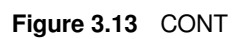


Figure 3.7 POLY1

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → CONT

CONT

Name	Description	Value	Unit
B1CT	CONT without NDIFF or PDIFF or POLY1 is not allowed (except dsb#, CATDIFF of davl#)	-	-
B1CTGA	CONT overlap of GATE is not allowed (except davl#)	-	-
B1CTSB	CONT overlap of SBLK is not allowed	-	-
B3CT	CONT stripes are only allowed to bend at 135 degrees	-	-
W2CT	Fixed CONT stripe width	0.22	μm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1CT	Fixed CONT size (except dphoc#, dphod#, DIFF within PHODIO)	0.22 x 0.22	μm x μm
S1CT	Minimum CONT spacing	0.25	μm
S2CT	Minimum CONT spacing (for contact arrays containing more than 4x4 elements)	0.28	μm
	Note: Two contact regions within 0.3 μm spacing are considered to be in the same array.		
S3CT	Minimum CONT stripe to CONT spacing	0.39	μm
S4CT	Minimum CONT stripe spacing/notch	0.67	μm
S1CTDF	Minimum CONT spacing to DIFF	0.2	μm
S1CTP1	Minimum CONT spacing to POLY1 (except davl#)	0.16	μm
E1CTIN	Minimum NIMP enclosure of DIFFCON	0.12	μm
E1CTIP	Minimum PIMP enclosure of DIFFCON	0.12	μm
E1DFCT	Minimum DIFF enclosure of CONT	0.1	μm
E1P1CT	Minimum POLY1 enclosure of CONT (except davl#)	0.1	μm
R1CT	Maximum ratio of CONT area to connected GATE area	10.0	-



3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → MET1

MET1

This layer is also relevant to the NLEAK/ PLEAK guidelines. It is required to follow the NLEAK/ PLEAK guidelines to ensure functionality of the circuit design.

Name	Description	Value	Unit
B1CTM1	CONT must be covered by MET1	-	-
B1M1	All MET1 tracks > 35µm wide to be slotted (except Pads)	-	-
W1M1	Minimum MET1 width	0.23	µm
<u>W4M1</u>	Minimum MET1 width joining wide MET1 track (> 35 µm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 µm from the main track prior to narrowing it.	10.0	µm
<u>W5M1</u>	Maximum MET1 region size Note: MET1 regions are defined as MET1 shapes (single MET1 shapes or a bundle of MET1 shapes, with width > 2.0µm, inclusive of the spacing if the spacing is ≤ 1.0µm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on "my X-FAB". Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	µm x µm
S1M1	Minimum MET1 spacing/notch	0.23	µm
S4M1	Minimum MET1 spacing (different net, tag_60v) Note: Valid for spacing of all MET1 shapes with label tag_60v to all MET1 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.38	µm
S2M1	Minimum MET1 spacing to WIDE_MET1	0.6	µm
E1M1CT	Minimum MET1 enclosure of CONT	0.005	µm
E2M1CT	Minimum MET1 enclosure of CONT (in one direction of CONT corner)	0.06	µm
E3M1CT	Minimum MET1 enclosure of CONT stripe	0.07	µm
A1M1	Minimum MET1 area	0.202	µm ²
<u>R1M1</u>	Minimum ratio of MET1 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2M1</u>	Maximum ratio of MET1 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1M1P1	Maximum ratio of MET1 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
R2M1P1	Maximum ratio of MET1 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
Q1M1	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ MET1

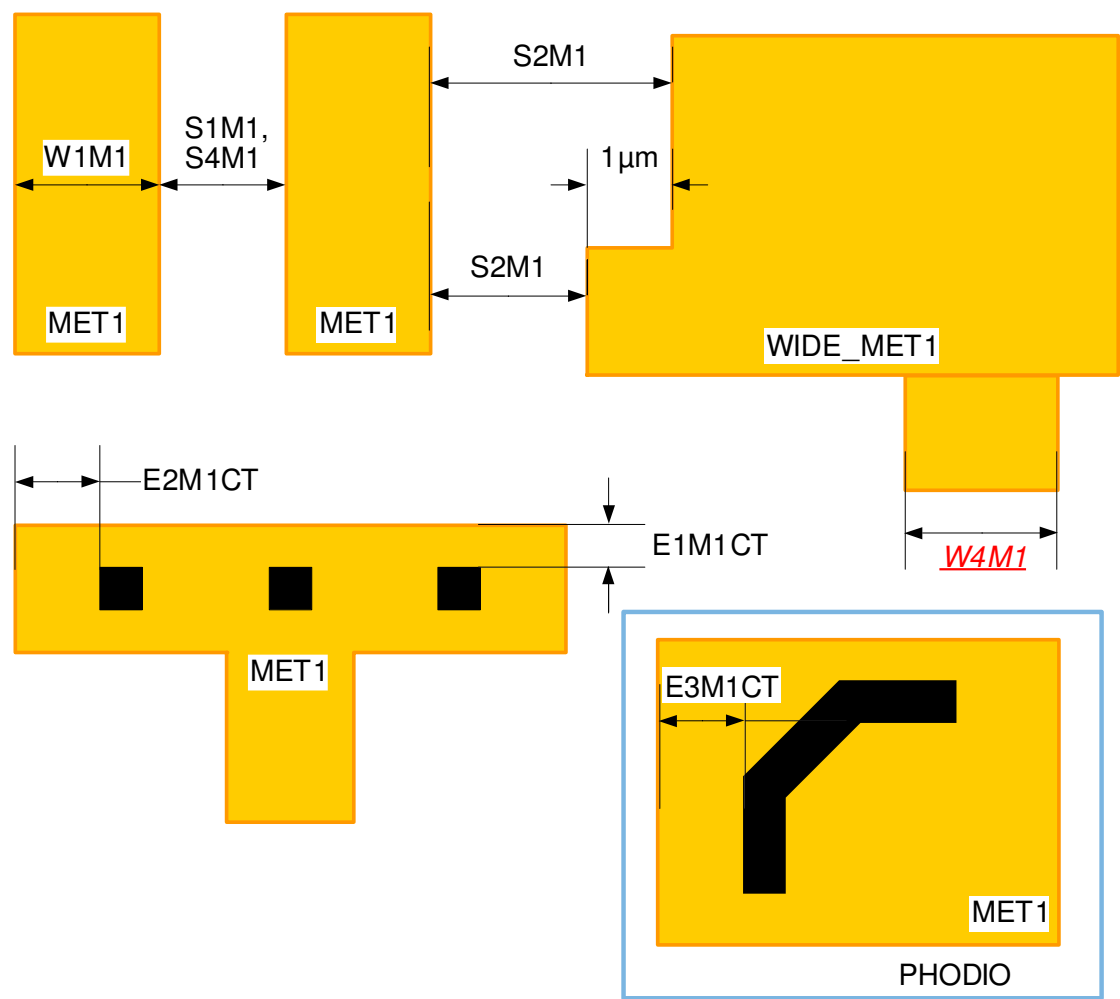


Figure 3.14 MET1

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ M1SLOT

M1SLOT

Name	Description	Value	Unit
W2M1	Minimum M1SLOT width	0.6	μm
W3M1	Minimum M1SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M1	Minimum M1SLOT spacing/notch	10.0	μm
E1M1M1	Minimum MET1 enclosure of M1SLOT	10.0	μm
	Note: M1SLOT without MET1 is not allowed.		

Note: Insert M1SLOTS in direction of current flow.

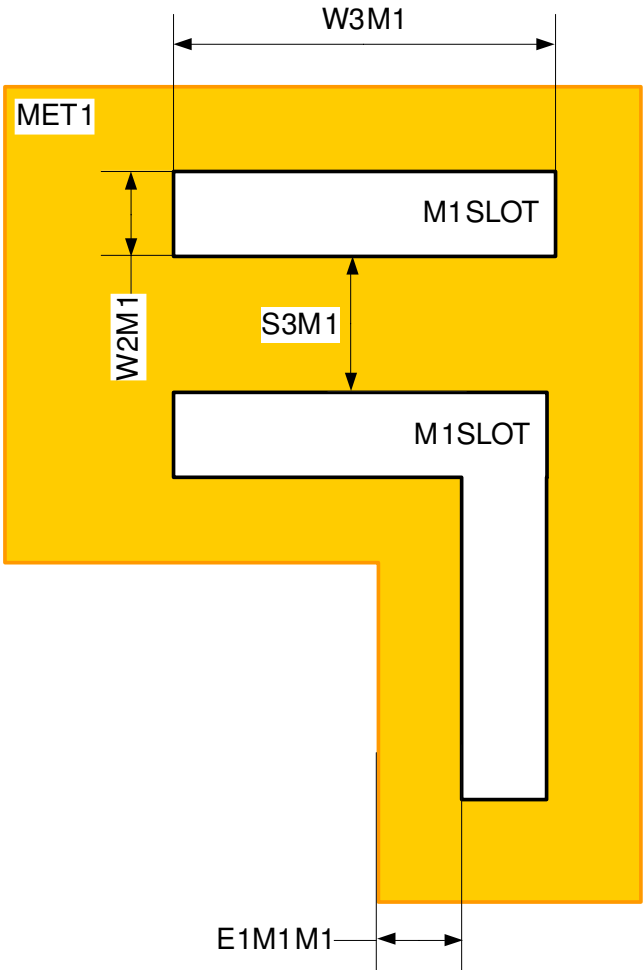


Figure 3.15 M1SLOT

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → M1DUMMY

M1DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y1	Only rectangular M1DUMMY is allowed	-	-
B2Y1	M1DUMMY overlap of MET1 is not allowed	-	-
B3Y1	M1DUMMY overlap of CONT or VIA1 is not allowed	-	-
B4Y1	M1DUMMY overlap of HRES is not allowed	-	-
W1Y1	Minimum M1DUMMY width	2.0	μm
W2Y1	Maximum M1DUMMY edge length	20.0	μm
S1Y1	Minimum M1DUMMY spacing	2.0	μm
S10Y1	Minimum M1DUMMY spacing to LOCKED	4.0	μm
S11Y1	Minimum M1DUMMY spacing to LOCKED1	4.0	μm
S12Y1	Minimum M1DUMMY spacing to LOCKED2	4.0	μm
S13Y1	Minimum M1DUMMY spacing to LOCKED3	4.0	μm
S14Y1	Minimum M1DUMMY spacing to LOCKED4	4.0	μm
S1Y1HR	Minimum M1DUMMY spacing to HRES	5.0	μm
S1Y1M1	Minimum M1DUMMY spacing to MET1	4.0	μm

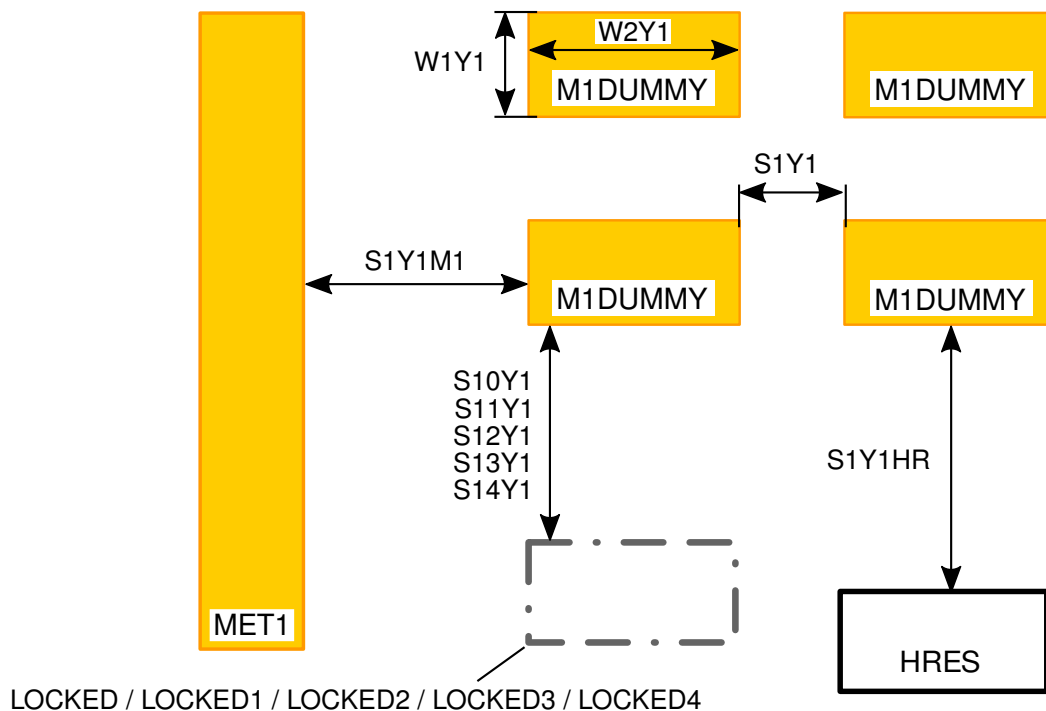


Figure 3.16 M1DUMMY

3. Layer and Device rules → 3.1 LPMOS main module → 3.1.1 Layer rules → VIA1

VIA1

Name	Description	Value	Unit
B1V1	VIA1 must be covered by MET1 and MET2	-	-
B2V1	VIA1 stripes are only allowed to bend at 135 degrees	-	-
W2V1	Fixed VIA1 stripe width	0.26	μm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1V1	Fixed VIA1 size (except dphoc#, dphod#, OPTOVIA1, PHODIO)	0.26 x 0.26	μm x μm
S1V1	Minimum VIA1 spacing	0.26	μm
S3V1	Minimum VIA1 stripe to VIA1 spacing	1.0	μm
E1M1V1	Minimum MET1 enclosure of VIA1	0.01	μm
E1M2V1	Minimum MET2 enclosure of VIA1	0.01	μm
E2M1V1	Minimum MET1 enclosure of VIA1 (in one direction of VIA1 corner)	0.06	μm
E2M2V1	Minimum MET2 enclosure of VIA1 (in one direction of VIA1 corner)	0.06	μm
E4M1V1	Minimum MET1 enclosure of VIA1 stripe	0.07	μm
E7M2V1	Minimum MET2 enclosure of VIA1 stripe	0.07	μm
R1V1	Maximum ratio of VIA1 area to connected GATE area	20.0	-
Q20V1	Check for the right usage of OPTOVIA1	-	-
	Note: VIA1 stripes must only be used for light shielding purpose.		
	Note: DRC reports this check request whenever this layer is present.		

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.1 LPMOS main module→ 3.1.1 Layer rules→ VIA1

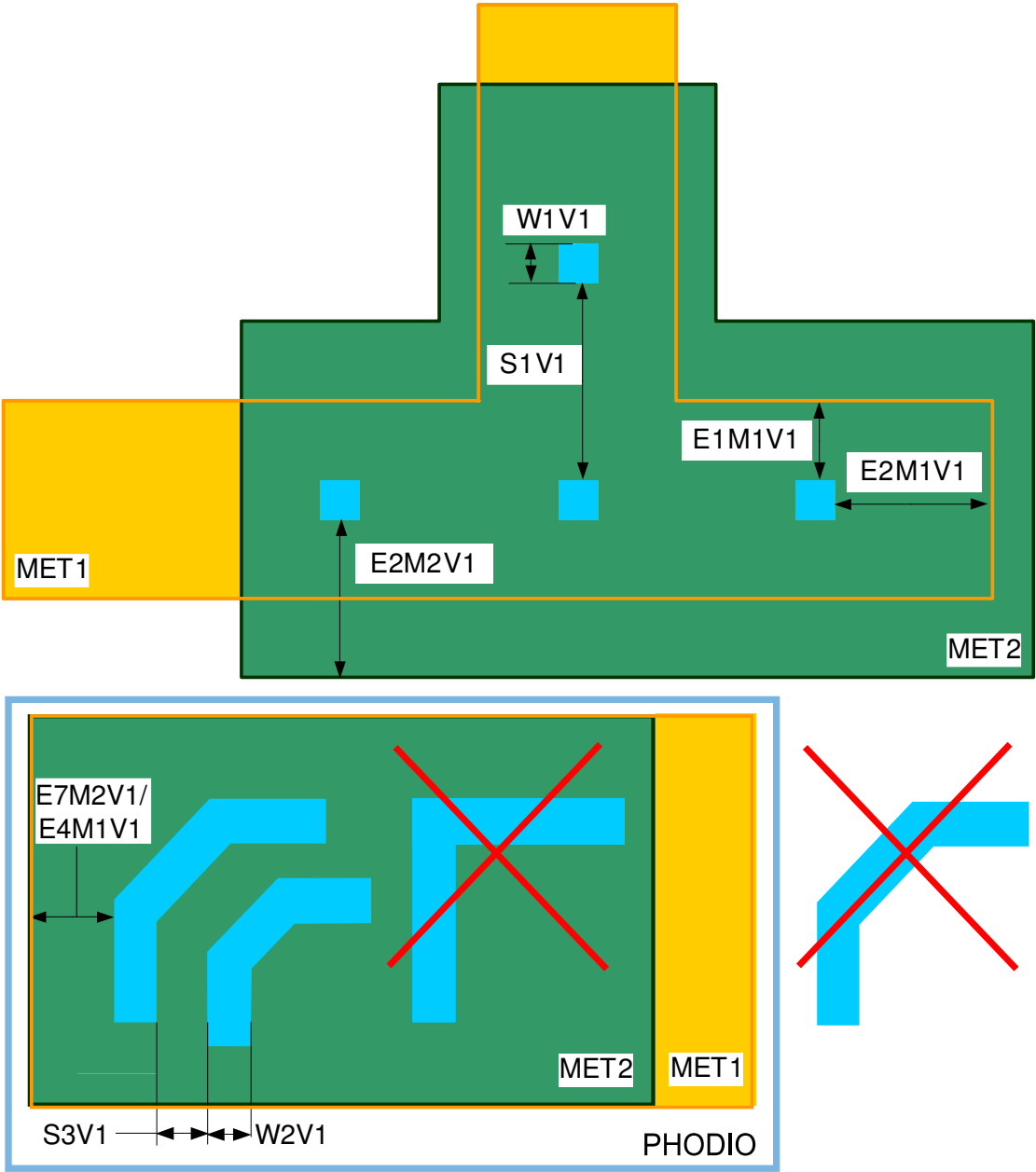


Figure 3.17 VIA1

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ M1SLOT

M1SLOT

Name	Description	Value	Unit
W2M1	Minimum M1SLOT width	0.6	μm
W3M1	Minimum M1SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M1	Minimum M1SLOT spacing/notch	10.0	μm
E1M1M1	Minimum MET1 enclosure of M1SLOT	10.0	μm
	Note: M1SLOT without MET1 is not allowed.		

Note: Insert M1SLOTS in direction of current flow.

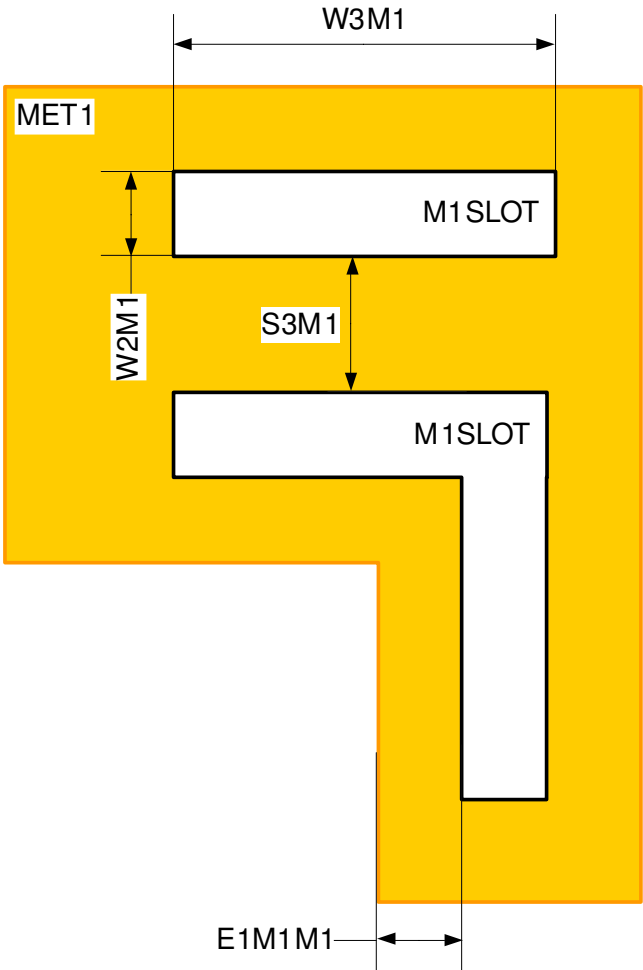


Figure 3.57 M1SLOT

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ M1DUMMY

M1DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y1	Only rectangular M1DUMMY is allowed	-	-
B2Y1	M1DUMMY overlap of MET1 is not allowed	-	-
B3Y1	M1DUMMY overlap of CONT or VIA1 is not allowed	-	-
B4Y1	M1DUMMY overlap of HRES is not allowed	-	-
W1Y1	Minimum M1DUMMY width	2.0	μm
W2Y1	Maximum M1DUMMY edge length	20.0	μm
S1Y1	Minimum M1DUMMY spacing	2.0	μm
S10Y1	Minimum M1DUMMY spacing to LOCKED	4.0	μm
S11Y1	Minimum M1DUMMY spacing to LOCKED1	4.0	μm
S12Y1	Minimum M1DUMMY spacing to LOCKED2	4.0	μm
S13Y1	Minimum M1DUMMY spacing to LOCKED3	4.0	μm
S14Y1	Minimum M1DUMMY spacing to LOCKED4	4.0	μm
S1Y1HR	Minimum M1DUMMY spacing to HRES	5.0	μm
S1Y1M1	Minimum M1DUMMY spacing to MET1	4.0	μm

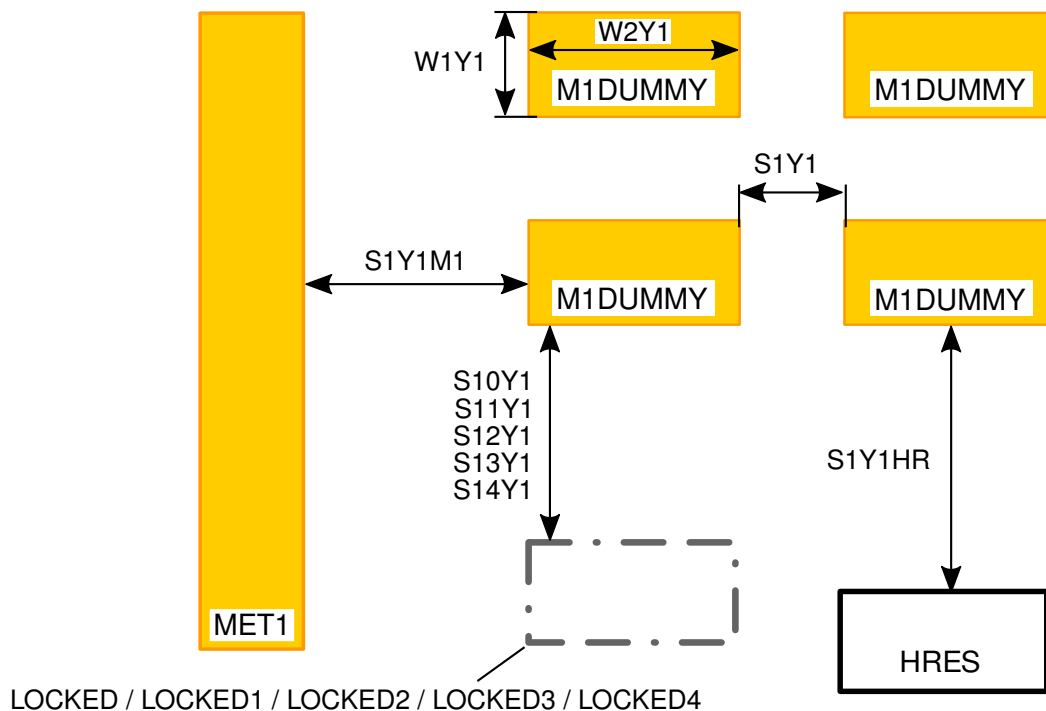


Figure 3.58 M1DUMMY

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ VIA1

VIA1

Name	Description	Value	Unit
B1V1	VIA1 must be covered by MET1 and MET2	-	-
B2V1	VIA1 stripes are only allowed to bend at 135 degrees	-	-
W2V1	Fixed VIA1 stripe width	0.26	μm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1V1	Fixed VIA1 size (except dphoc#, dphod#, OPTOVIA1, PHODIO)	0.26 x 0.26	μm x μm
S1V1	Minimum VIA1 spacing	0.26	μm
S3V1	Minimum VIA1 stripe to VIA1 spacing	1.0	μm
E1M1V1	Minimum MET1 enclosure of VIA1	0.01	μm
E1M2V1	Minimum MET2 enclosure of VIA1	0.01	μm
E2M1V1	Minimum MET1 enclosure of VIA1 (in one direction of VIA1 corner)	0.06	μm
E2M2V1	Minimum MET2 enclosure of VIA1 (in one direction of VIA1 corner)	0.06	μm
E4M1V1	Minimum MET1 enclosure of VIA1 stripe	0.07	μm
E7M2V1	Minimum MET2 enclosure of VIA1 stripe	0.07	μm
R1V1	Maximum ratio of VIA1 area to connected GATE area	20.0	-
Q20V1	Check for the right usage of OPTOVIA1	-	-
	Note: VIA1 stripes must only be used for light shielding purpose.		
	Note: DRC reports this check request whenever this layer is present.		

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ VIA1

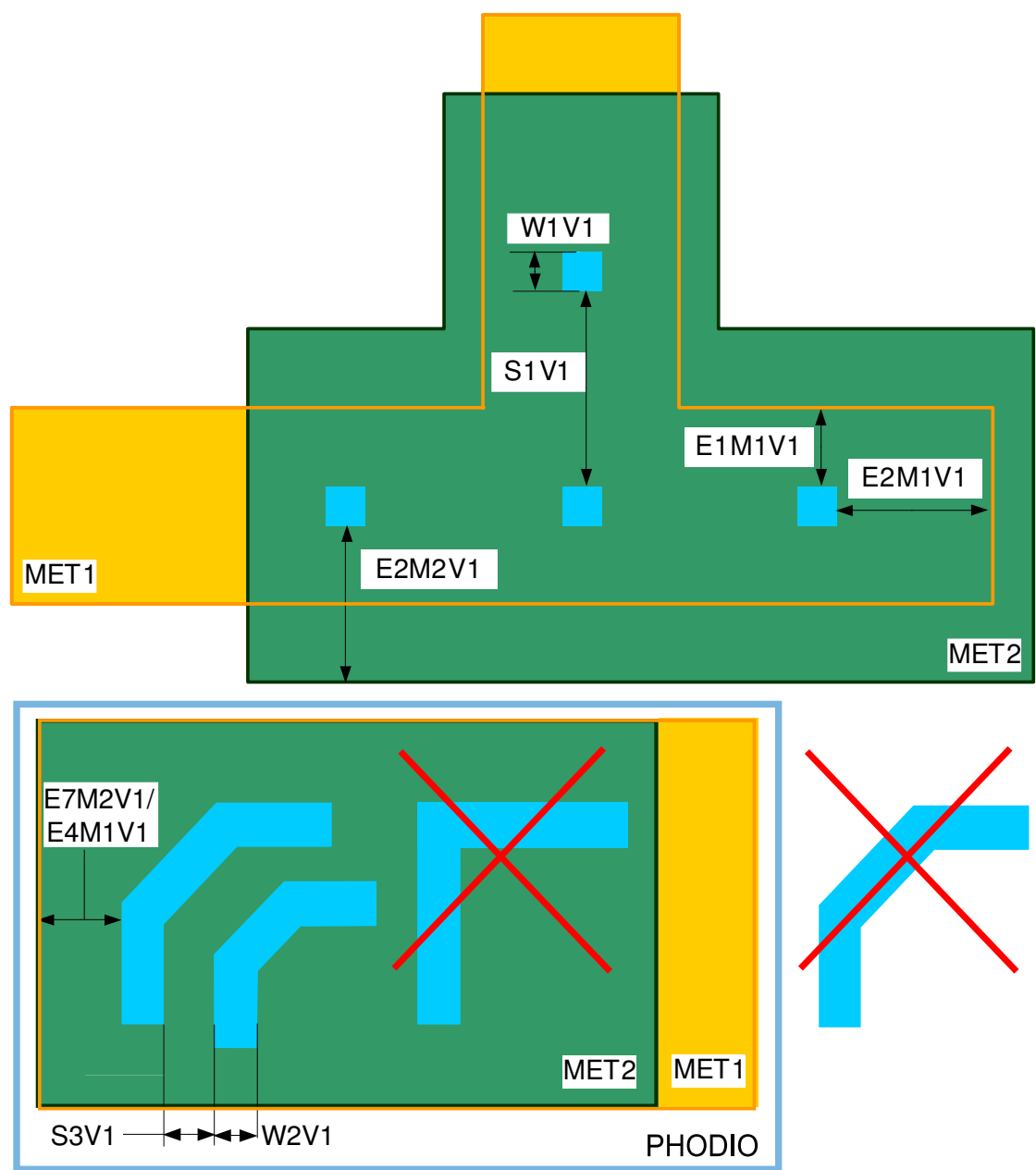


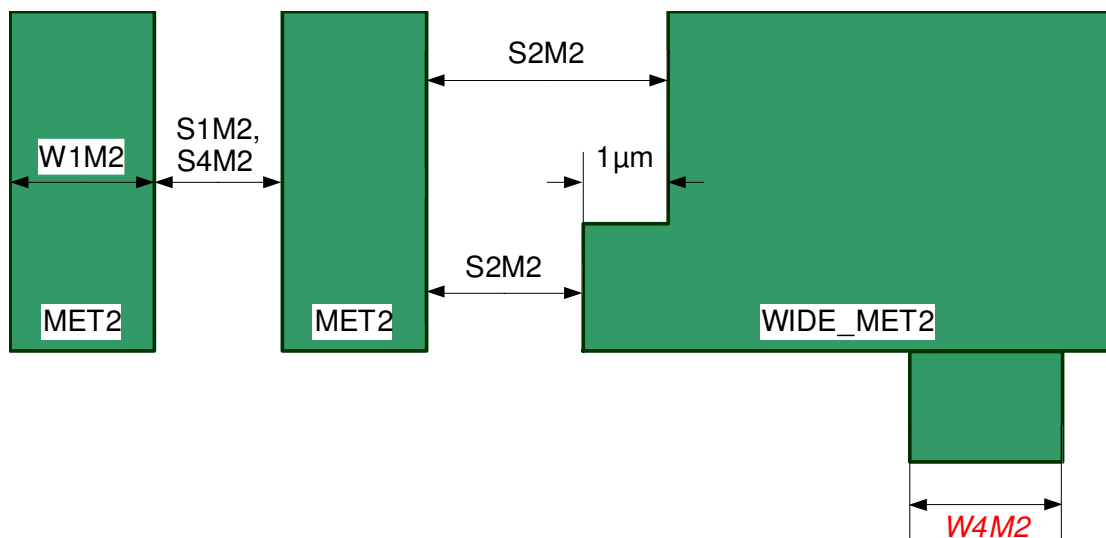
Figure 3.59 VIA1

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ MET2

MET2

This layer is also relevant to the NLEAK/ PLEAK guidelines. It is required to follow the NLEAK/ PLEAK guidelines to ensure functionality of the circuit design.

Name	Description	Value	Unit
B1M2	All MET2 tracks > 35μm wide to be slotted (except Pads, dphoc0, dphod0)	-	-
W1M2	Minimum MET2 width	0.28	μm
<u>W4M2</u>	Minimum MET2 width joining wide MET2 track (> 35 μm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 μm from the main track prior to narrowing it.	10.0	μm
<u>W5M2</u>	Maximum MET2 region size Note: MET2 regions are defined as MET2 shapes (single MET2 shapes or a bundle of MET2 shapes, with width > 2.0μm, inclusive of the spacing if the spacing is ≤ 1.0μm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on “my X-FAB”. Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	μm x μm
S1M2	Minimum MET2 spacing/notch	0.28	μm
S4M2	Minimum MET2 spacing (different net, tag_60v) Note: Valid for spacing of all MET2 shapes with label tag_60v to all MET2 shapes having lower voltage classes. Refer to the design related guideline “Voltage class definitions”.	0.4	μm
S2M2	Minimum MET2 spacing to WIDE_MET2	0.6	μm
A1M2	Minimum MET2 area	0.202	μm ²
<u>R1M2</u>	Minimum ratio of MET2 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2M2</u>	Maximum ratio of MET2 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1M2P1	Maximum ratio of MET2 area to connected GATE area Note: Refer to section “Antenna Rule definitions” as well.	400.0	-
R2M2P1	Maximum ratio of MET2 area to connected GATE area Note: Refer to section “Antenna Rule definitions” as well.	400.0	-
Q1M2	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline “Voltage class definitions”.	-	-

**Figure 3.60** MET2

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ M2SLOT

M2SLOT

Name	Description	Value	Unit
W2M2	Minimum M2SLOT width	0.6	μm
W3M2	Minimum M2SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M2	Minimum M2SLOT spacing/notch	10.0	μm
S1M2M1	Minimum M2SLOT spacing to M1SLOT	2.0	μm
	Note: M2SLOT is not allowed over M1SLOT.		
E1M2M2	Minimum MET2 enclosure of M2SLOT	10.0	μm
	Note: M2SLOT without MET2 is not allowed		

Note: Insert M2SLOTS in direction of current flow.

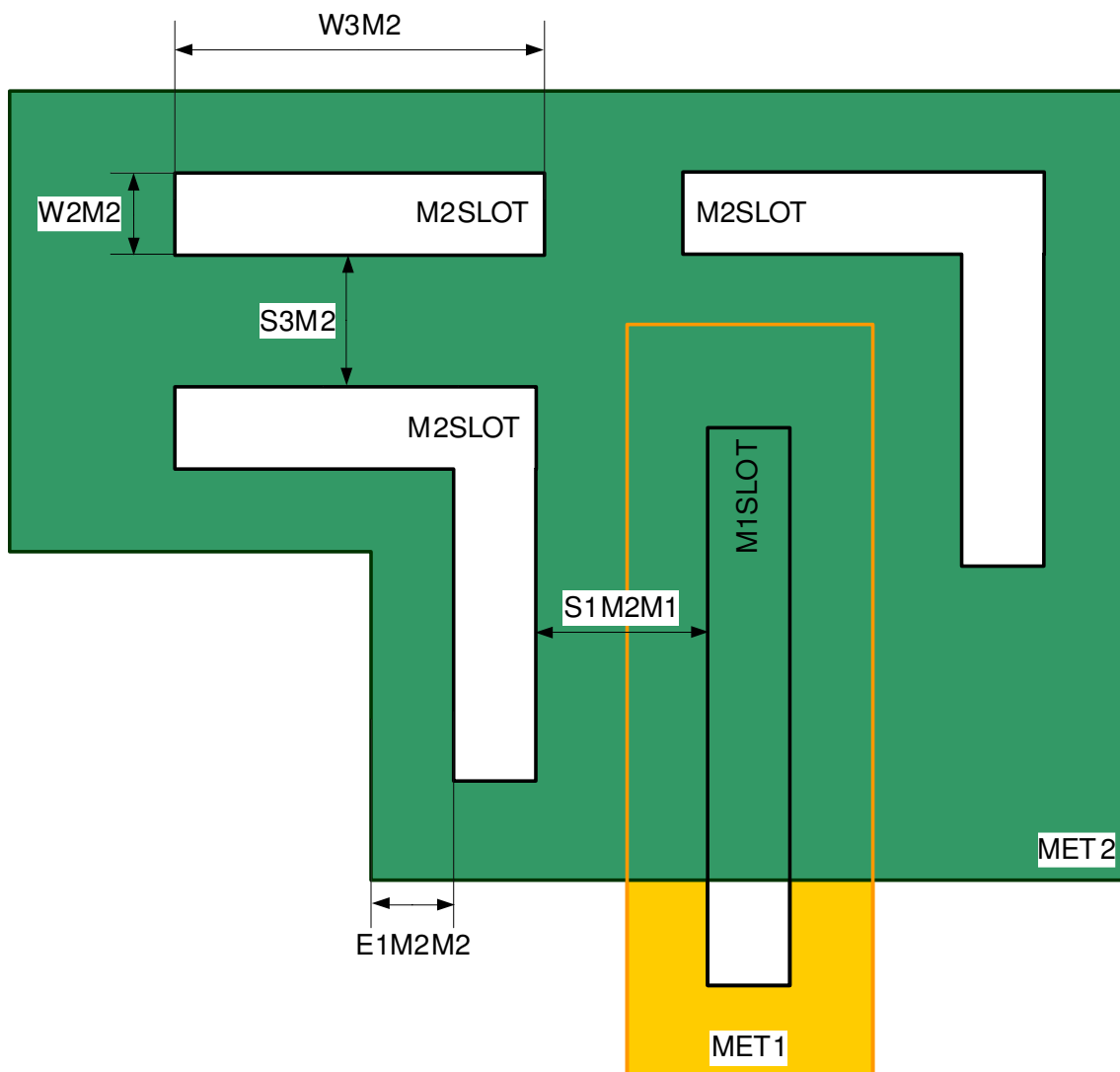


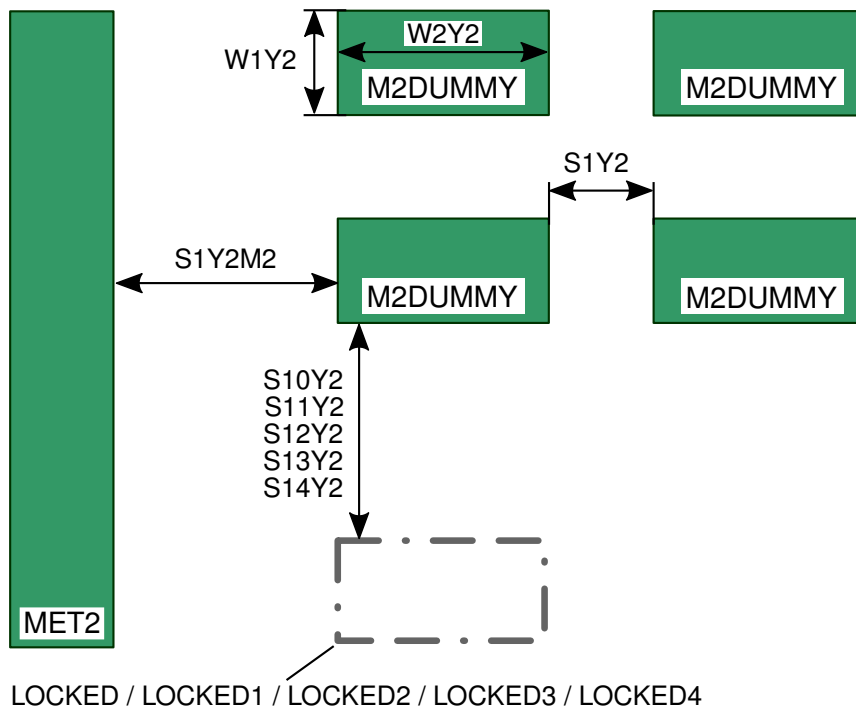
Figure 3.61 M2SLOT

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ M2DUMMY

M2DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y2	Only rectangular M2DUMMY is allowed	-	-
B2Y2	M2DUMMY overlap of MET2 is not allowed	-	-
B3Y2V1	M2DUMMY overlap of VIA1 is not allowed	-	-
B3Y2V2	M2DUMMY overlap of VIA2 is not allowed	-	-
B3Y2VT	M2DUMMY overlap of VIATP is not allowed	-	-
	Note: Only valid if module MET3 is not selected.		
W1Y2	Minimum M2DUMMY width	2.0	μm
W2Y2	Maximum M2DUMMY edge length	20.0	μm
S1Y2	Minimum M2DUMMY spacing	2.0	μm
S10Y2	Minimum M2DUMMY spacing to LOCKED	4.0	μm
S11Y2	Minimum M2DUMMY spacing to LOCKED1	4.0	μm
S12Y2	Minimum M2DUMMY spacing to LOCKED2	4.0	μm
S13Y2	Minimum M2DUMMY spacing to LOCKED3	4.0	μm
S14Y2	Minimum M2DUMMY spacing to LOCKED4	4.0	μm
S1Y2M2	Minimum M2DUMMY spacing to MET2	4.0	μm

**Figure 3.62** M2DUMMY**LOCKED, LOCKED1, LOCKED2**

This layer protects the IP core against routing (of specific layers) and is used to increase the safety of designs during IP replacement. LOCKED* layers are part of the X-FAB provided IP LEF file. ⇒

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ LOCKED, LOCKED1, LOC...

Name	Description	Value	Unit
BDLOCK	Any structure over LOCKED (ALL) or LOCKED (boundary) is not allowed	-	-
B1LOCK	Any structure over LOCKED (LOCKED1) or LOCKED1 (boundary) is not allowed Note: Except: - METTPL if (MET4 and METMID) modules are selected or MET5 module is selected - METTP if MET5 module is selected	-	-
B2LOCK	Any structure over LOCKED (LOCKED2) or LOCKED2 (boundary) is not allowed Note: Except: - METTPL if (MET3 and METMID) modules are selected or MET4 module is selected - METTP if MET4 module is selected - MET5	-	-

LOCKED3

This layer protects the IP core against routing (of specific layers) and is used to increase the safety of designs during IP replacement. LOCKED* layers are part of the X-FAB provided IP LEF file.

Name	Description	Value	Unit
B3LOCK	Any structure over LOCKED3 is not allowed Note: Except: - METTPL if METMID module is selected or MET3 module is selected - METTP if MET3 module is selected - MET5 - MET4	-	-

LOCKED4

This layer protects the IP core against routing (of specific layers) and is used to increase the safety of designs during IP replacement. LOCKED* layers are part of the X-FAB provided IP LEF file.

Name	Description	Value	Unit
B4LOCK	Any structure over LOCKED4 is not allowed Note: Except: - METTPL, METTP, MET5, MET4, MET3	-	-

CAPDEF

This layer is only used to define certain capacitors.

Name	Description	Value	Unit
B1CDEF	CAPDEF overlap of PAD is not allowed	-	-

DIODEF

This layer is used to define diodes that are to be extracted into the netlist. All diodes without DIODEF are extracted as parasitic components. DIODEF must encompass the entire P/N junction forming the diode.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ PHODEF

PHODEF

Important note: PHODEF is necessary to identify photodiodes derived from the basic diode constructions. Please make sure to draw it only for the light sensitive area. It must only be used for this purpose.

Name	Description	Value	Unit
B1D1	PHODEF without DIODEF is not allowed (except dphoa, dphob)	-	-
B1D1DF	PHODEF and DIFF without NIMP or PIMP or SBLK is not allowed (except dphoa, dphob, CATDIFF of davl#)	-	-
B2D1	PHODEF without BLKALL is not allowed (except davl#)	-	-
B3D1	PHODEF and DIFF without SBLK is not allowed (except DIFFCON sized by 0.5μm, davl#)	-	-
W1DFD1	Maximum (DIFF overlap of PHODEF) bounding box size	505.0 x 505.0	μm x μm
	Note: The bounding box is the generated minimum rectangle enclosing the polygon.		
O1SBIN	Minimum PHODEF and SBLK overlap of NIMP (except davl#)	0.22	μm
O1SBIP	Minimum PHODEF and SBLK overlap of PIMP (except davl#)	0.22	μm
Q1D1	Check for the right usage of the layer PHODEF	-	-
	Note: DRC reports this check request whenever this layer is present.		

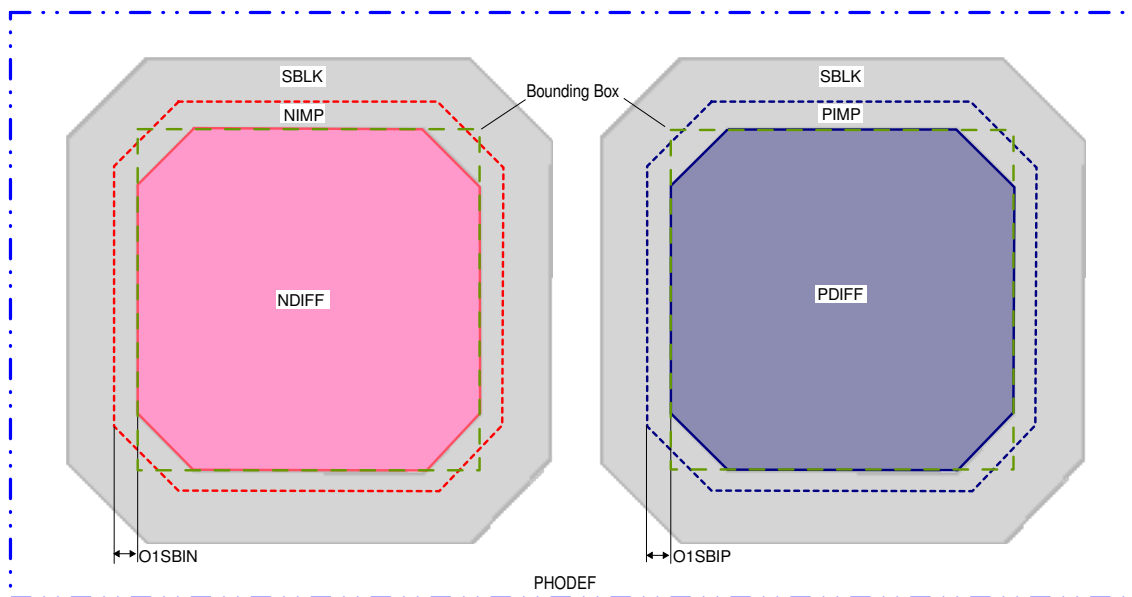


Figure 3.63 PHODEF

XFLAY

This layer is a reserved layer.

Name	Description	Value	Unit
BDXF	Not allowed to be used by customers	-	-

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.1 Layer rules→ SUBCUT

SUBCUT

Refer to the corresponding [application note](#) available on "my X-FAB" for information about this layer.

Name	Description	Value	Unit
B1SU	DIFF or *WELL crossing SUBCUT edge is not allowed	-	-
	Note: Not valid for DIFFDUMMY.		
Q1SU	Check for the right usage of the layer SUBCUT	-	-
	Note: DRC reports this check request whenever this layer is present.		

3.2.2 Device rules

nn3

Name	Description	Value	Unit
B3GA	Only rectangular GATE inside PWBLK is allowed	-	-
W4DF	Minimum GATE width	1.0	µm
W5P1	Minimum GATE length	1.0	µm
Q1GA	Recommended minimum GATE length	2.0	µm
	Note: It is strongly recommended to use a channel length of $\geq 2\mu\text{m}$ when nn3 is operated in the sub-threshold region.		

Note: MV is necessary for nn3.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ nn3

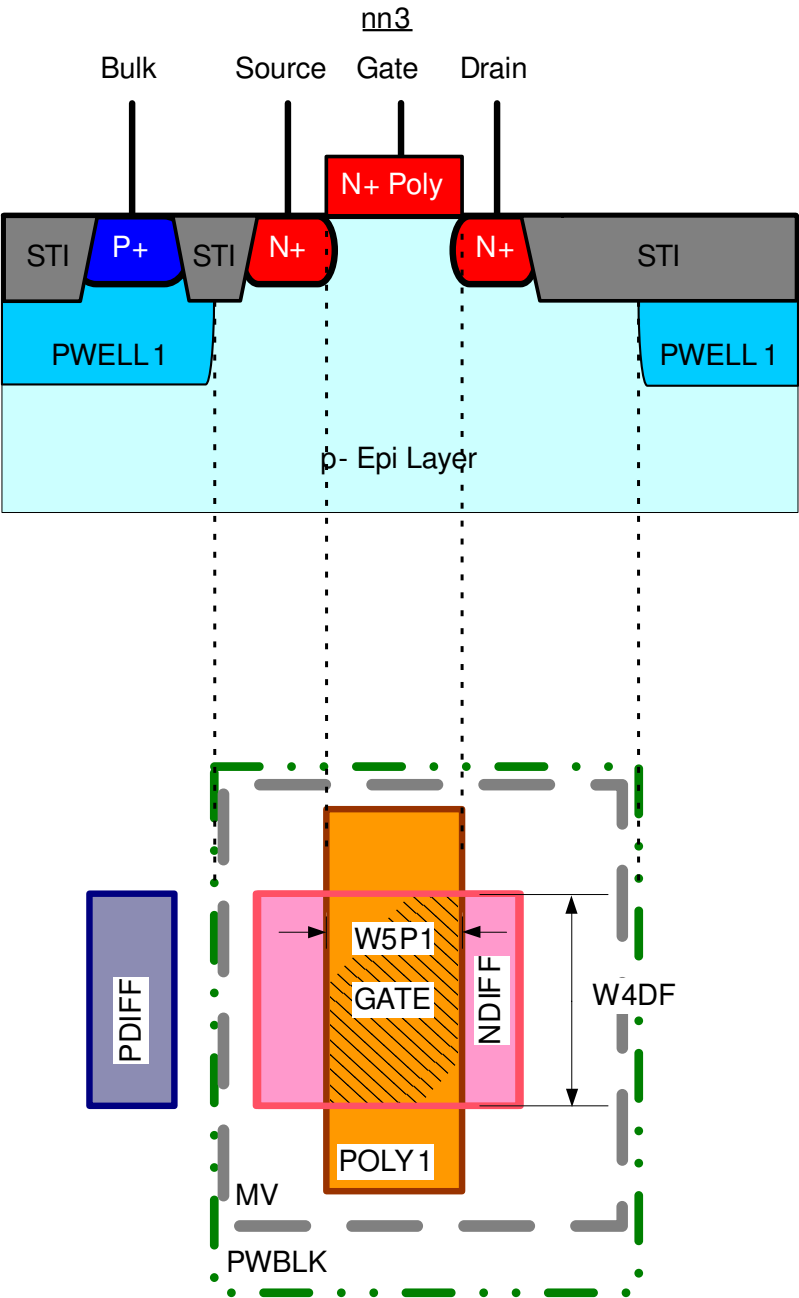


Figure 3.64 nn3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ ne3, pe3

ne3, pe3

Name	Description	Value	Unit
W3DF	Minimum GATE width	0.22	μm
	Note: The design rule check error message W3DF is also used generally for GATE areas (of any devices or shapes) having smaller GATE width than 0.22μm.		
W6P1	Minimum GATE length	0.35	μm
W7P1	Minimum GATE length	0.3	μm

Note: MV is necessary for ne3 and pe3.

Note: For more extensive LVS, it is recommended to use the related 5 or 6 terminal device.

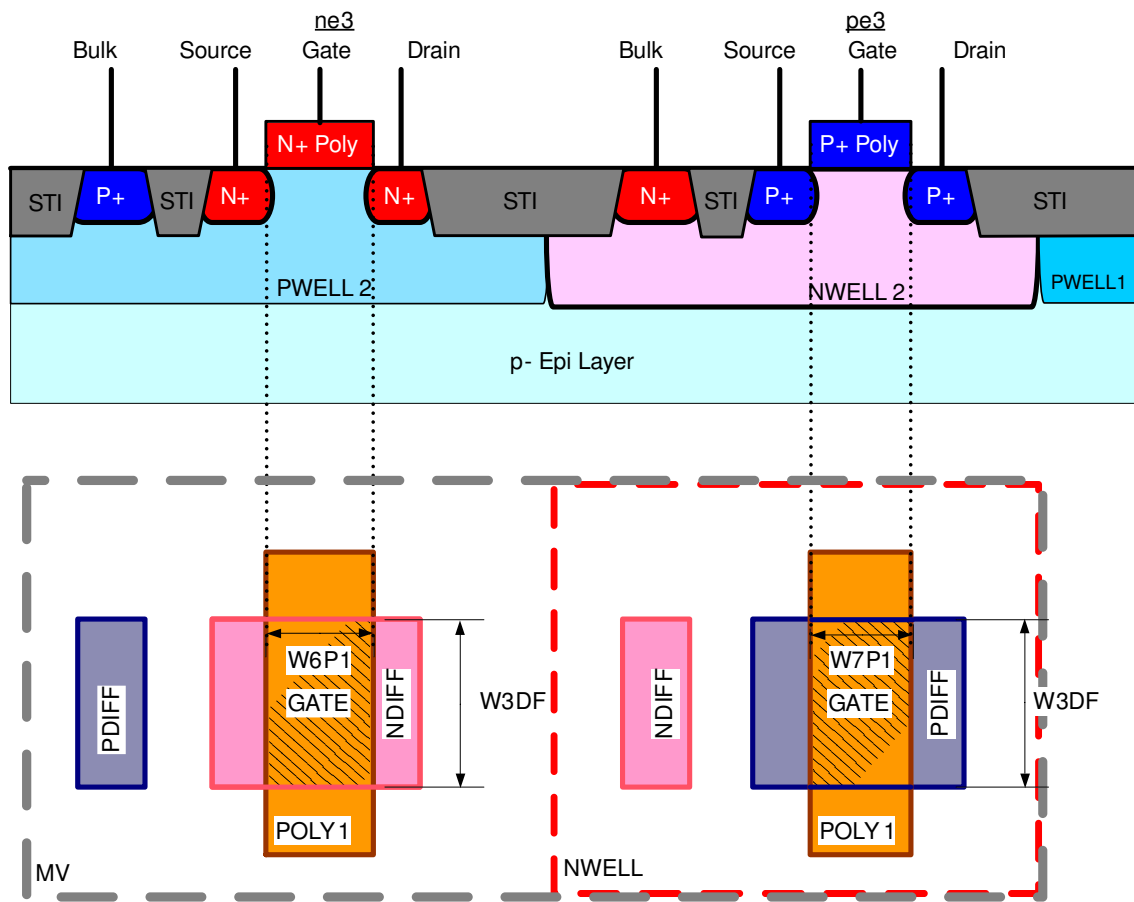


Figure 3.65 ne3, pe3

pe3_5

Note: pe3_5 device must be labeled “5T” using POLY1 (VERIFICATION) layer over the GATE.
Note: MV is necessary for pe3_5.



3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ qpva3, qpvb3, qpvc3

qpva3, qpvb3, qpvc3

This section describes the vertical PNP transistors qpva3, qpvb3 and qpvc3. These transistors use PDIFF as emitter, NWELL2 as base and PWELL2 as collector. Their layouts are predefined. They must not be changed.

qpva3

NDIFF base surrounding PDIFF emitter contact, emitter area: $2 \times 2 \mu\text{m}^2$

qpvb3

NDIFF base surrounding PDIFF emitter contact, emitter area: $5 \times 5 \mu\text{m}^2$

qpvc3

NDIFF base surrounding PDIFF emitter contact, emitter area: $10 \times 10 \mu\text{m}^2$

Note: MV is necessary for qpva3, qpvb3 and qpvc3.

Note: qpva3, qpvb3 and qpvc3 must not be drawn inside DNWELL, DNWELLMV, HVNWELL, HVPWELL, or DEPL.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ qpva3, qpvb3, qpvc3

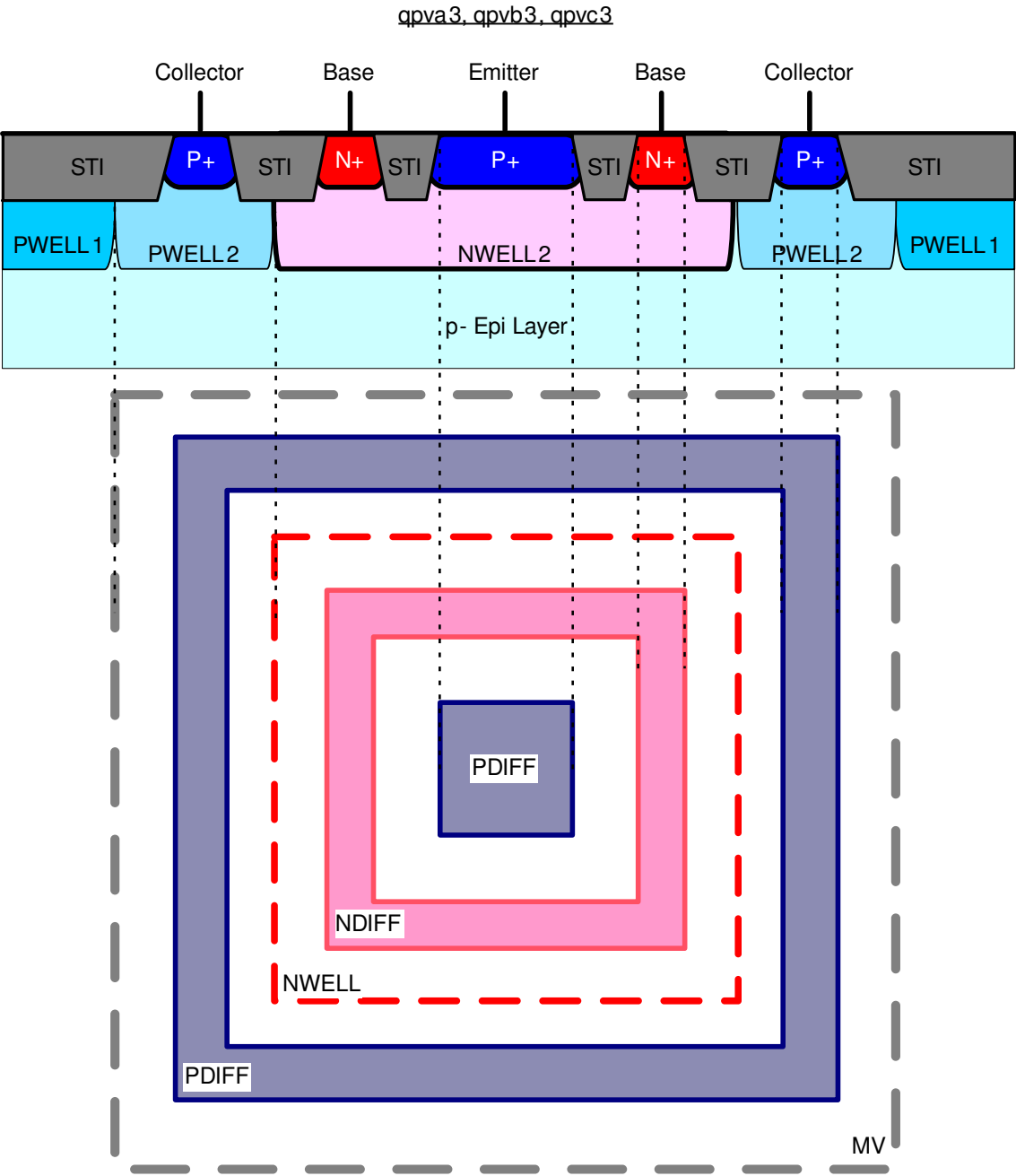


Figure 3.67 qpva3, qpvb3, qpvc3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rdn3, rdp3

rdn3, rdp3

Name	Description	Value	Unit
B1DN	rdn/rdn3 inside ISOPW, DEPL, SCI, HVPWELL, PDD or PDF is not allowed Note: rdn, rdn3 are only allowed in ISOPW in case on Gate in IO blocks and ESD.	-	-
W5DF	Minimum rdn/rdn3, rdp/rdp3 width	0.42	μm

Note: MV is necessary for rdn3 and rdp3.

Note: Recommended minimum number of squares is $L/W \geq 5$.

Note: rdn, rdn3 resistor definition: SBLK and PTYPE_WELL and NDIFF (except GATE, PHODIO).

Note: rdp, rdp3 resistor definition: SBLK and NWELL and PDIFF (except GATE, PHODIO).

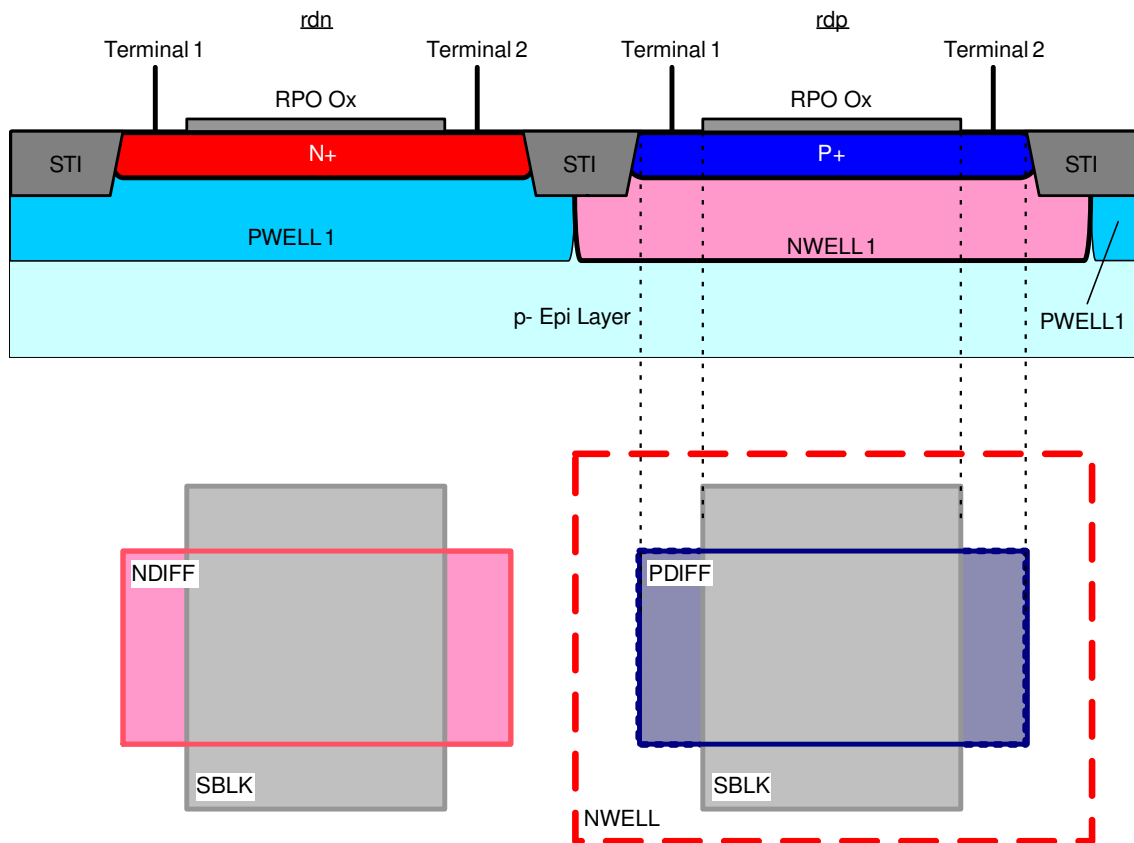


Figure 3.68 rdn, rdp, rdn3, rdp3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rnw3

rnw3

Name	Description	Value	Unit
B1NW	NW_VERIFY overlap of DIFF is not allowed	-	-
	Note: NW_VERIFY edge must touch rnw/rnw3 NDIFF-contact edge.		
B2NW	NW_VERIFY overlap of DNWELL or DNWELLMV is not allowed	-	-
B3GH	HVGX overlap of rnw is not allowed	-	-
W2NW	Minimum rnw, rnw3 width	2.0	μm

Note: Recommended minimum number of squares is $L/W \geq 5$.

Note: It is recommended to place DIFF stripes close to resistor.

Note: rnw and rnw3 resistor definition: NWELL and NW_VERIFY.

Note: MV is necessary for rnw3.

Note: Recommended maximum width is 10μm.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rnw3

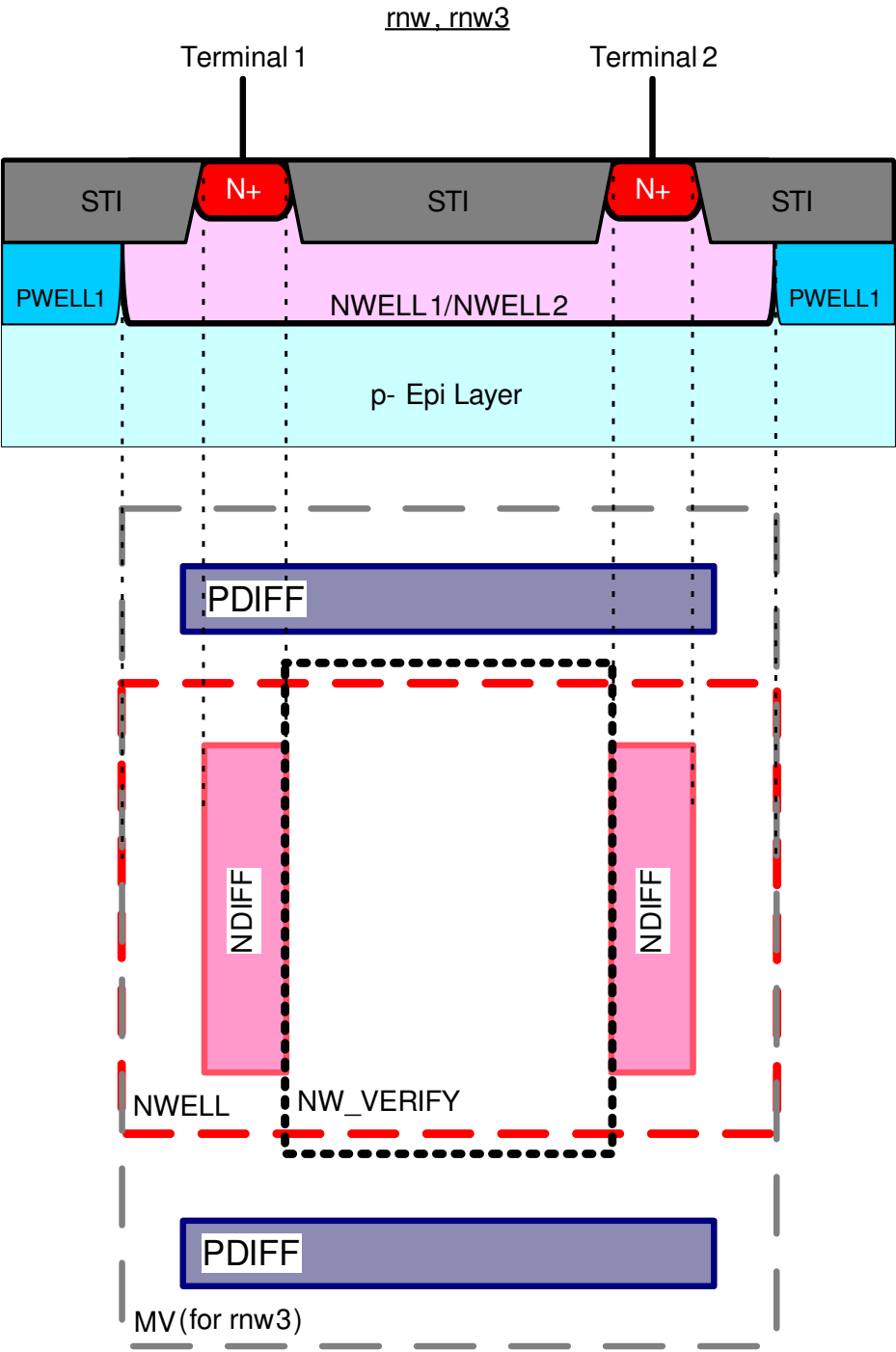


Figure 3.69 rnw, rnw3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rdn3s

rdn3s

Name	Description	Value	Unit
B2DN	rdn3s overlap of ISOPW, PHODEF or DIODEF is not allowed	-	-
B4CT	CONT is not allowed within rdn3s	-	-
B7DF	rdn3s overlap of POLY1 is not allowed	-	-
W6DF	Minimum width	0.7	μm
<i>W7DF</i>	Maximum width	2.0	μm
W8DF	Minimum length	5.0	μm
E1DNDF	Minimum NDIFF extension beyond rdn3s	0.32	μm
E3INDF	Minimum NIMP extension beyond rdn3s	0.18	μm

Note: Recommended minimum number of squares is $L/W \geq 5$.

Note: This device is not recommended to be used for precision circuits.

Note: MV is necessary for rdn3s.

Note: rdn3s resistor definition: NDIFF and DIFF_VERIFY and PWELL2 and not SBLK.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rdn3s

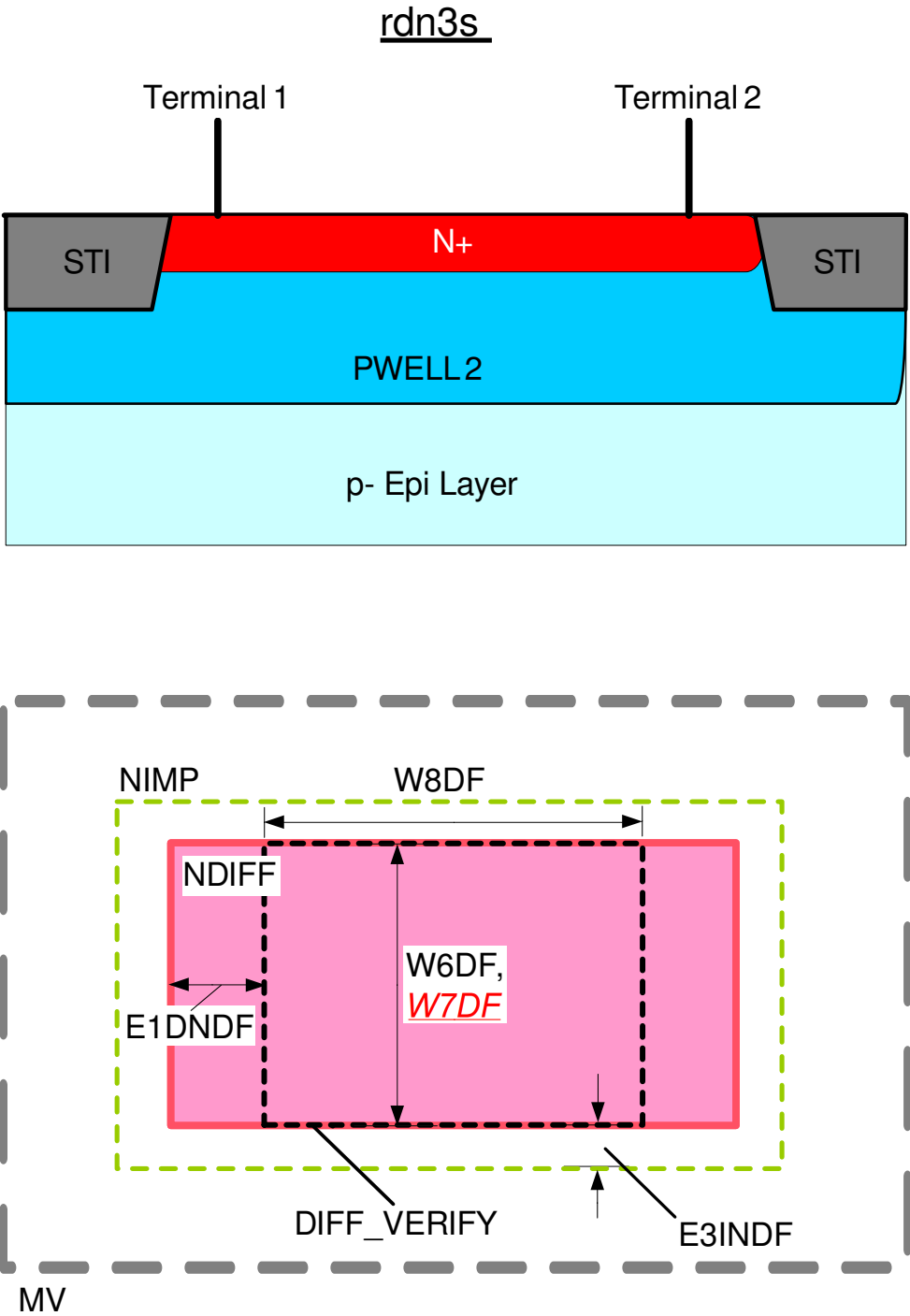


Figure 3.70 rdn3s

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rnp1, rnp1_3, rpp1, ...

rnp1, rnp1_3, rpp1, rpp1_3, rpp1s, rpp1s_3

Name	Description	Value	Unit
B10P1	rpp1s_3 overlap rnpw/rnw3/rdnwmv is not allowed	-	-
B2CT	CONT is not allowed within rpp1s#	-	-
B6P1	rnp1_3 overlap rnpw/rnw3/rdnwmv is not allowed	-	-
B8P1	rpp1_3 overlap rnpw/rnw3/rdnwmv is not allowed	-	-
B5P1	rnp1_3 crossing NTYPE_WELL or PTYPE_WELL or SUBCUT edge is not allowed	-	-
B7P1	rpp1_3 crossing NTYPE_WELL or PTYPE_WELL or SUBCUT edge is not allowed	-	-
B9P1	rpp1s_3 crossing NTYPE_WELL or PTYPE_WELL or SUBCUT edge is not allowed	-	-
W3P1	Minimum rnp1#, rpp1# width	0.42	μm
S1INP1	Minimum NIMP spacing to rnp1# or rpp1s#	0.26	μm
S1IPP1	Minimum PIMP spacing to rnp1#	0.26	μm
E1INP1	Minimum NIMP extension beyond rnp1#	0.18	μm
E1IPP1	Minimum PIMP extension beyond rpp1# or rpp1s#	0.18	μm

Note: Recommended minimum number of squares is $L/W \geq 5$.

Note: Do not use dog-bone at the end of POLY1 resistor for CONT pickup.

Note: CONT array for POLY1 resistor should be a single column.

Note: rnp1# resistor definition: POLY1 and SBLK and NIMP. (except GATE oversized by 0.22μm)

Note: rnp1_3 device must be labeled "rnp1_3" using POLY1 (VERIFICATION) layer.

Note: rpp1# resistor definition: POLY1 and SBLK and PIMP. (except GATE oversized by 0.22μm)

Note: rpp1_3 device must be labeled "rpp1_3" using POLY1 (VERIFICATION) layer.

Note: rpp1s# resistor definition: POLY1 and POLY1_VERIFY and PIMP. (except rpp1 oversized by 0.22 μm and not GATE)

Note: rpp1s_3 device must be labeled "rpp1s_3" using POLY1 (VERIFICATION) layer.

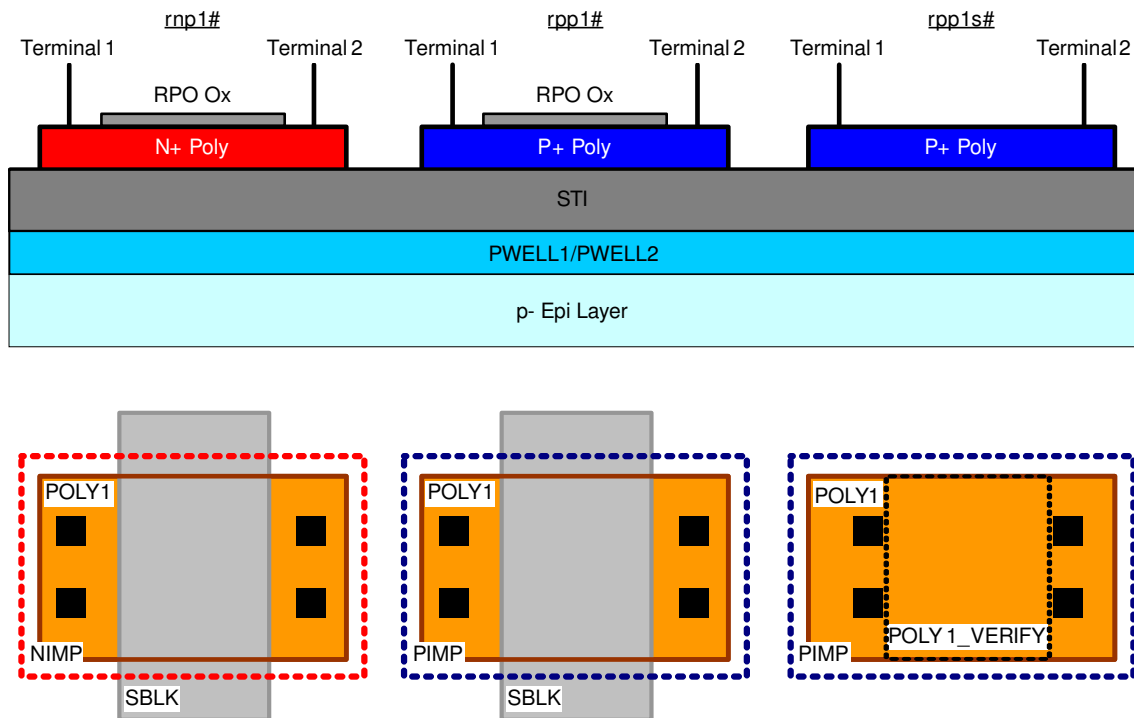


Figure 3.71 rnp1#, rpp1#, rpp1s#

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ rm1, rm2

rm1, rm2

Name	Description	Value	Unit
B2M1	CONT or VIA1 is not allowed within rm1	-	-
B2M2	VIA1 is not allowed within rm2	-	-
B4M2	VIA2 is not allowed within rm2	-	-
	Note: Valid if MET3 module is selected.		

Note: rm1 resistor definition: MET1 and M1VERIFY.

Note: rm2 resistor definition: MET2 and M2VERIFY.



Figure 3.72 rm1, rm2

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ mosvc3

mosvc3

Name	Description	Value	Unit
B4GA	Only rectangular GATE is allowed	-	-
W38GA	Minimum GATE length	2.0	µm
W39GA	Minimum GATE width	2.0	µm

Note: MV is necessary for mosvc3.

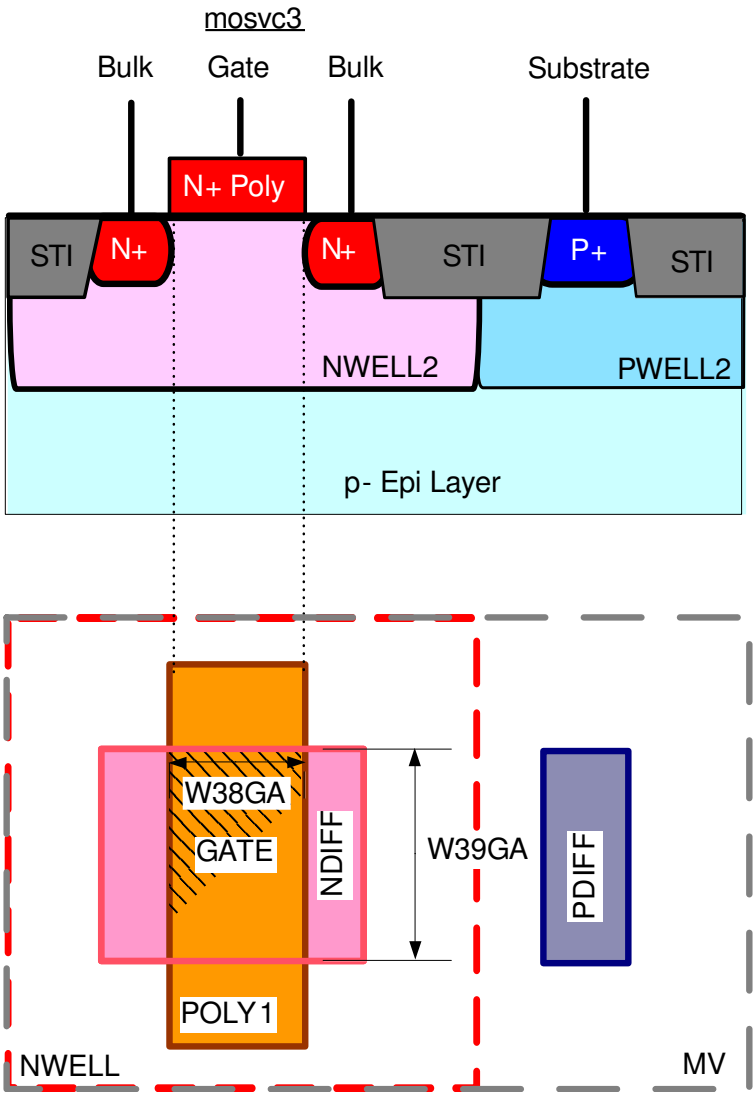


Figure 3.73 mosvc3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ csf2p

csf2p

The layout of the device csf2p is fixed and must not be changed. A single cell instance has an area of $4.48\text{ }\mu\text{m} \times 10.80\text{ }\mu\text{m}$.

Note: Higher values may be achieved by the formation of arrays.

Note: CAPDEF is necessary for csf2p.

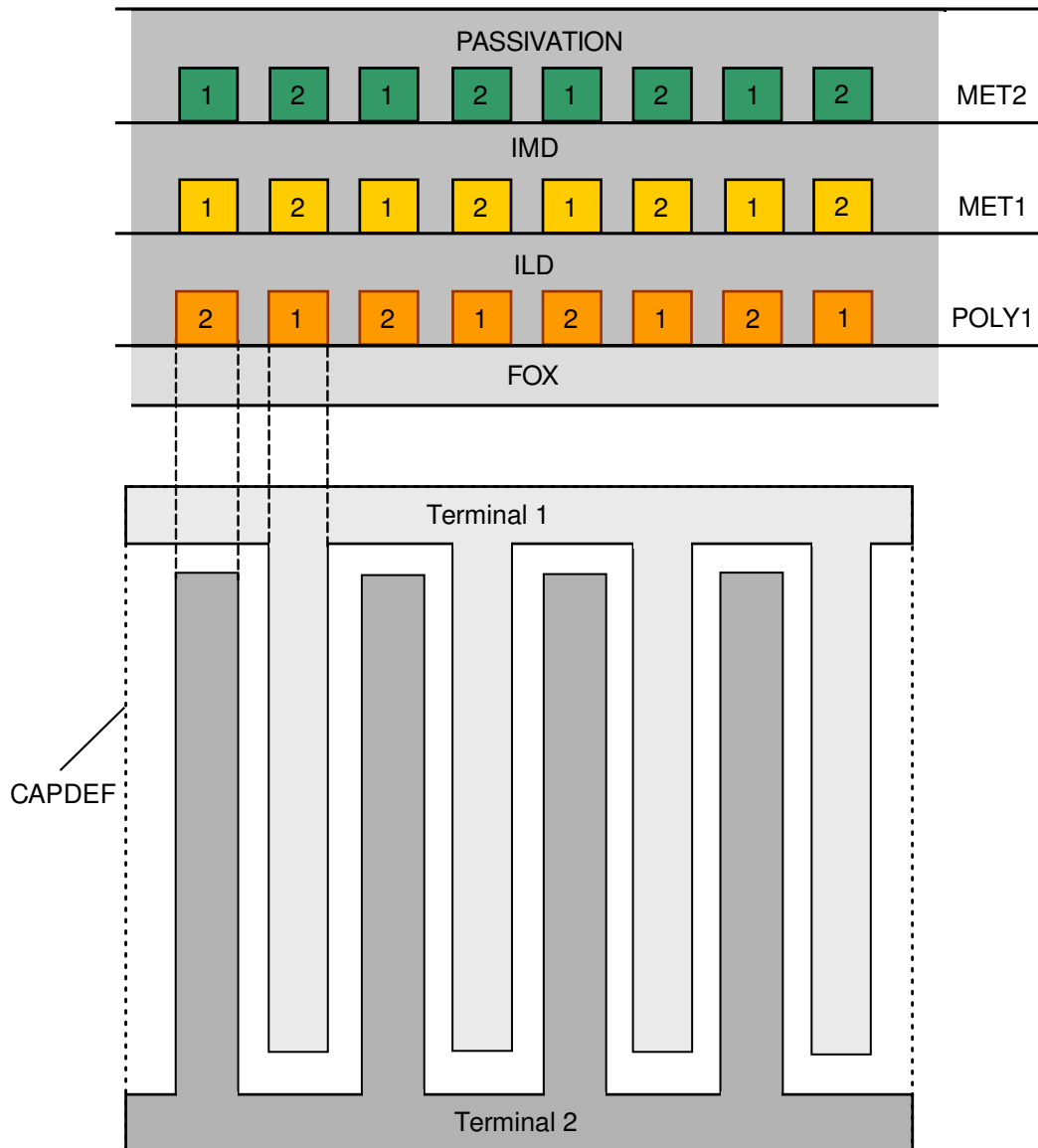


Figure 3.74 csf2p

3. Layer and Device rules → 3.2 MOS3LP main module → 3.2.2 Device rules → dn3, dp3, dnw3

dn3, dp3, dnw3

Note: MV is necessary for dn3, dp3 and dnw3.

Note: The layer DIODEF must enclose the pn junction, crossing the pn junction is not allowed.

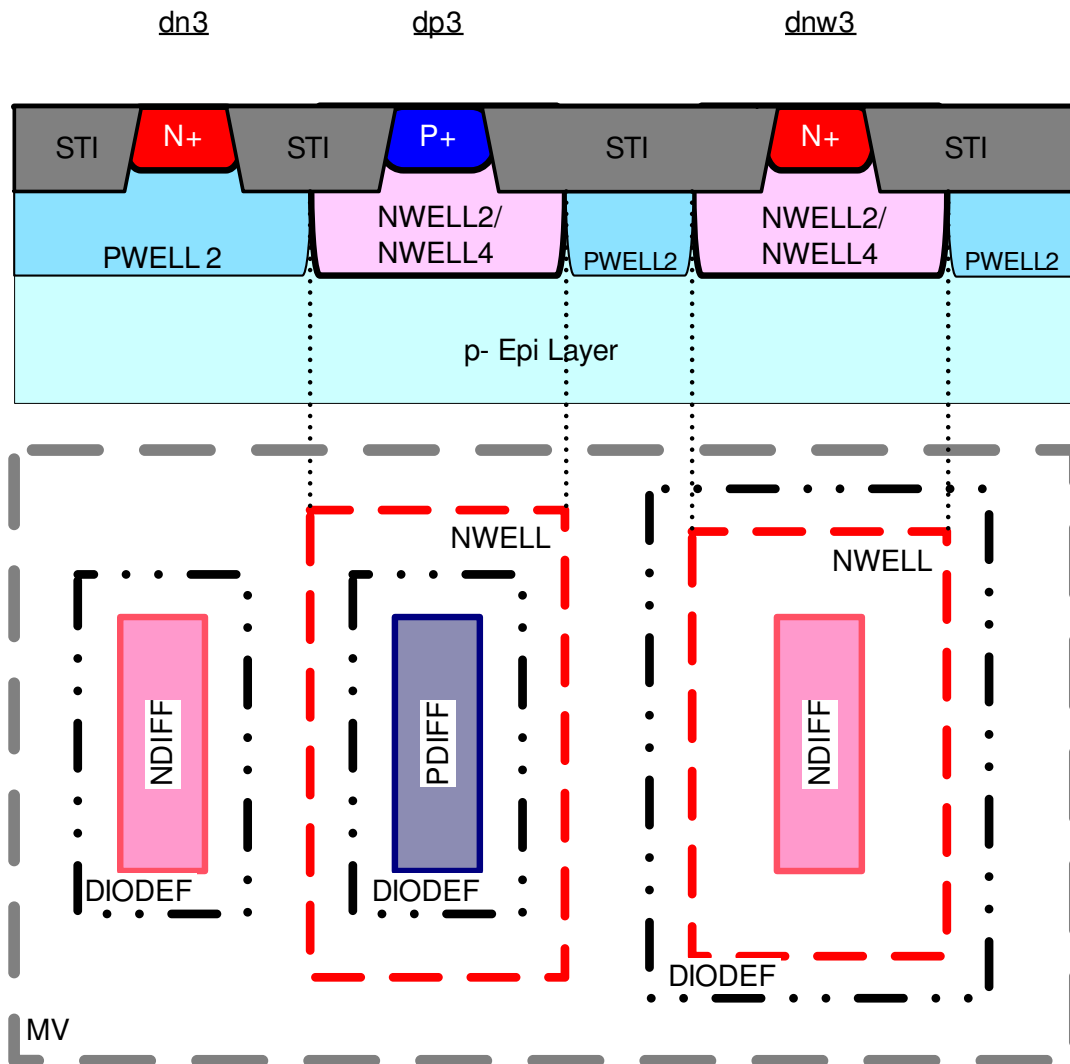


Figure 3.75 dn3, dp3, dnw3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ dnn3

dnn3

Note: MV is necessary for dnn3.

Note: The layer DIODEF must enclose the pn junction, crossing the pn junction is not allowed.

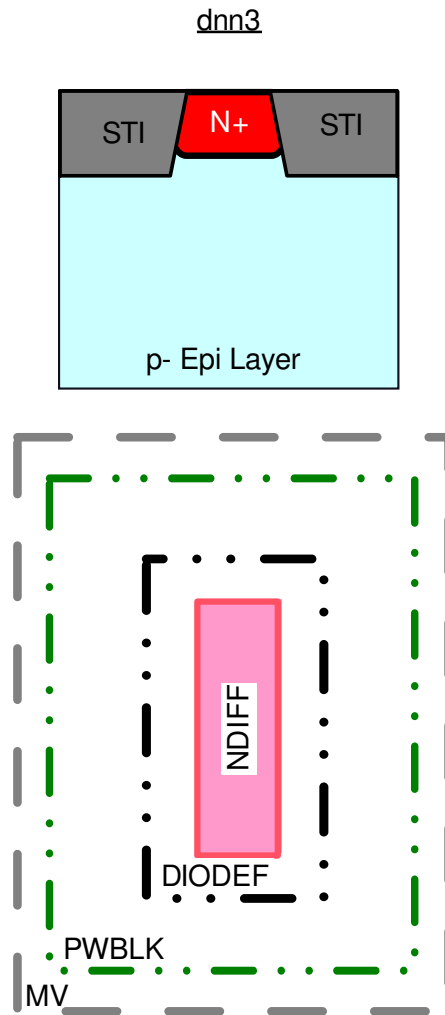


Figure 3.76 dnn3

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ dpol

dpol

The layout of dpol is predefined and scalable concerning device width only. All other dimensions must not be changed.

Name	Description	Value	Unit
B11P1	Only rectangular dpol body is allowed	-	-
B12P1	dpol body (oversized by 1.5 µm) overlap of DIFF is not allowed	-	-
B13P1	dpol body (oversized by 1.0 µm) overlap of MET1, MET2, MET3, MET4, MET5, METTP or METTPL is not allowed	-	-
B14P1	dpol body (oversized by 1.0 µm) without BLKALL is not allowed	-	-
W12P1	Minimum dpol width	1.0	µm
W13P1	Maximum dpol width	200.0	µm
W14P1	Fixed dpol body length	0.85	µm
E3IPP1	Minimum PIMP extension beyond POLY1	0.18	µm
E4INP1	Minimum NIMP extension beyond POLY1	0.18	µm
O1INSB	Fixed SBLK overlap of (POLY and NIMP)	1.0	µm
O1IPSB	Fixed SBLK overlap of (POLY1 and PIMP)	1.0	µm

Note: dpol body is defined as (POLY and SBLK) and not (NIMP or PIMP).

Note: The layer DIODEF must enclose the pn junction, crossing the pn junction is not allowed.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ dpol

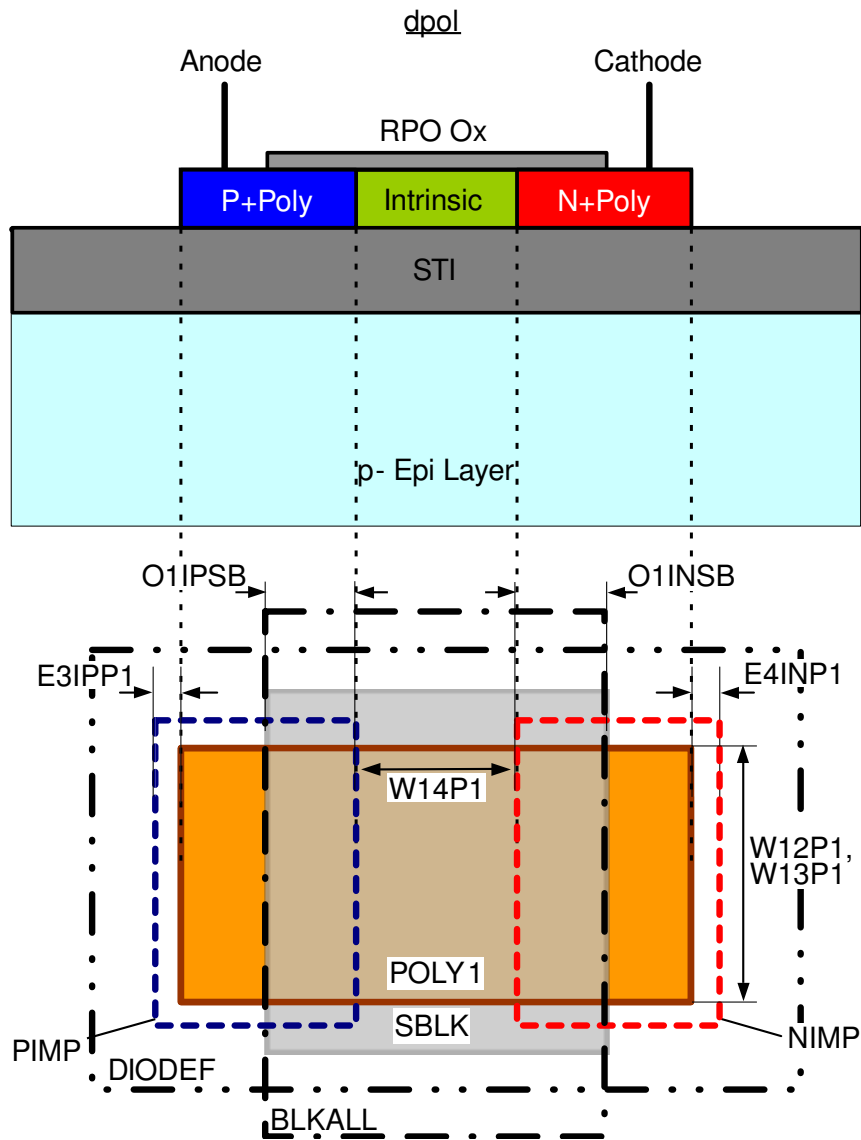


Figure 3.77 dpol

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ pfuse

pfuse

The layout of pfuse is predefined. It must not be changed. The other rules support the fixed dimensions of the layout or describe the relation to adjacent structures.

Name	Description	Value	Unit
B2P1	pfuse overlap of MET1, MET2, MET3, MET4, MET5, METTP or METTPL is not allowed	-	-
B3P1	pfuse overlap of SBLK or DIFF is not allowed	-	-
B4P1	pfuse without NIMP and BLKALL is not allowed	-	-
S1DFP1	Minimum DIFF spacing to pfuse Note: Valid in pfuse width direction only.	2.0	µm
S1M1P1	Minimum MET1 spacing to pfuse Note: Valid in pfuse width direction only.	0.8	µm
S1M2P1	Minimum MET2 spacing to pfuse Note: Valid in pfuse width direction only.	0.8	µm
S1M3P1	Minimum MET3 spacing to pfuse Note: Valid in pfuse width direction only.	1.7	µm
S1M4P1	Minimum MET4 spacing to pfuse Note: Valid in pfuse width direction only.	1.7	µm
S1M5P1	Minimum MET5 spacing to pfuse Note: Valid in pfuse width direction only.	1.7	µm
S1MLP1	Minimum METTPL spacing to pfuse Note: Valid in pfuse width direction only.	1.7	µm
S1MTP1	Minimum METTP spacing to pfuse Note: Valid in pfuse width direction only.	1.7	µm
S2P1	Minimum POLY1 spacing to pfuse Note: Valid in pfuse width direction only.	2.0	µm
S2SBP1	Minimum SBLK spacing to pfuse Note: Valid in pfuse width direction only.	2.0	µm
E1BAP1	Minimum BLKALL enclosure of pfuse	1.7	µm
E1NWP1	Fixed NWELL enclosure of pfuse Note: NWELL is an electrically floating area without any connections.	1.7	µm
E3INP1	Minimum NIMP enclosure of POLY1	0.3	µm

Note: pfuse device must be labeled "PFUSE" using POLY1 (VERIFICATION) layer.

Note: pfuse device must have POLY1 (VERIFICATION) layer.

3. Layer and Device rules → 3.2 MOS3LP main module→ 3.2.2 Device rules→ pfuse

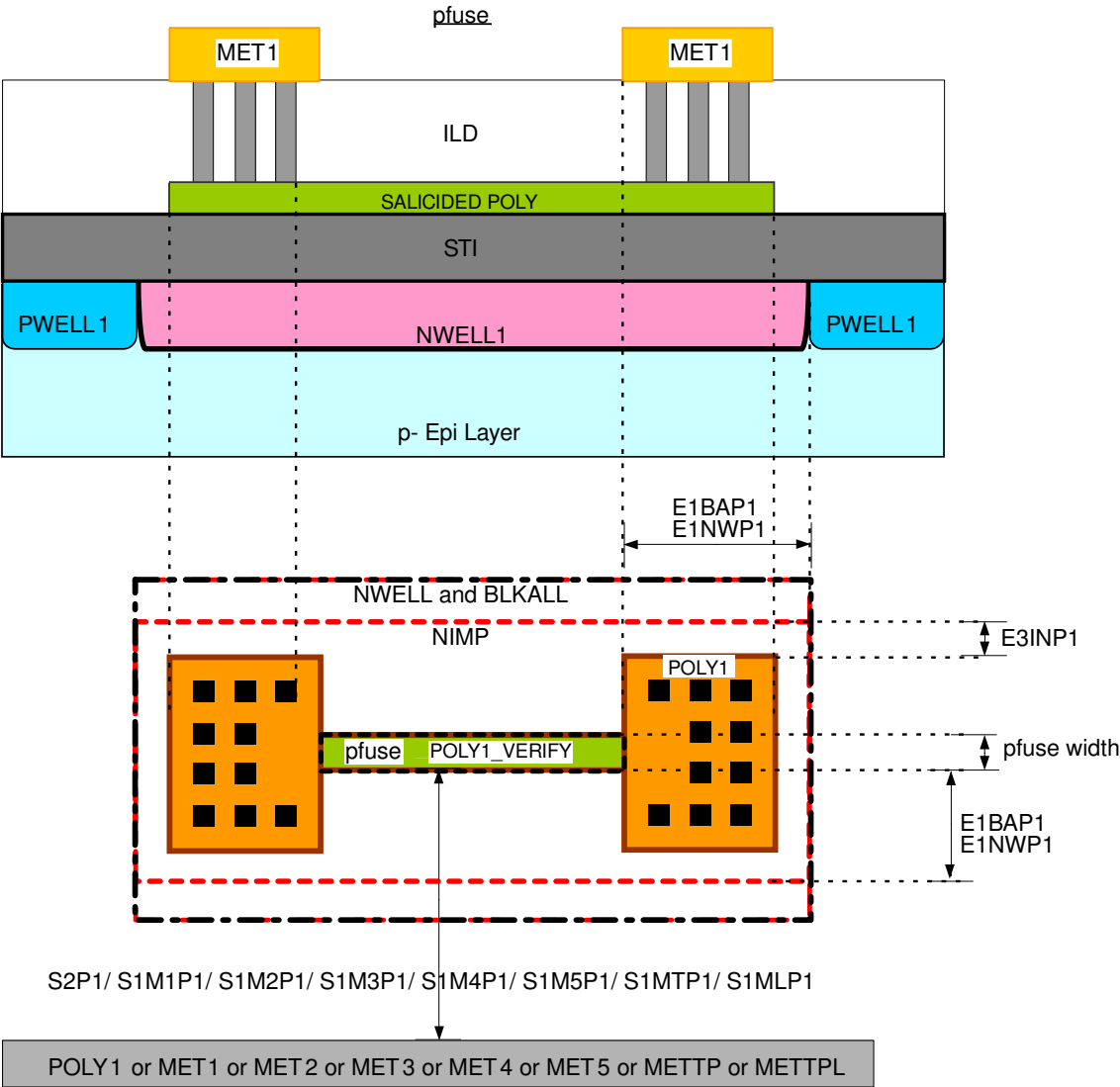


Figure 3.78 pfuse

3. Layer and Device rules → 3.3 MET3 module

3.3 MET3 module**3.3.1 Layer rules****VIA2**

Name	Description	Value	Unit
B1V2	VIA2 must be covered by MET2 and MET3	-	-
B2V2	VIA2 stripes are only allowed to bend at 135 degrees	-	-
W2V2	Fixed VIA2 stripe width	0.26	µm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1V2	Fixed VIA2 size (except dphoc#, dphod#, OPTOVIA2, PHODIO)	0.26 x 0.26	µm x µm
S1V2	Minimum VIA2 spacing	0.26	µm
S9V2	Minimum VIA2 stripe to VIA2 spacing	1.0	µm
E1M2V2	Minimum MET2 enclosure of VIA2	0.01	µm
E1M3V2	Minimum MET3 enclosure of VIA2	0.01	µm
E2M2V2	Minimum MET2 enclosure of VIA2 (in one direction of VIA2 corner)	0.06	µm
E2M3V2	Minimum MET3 enclosure of VIA2 (in one direction of VIA2 corner)	0.06	µm
E3M3V2	Minimum MET3 enclosure of VIA2 stripe	0.1	µm
E8M2V2	Minimum MET2 enclosure of VIA2 stripe	0.1	µm
R1V2	Maximum ratio of VIA2 area to connected GATE area	20.0	-
Q20V2	Check for the right usage of OPTOVIA2	-	-
	Note: VIA2 stripes must only be used for light shielding purpose.		
	Note: DRC reports this check request whenever this layer is present.		

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.3 MET3 module→ 3.3.1 Layer rules→ VIA2

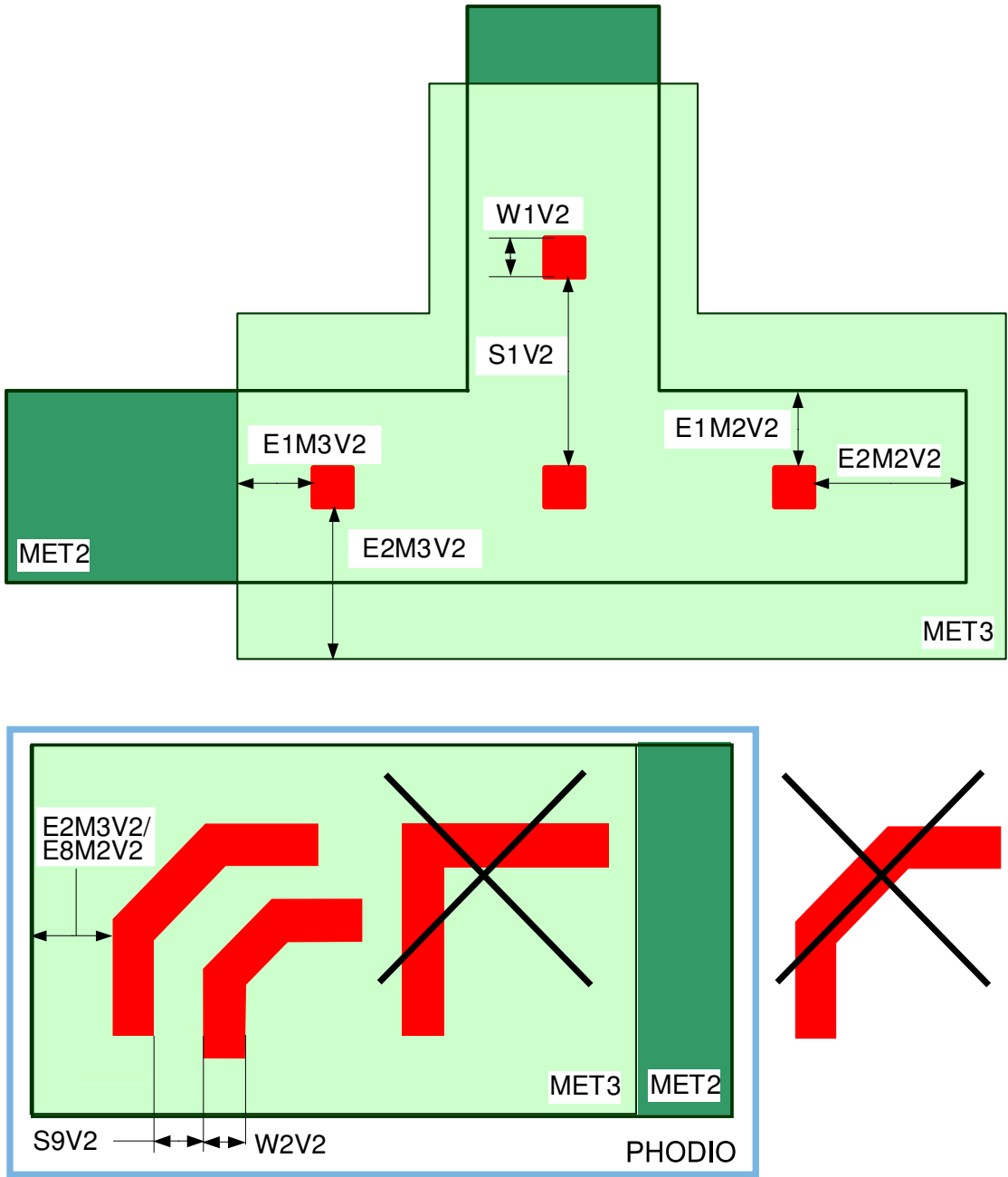
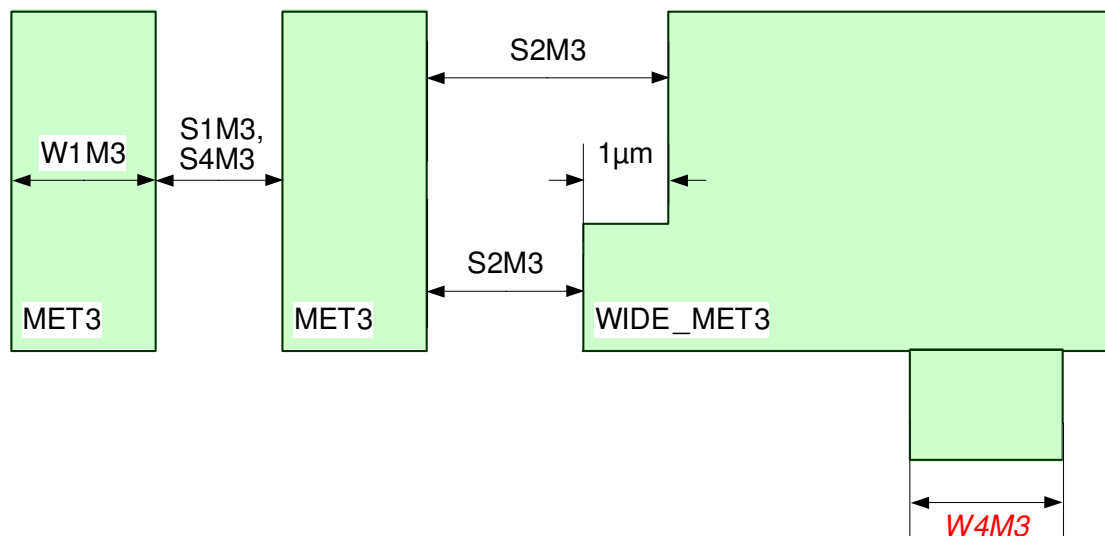


Figure 3.79 VIA2

3. Layer and Device rules → 3.3 MET3 module → 3.3.1 Layer rules → MET3

MET3

Name	Description	Value	Unit
B1M3	All MET3 tracks > 35µm wide to be slotted (except Pads)	-	-
W1M3	Minimum MET3 width	0.28	µm
<u>W4M3</u>	Minimum MET3 width joining wide MET3 track (> 35 µm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 µm from the main track prior to narrowing it.	10.0	µm
<u>W5M3</u>	Maximum MET3 region size Note: MET3 regions are defined as MET3 shapes (single MET3 shapes or a bundle of MET3 shapes, with width > 2.0µm, inclusive of the spacing if the spacing is ≤ 1.0µm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on "my X-FAB". Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	µm x µm
S1M3	Minimum MET3 spacing/notch	0.28	µm
S4M3	Minimum MET3 spacing (different net, tag_60v) Note: Valid for spacing of all MET3 shapes with label tag_60v to all MET3 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.4	µm
S2M3	Minimum MET3 spacing to WIDE_MET3	0.6	µm
A1M3	Minimum MET3 area	0.202	µm ²
<u>R1M3</u>	Minimum ratio of MET3 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2M3</u>	Maximum ratio of MET3 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1M3P1	Maximum ratio of MET3 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
R2M3P1	Maximum ratio of MET3 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
Q1M3	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

**Figure 3.80** MET3

3. Layer and Device rules → 3.3 MET3 module → 3.3.1 Layer rules → M3SLOT

M3SLOT

Name	Description	Value	Unit
W2M3	Minimum M3SLOT width	0.6	μm
W3M3	Minimum M3SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M3	Minimum M3SLOT spacing/notch	10.0	μm
S1M3M2	Minimum M3SLOT spacing to M2SLOT	2.0	μm
	Note: M3SLOT is not allowed over M2SLOT.		
E1M3M3	Minimum MET3 enclosure of M3SLOT	10.0	μm
	Note: M3SLOT without MET3 is not allowed.		

Note: Insert M3SLOTS in direction of current flow.

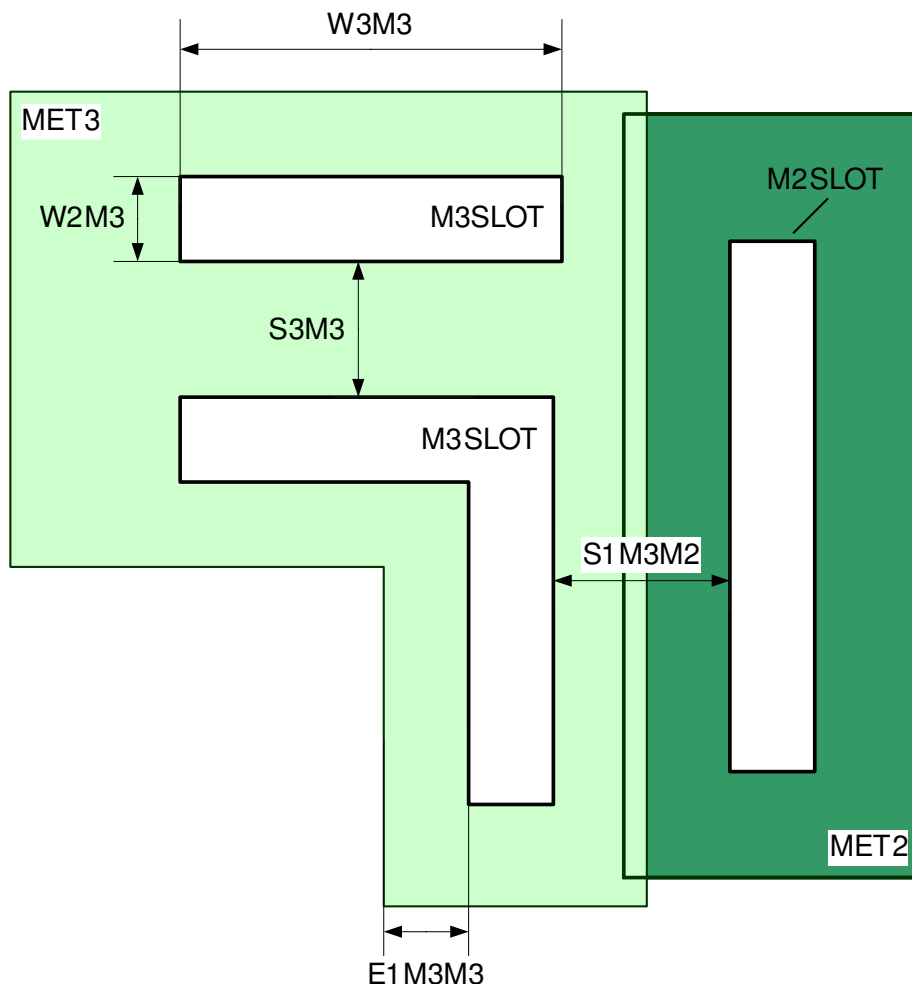


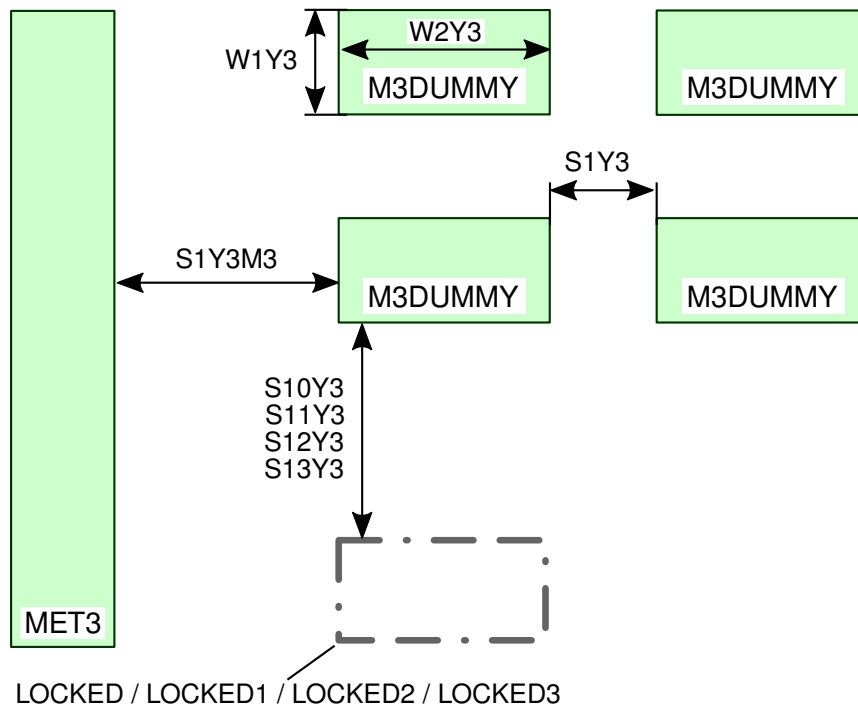
Figure 3.81 M3SLOT

3. Layer and Device rules → 3.3 MET3 module → 3.3.1 Layer rules → M3DUMMY

M3DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y3	Only rectangular M3DUMMY is allowed	-	-
B2Y3	M3DUMMY overlap of MET3 is not allowed	-	-
B3Y3V2	M3DUMMY overlap of VIA2 is not allowed	-	-
B3Y3V3	M3DUMMY overlap of VIA3 is not allowed	-	-
B3Y3VT	M3DUMMY overlap of VIATP is not allowed	-	-
	Note: Only valid if module MET4 is not selected.		
W1Y3	Minimum M3DUMMY width	2.0	μm
W2Y3	Maximum M3DUMMY edge length	20.0	μm
S1Y3	Minimum M3DUMMY spacing	2.0	μm
S10Y3	Minimum M3DUMMY spacing to LOCKED	4.0	μm
S11Y3	Minimum M3DUMMY spacing to LOCKED1	4.0	μm
S12Y3	Minimum M3DUMMY spacing to LOCKED2	4.0	μm
S13Y3	Minimum M3DUMMY spacing to LOCKED3	4.0	μm
S1Y3M3	Minimum M3DUMMY spacing to MET3	4.0	μm

**Figure 3.82** M3DUMMY**3.3.2 Device rules****rm3**

Name	Description	Value	Unit
B2M3	VIA2 is not allowed within rm3	-	-

⇒

3. Layer and Device rules → 3.3 MET3 module→ 3.3.2 Device rules→ rm3

Name	Description	Value	Unit
B3M3	VIATP is not allowed within rm3	-	-
	Note: Valid if MET4 module is not selected.		
B4M3	VIA3 is not allowed within rm3	-	-
	Note: Valid if MET4 module is selected.		

Note: rm3 resistor definition: MET3 and M3VERIFY.

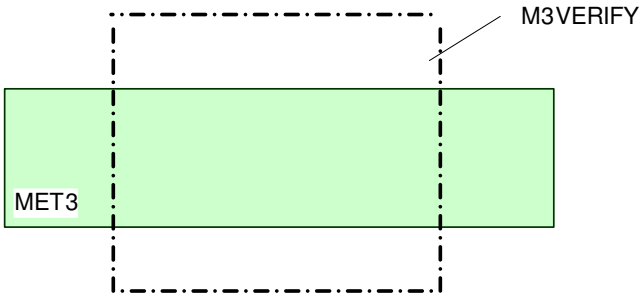


Figure 3.83 rm3

csandwt3

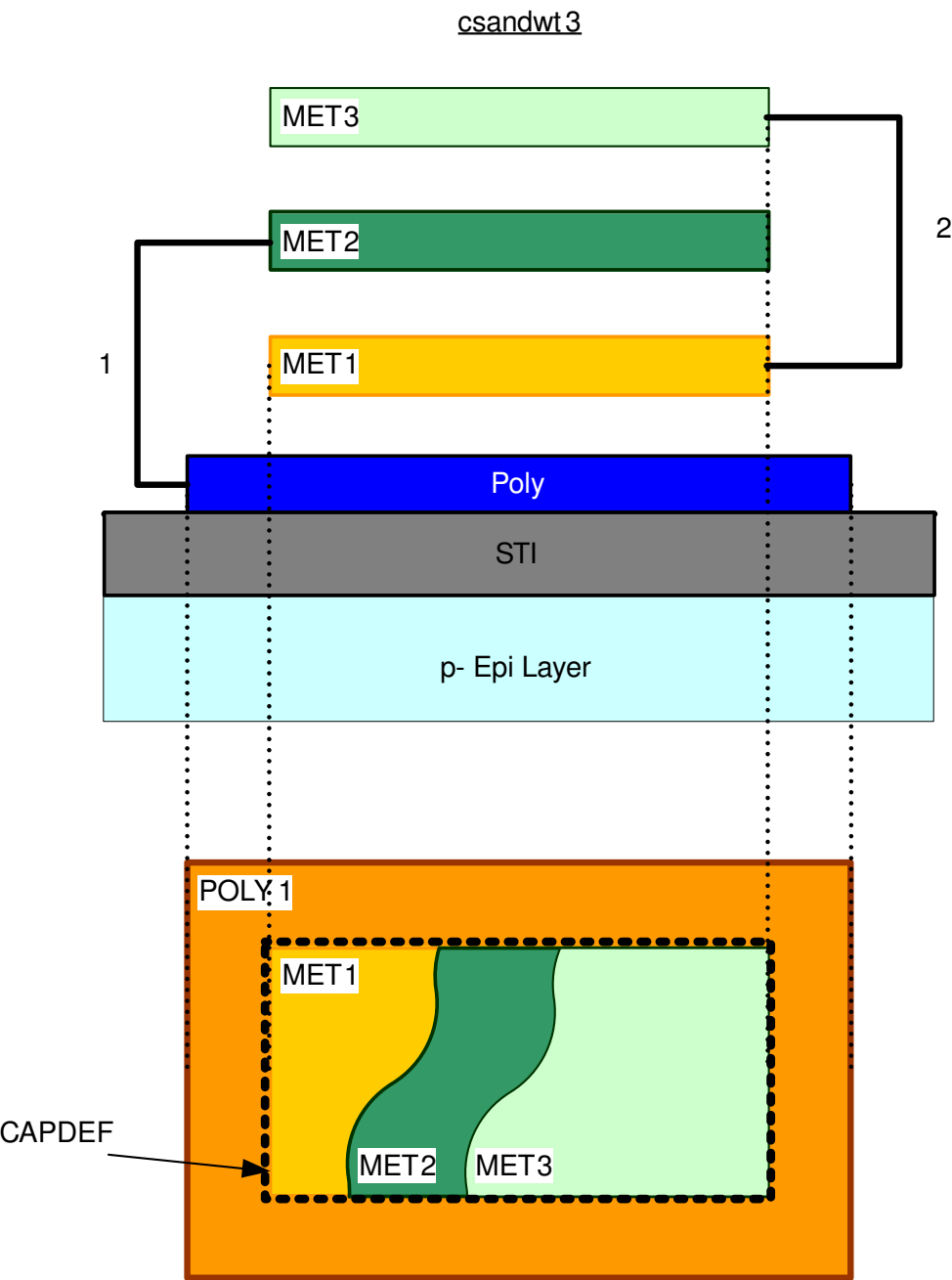


Figure 3.84 csandwt3

3. Layer and Device rules → 3.3 MET3 module → 3.3.2 Device rules → csf3p, csf3, csf3a

csf3p, csf3, csf3a

The layouts of the devices csf3p, csf3 and csf3a are fixed and must not be changed. For csf3p and csf3, a single cell instance has an area of $4.48\text{ }\mu\text{m} \times 10.80\text{ }\mu\text{m}$. For csf3a, a single cell instance has an area of $5.76\text{ }\mu\text{m} \times 11.10\text{ }\mu\text{m}$.

Note: CAPDEF is necessary for csf3p, csf3 and csf3a.

Note: Higher values may be achieved by the formation of arrays.

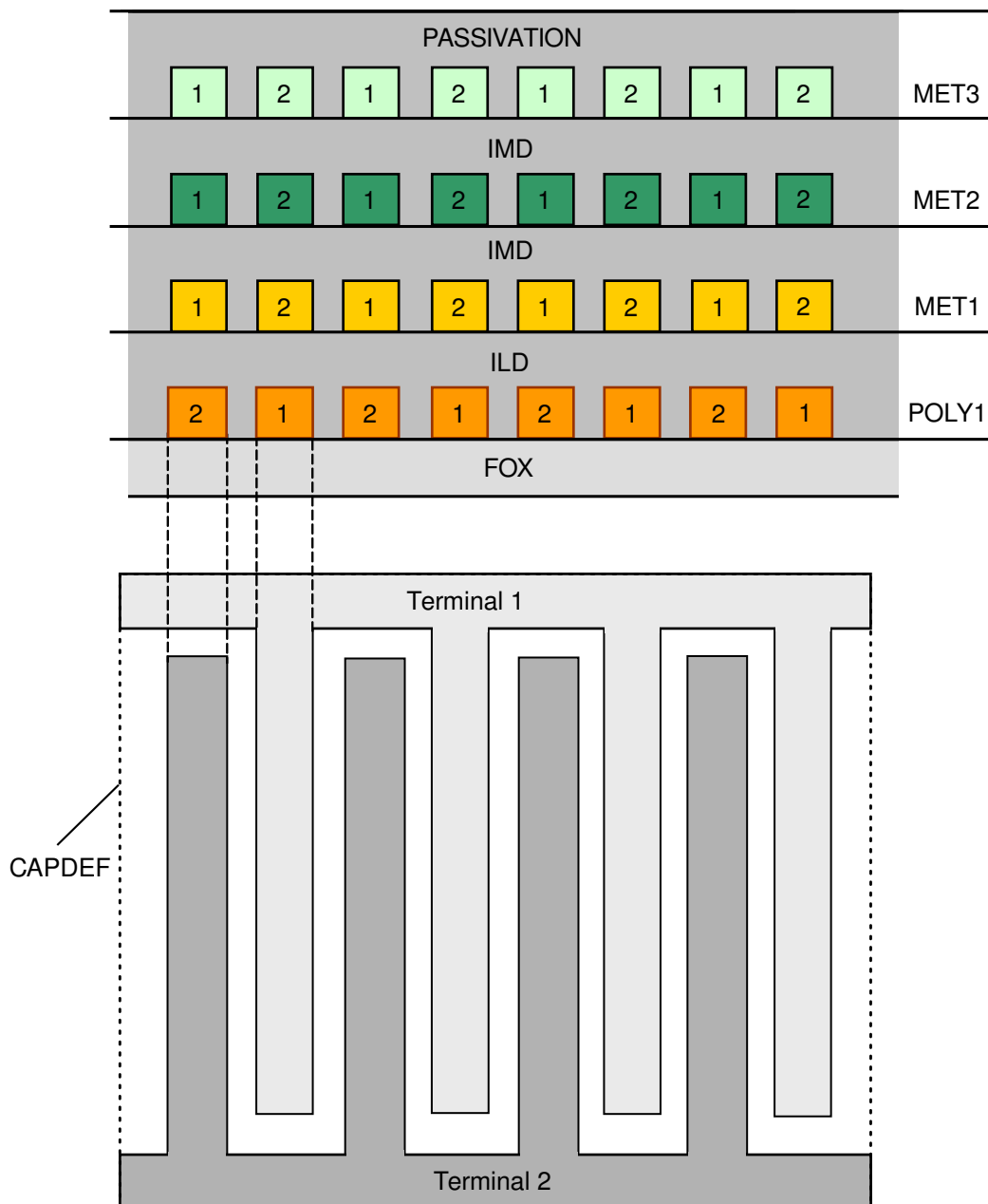


Figure 3.85 csf3p

3. Layer and Device rules → 3.3 MET3 module→ 3.3.2 Device rules→ csf3p, csf3, csf3a

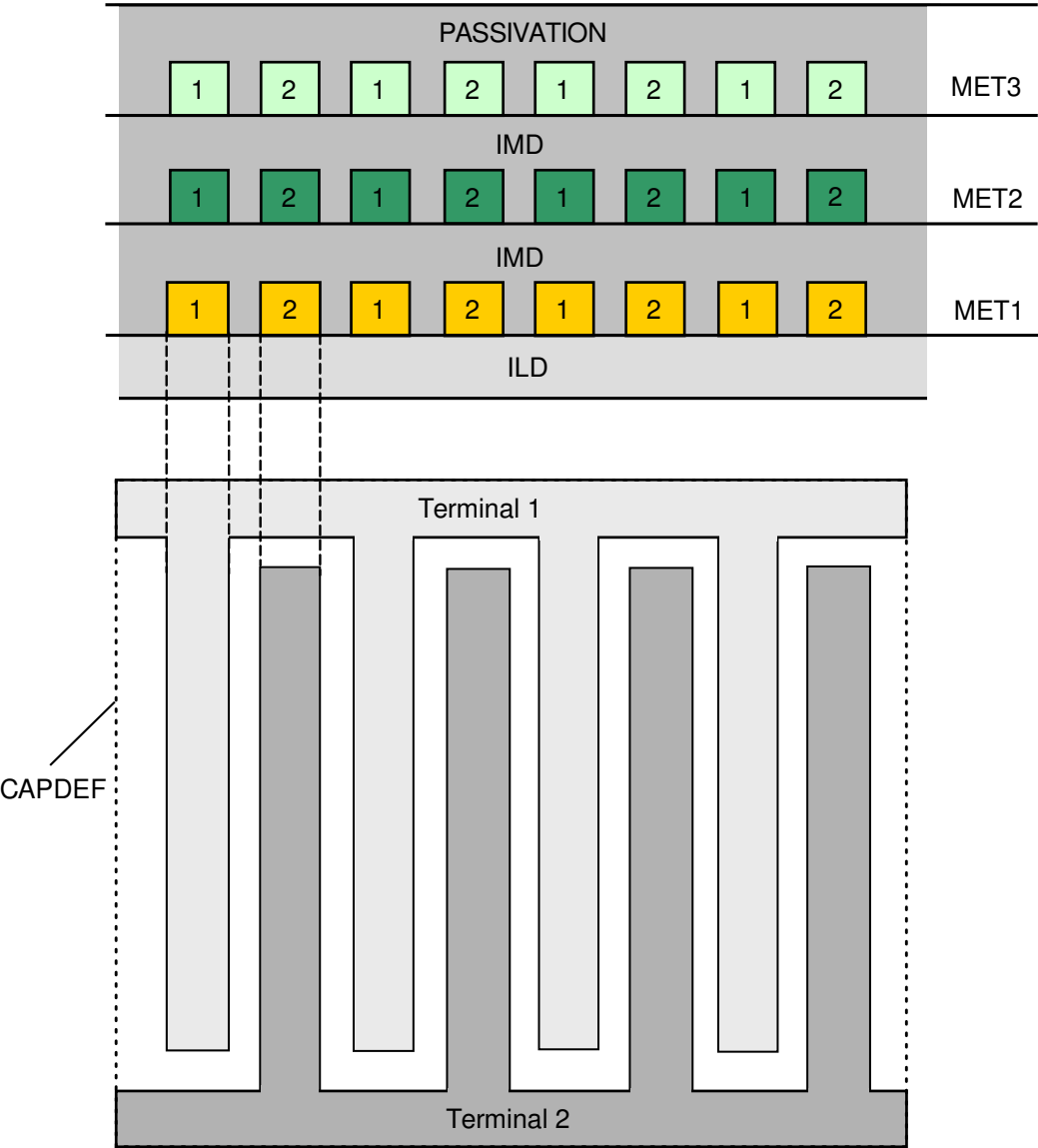


Figure 3.86 csf3, csf3a

3. Layer and Device rules → 3.3 MET3 module→ 3.3.2 Device rules→ cif3a

cif3a

The layout of the device cif3a is fixed and must not be changed. A single cell instance has an area of 4.48 μm x 10.80 μm.

Note: Higher values may be achieved by the formation of arrays.

Note: CAPDEF is necessary for cif3a.

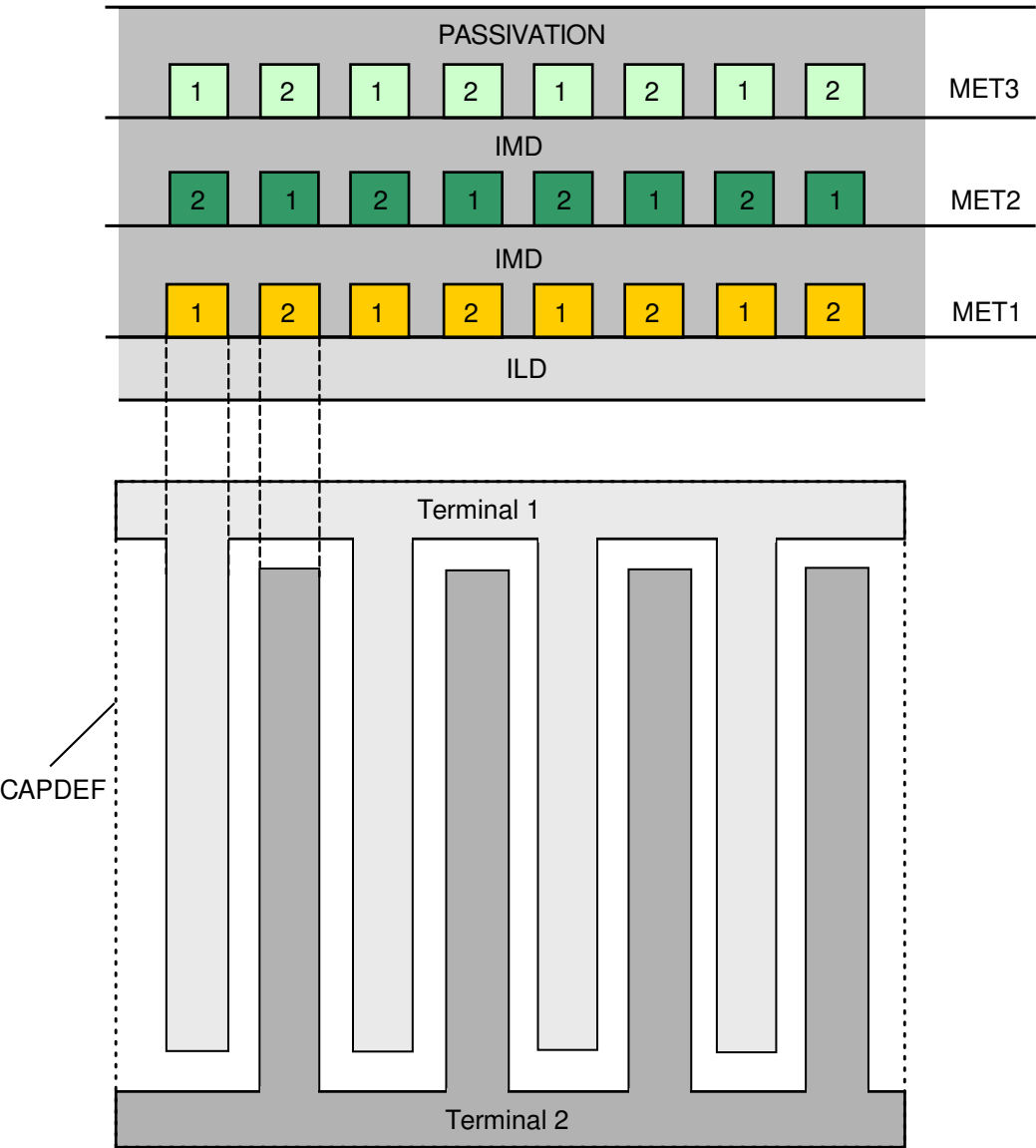


Figure 3.87 cif3a

3. Layer and Device rules → 3.4 MET4 module

3.4 MET4 module**3.4.1 Layer rules****VIA3**

Name	Description	Value	Unit
B1V3	VIA3 must be covered by MET3 and MET4	-	-
B2V3	VIA3 stripes are only allowed to bend at 135 degrees	-	-
W2V3	Fixed VIA3 stripe width	0.26	μm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1V3	Fixed VIA3 size (except dphoc#, dphod#, OPTOVIA3, PHODIO)	0.26 x 0.26	μm x μm
S1V3	Minimum VIA3 spacing	0.26	μm
S9V3	Minimum VIA3 stripe to VIA3 spacing	1.0	μm
E1M3V3	Minimum MET3 enclosure of VIA3	0.01	μm
E1M4V3	Minimum MET4 enclosure of VIA3	0.01	μm
E2M3V3	Minimum MET3 enclosure of VIA3 (in one direction of VIA3 corner)	0.06	μm
E2M4V3	Minimum MET4 enclosure of VIA3 (in one direction of VIA3 corner)	0.06	μm
E4M4V3	Minimum MET4 enclosure of VIA3 stripe	0.1	μm
E8M3V3	Minimum MET3 enclosure of VIA3 stripe	0.1	μm
R1V3	Maximum ratio of VIA3 area to connected GATE area	20.0	-
Q20V3	Check for the right usage of OPTOVIA3	-	-
	Note: VIA3 stripes must only be used for light shielding purpose.		
	Note: DRC reports this check request whenever this layer is present.		

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.4 MET4 module→ 3.4.1 Layer rules→ VIA3

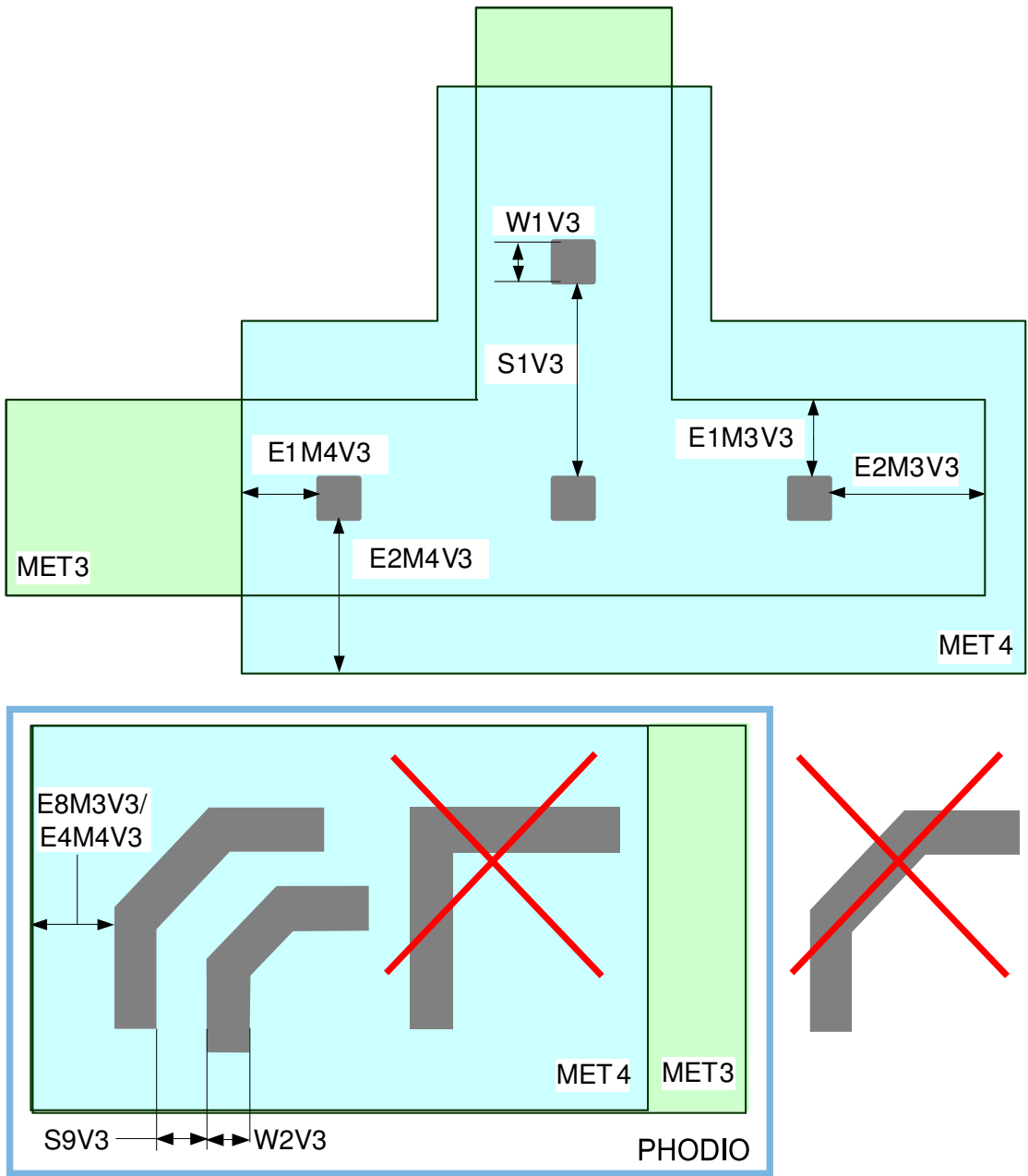
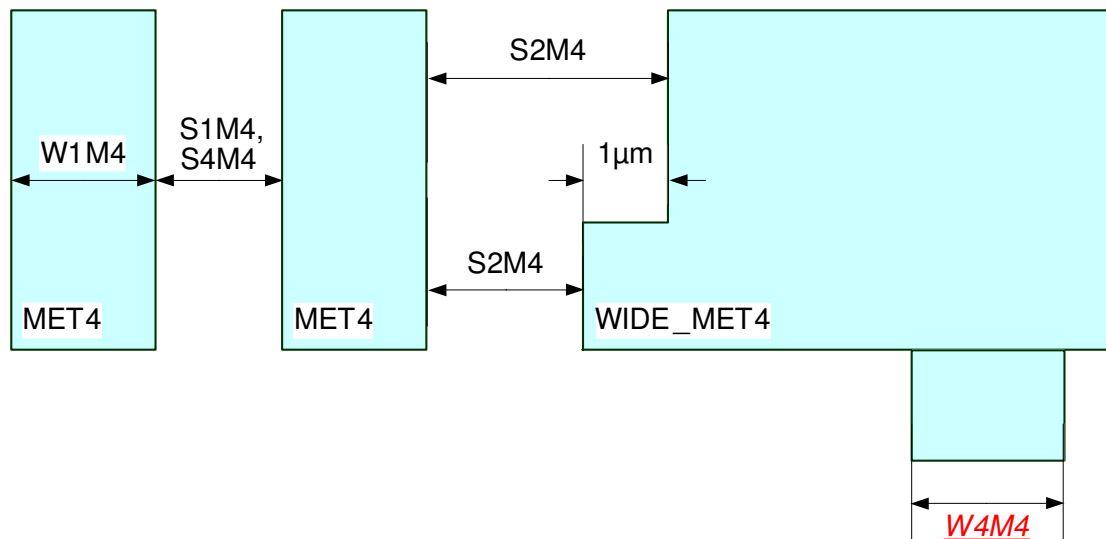


Figure 3.88 VIA3

3. Layer and Device rules → 3.4 MET4 module → 3.4.1 Layer rules → MET4

MET4

Name	Description	Value	Unit
B1M4	All MET4 tracks > 35µm wide to be slotted (except Pads)	-	-
W1M4	Minimum MET4 width	0.28	µm
<u>W4M4</u>	Minimum MET4 width joining wide MET4 track (> 35 µm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 µm from the main track prior to narrowing it.	10.0	µm
<u>W5M4</u>	Maximum MET4 region size Note: MET4 regions are defined as MET4 shapes (single MET4 shapes or a bundle of ME4 shapes, with width > 2.0µm, inclusive of the spacing if the spacing is ≤ 1.0µm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on "my X-FAB". Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	µm x µm
S1M4	Minimum MET4 spacing/notch	0.28	µm
S4M4	Minimum MET4 spacing (different net, tag_60v) Note: Valid for spacing of all MET4 shapes with label tag_60v to all MET4 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.4	µm
S2M4	Minimum MET4 spacing to WIDE_MET4	0.6	µm
A1M4	Minimum MET4 area	0.202	µm ²
<u>R1M4</u>	Minimum ratio of MET4 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2M4</u>	Maximum ratio of MET4 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1M4P1	Maximum ratio of MET4 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
R2M4P1	Maximum ratio of MET4 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
Q1M4	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

**Figure 3.89** MET4

3. Layer and Device rules → 3.4 MET4 module → 3.4.1 Layer rules → M4SLOT

M4SLOT

Name	Description	Value	Unit
W2M4	Minimum M4SLOT width	0.6	μm
W3M4	Minimum M4SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M4	Minimum M4SLOT spacing/notch	10.0	μm
S1M4M3	Minimum M4SLOT spacing to M3SLOT	2.0	μm
	Note: M4SLOT is not allowed over M3SLOT.		
E1M4M4	Minimum MET4 enclosure of M4SLOT	10.0	μm
	Note: M4SLOT without MET4 is not allowed.		

Note: Insert M4SLOTS in direction of current flow.

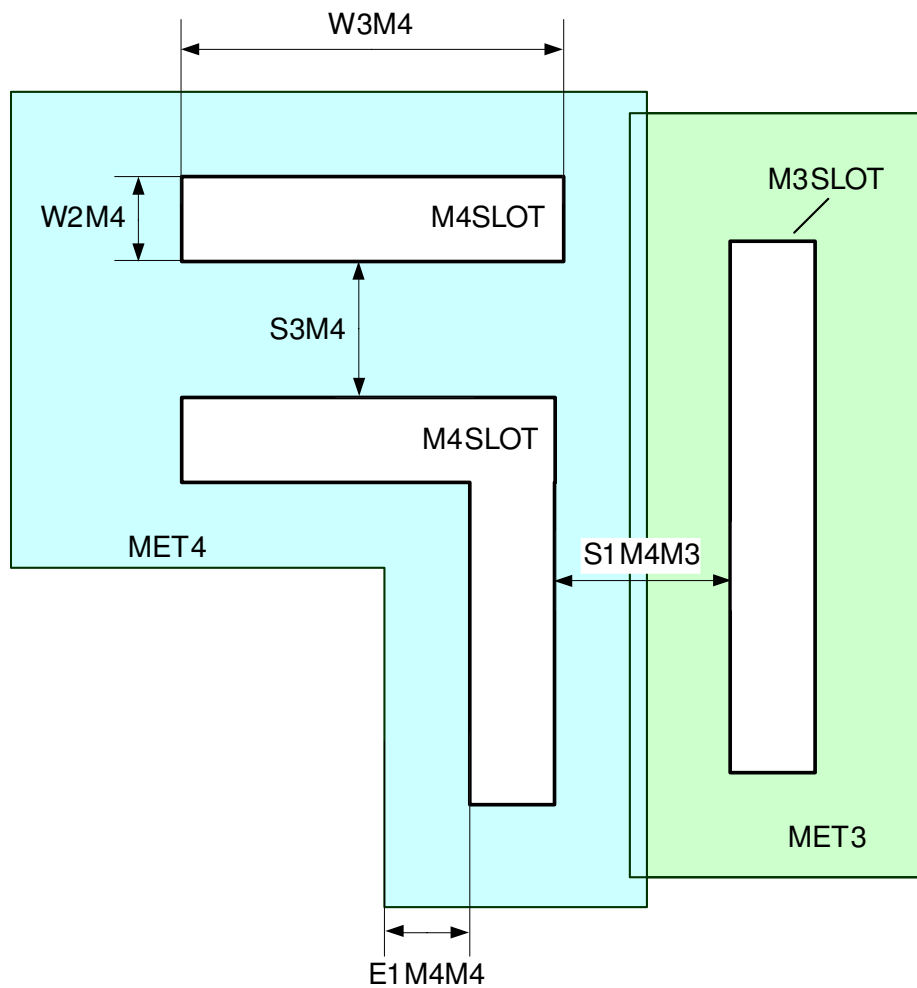


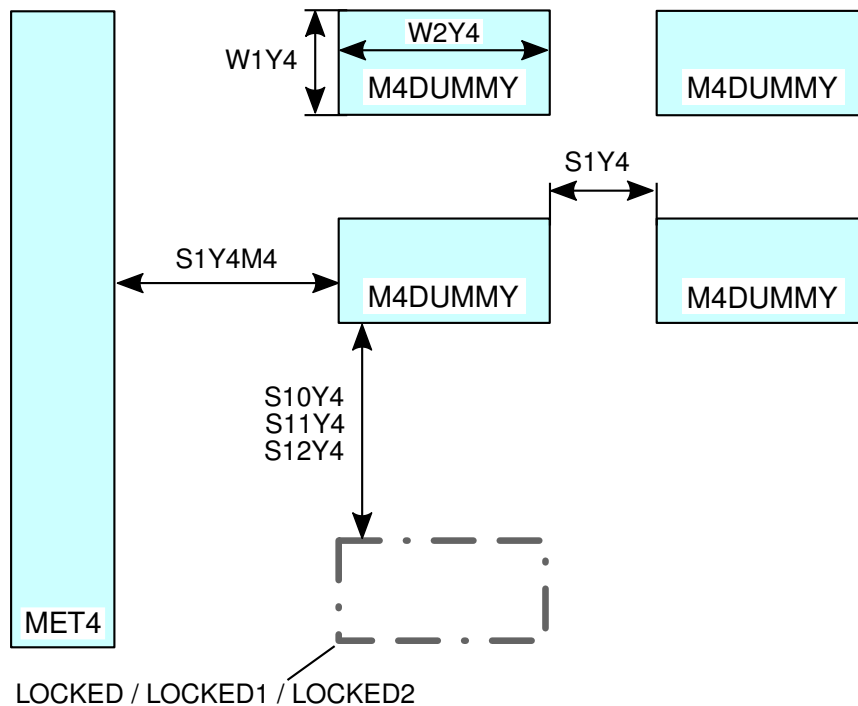
Figure 3.90 M4SLOT

3. Layer and Device rules → 3.4 MET4 module → 3.4.1 Layer rules → M4DUMMY

M4DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y4	Only rectangular M4DUMMY is allowed	-	-
B2Y4	M4DUMMY overlap of MET4 is not allowed	-	-
B3Y4V3	M4DUMMY overlap of VIA3 is not allowed	-	-
B3Y4V4	M4DUMMY overlap of VIA4 is not allowed	-	-
B3Y4VT	M4DUMMY overlap of VIATP is not allowed Note: Only valid if module MET5 is not selected.	-	-
W1Y4	Minimum M4DUMMY width	2.0	μm
W2Y4	Maximum M4DUMMY edge length	20.0	μm
S1Y4	Minimum M4DUMMY spacing	2.0	μm
S10Y4	Minimum M4DUMMY spacing to LOCKED	4.0	μm
S11Y4	Minimum M4DUMMY spacing to LOCKED1	4.0	μm
S12Y4	Minimum M4DUMMY spacing to LOCKED2	4.0	μm
S1Y4M4	Minimum M4DUMMY spacing to MET4	4.0	μm

**Figure 3.91** M4DUMMY**3.4.2 Device rules****rm4**

Name	Description	Value	Unit
B2M4	VIA3 is not allowed within rm4	-	-

⇒

3. Layer and Device rules → 3.4 MET4 module→ 3.4.2 Device rules→ rm4

Name	Description	Value	Unit
B3M4	VIATP is not allowed within rm4	-	-
	Note: Valid if module METMID is selected and not MET5		
B4M4	VIA4 is not allowed within rm4	-	-
	Note: Valid if MET5 module is selected.		

Note: rm4 resistor definition: MET4 and M4VERIFY.

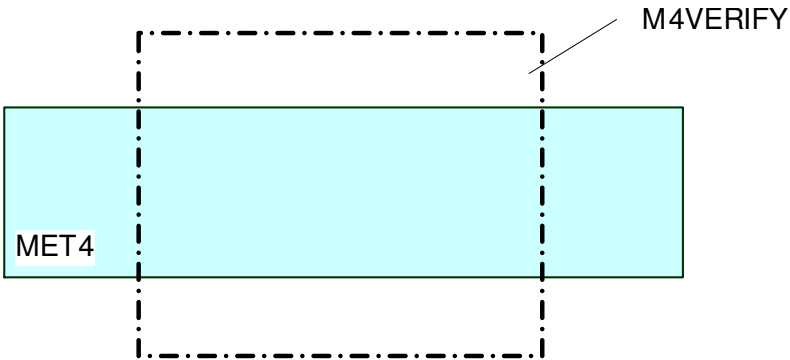


Figure 3.92 rm4

3. Layer and Device rules → 3.4 MET4 module→ 3.4.2 Device rules→ csandwt4

csandwt4

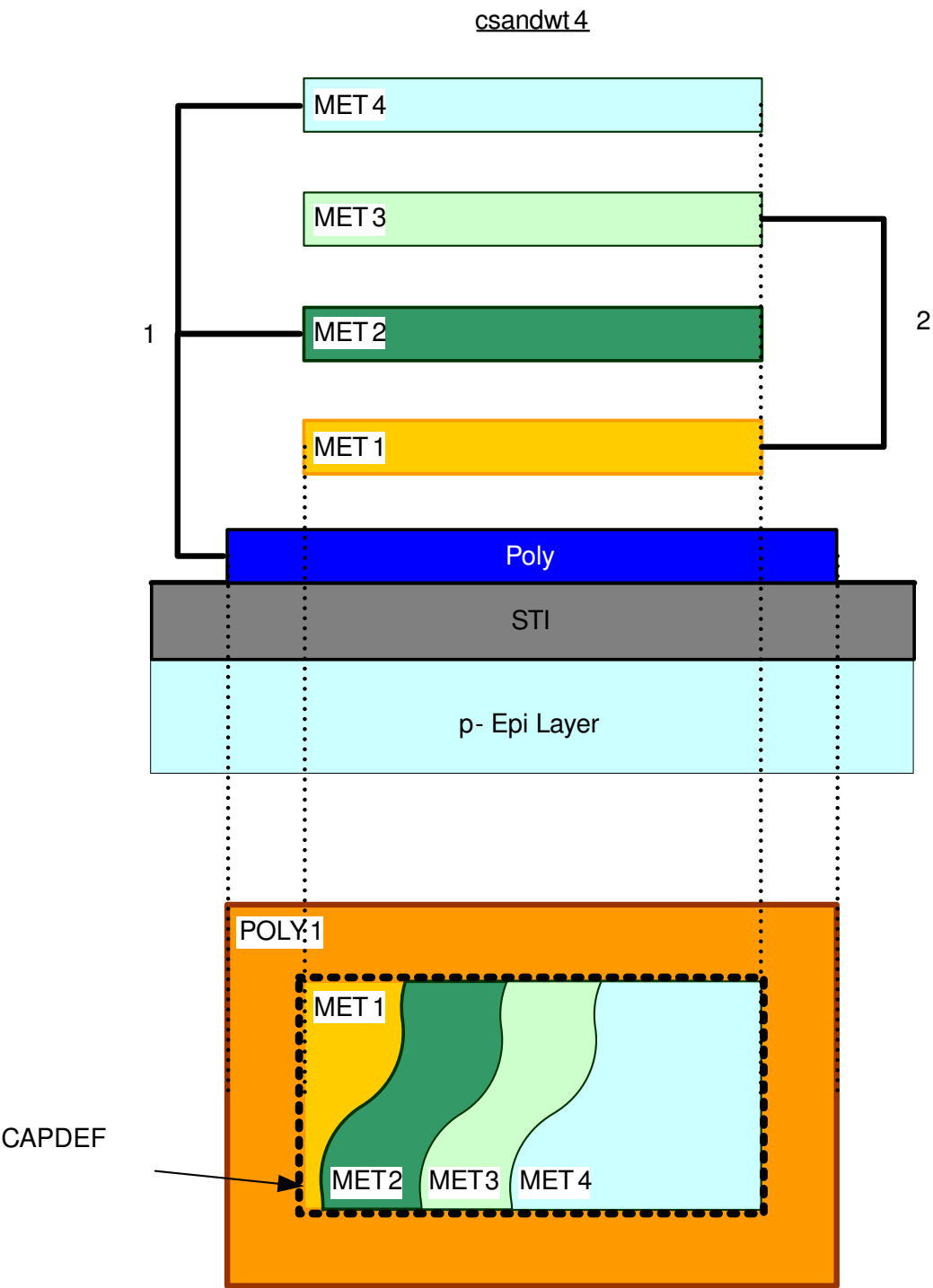


Figure 3.93 csandwt4

3. Layer and Device rules → 3.4 MET4 module→ 3.4.2 Device rules→ cif4a

cif4a

The layout of the device cif4a is fixed and must not be changed. A single cell instance has an area of 4.48 μm x 10.80 μm.

Note: Higher values may be achieved by the formation of arrays.

Note: CAPDEF is necessary for cif4a.

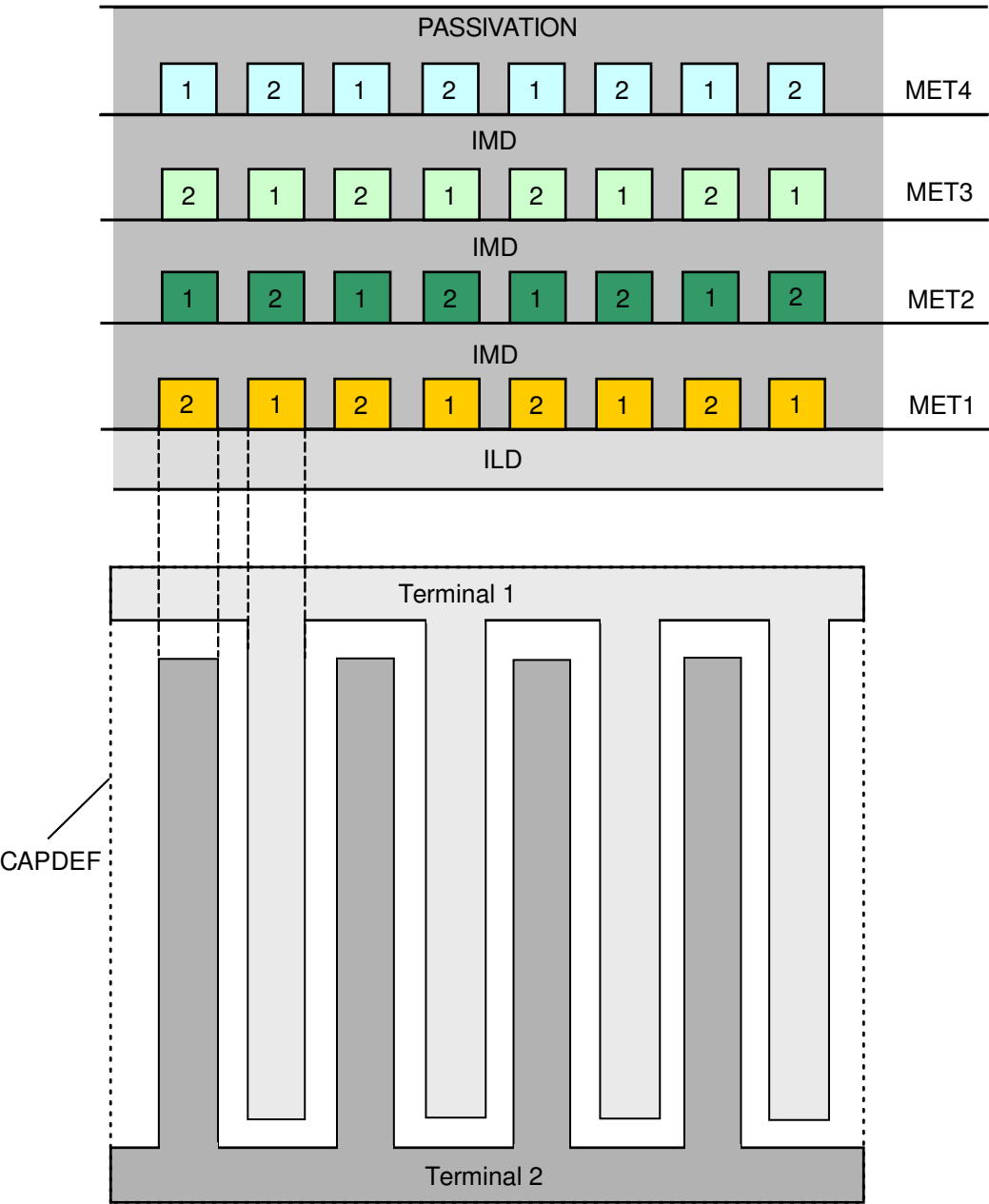


Figure 3.94 cif4a

3. Layer and Device rules → 3.4 MET4 module → 3.4.2 Device rules → csf4, csf4a

csf4, csf4a

The layouts of the devices csf4 and csf4a are fixed and must not be changed. For csf4, a single cell instance has an area of $4.48\ \mu\text{m} \times 10.80\ \mu\text{m}$. For csf4a, a single cell instance has an area of $5.76\ \mu\text{m} \times 11.10\ \mu\text{m}$.

Note: CAPDEF is necessary for csf4 and csf4a.

Note: Higher values may be achieved by the formation of arrays.

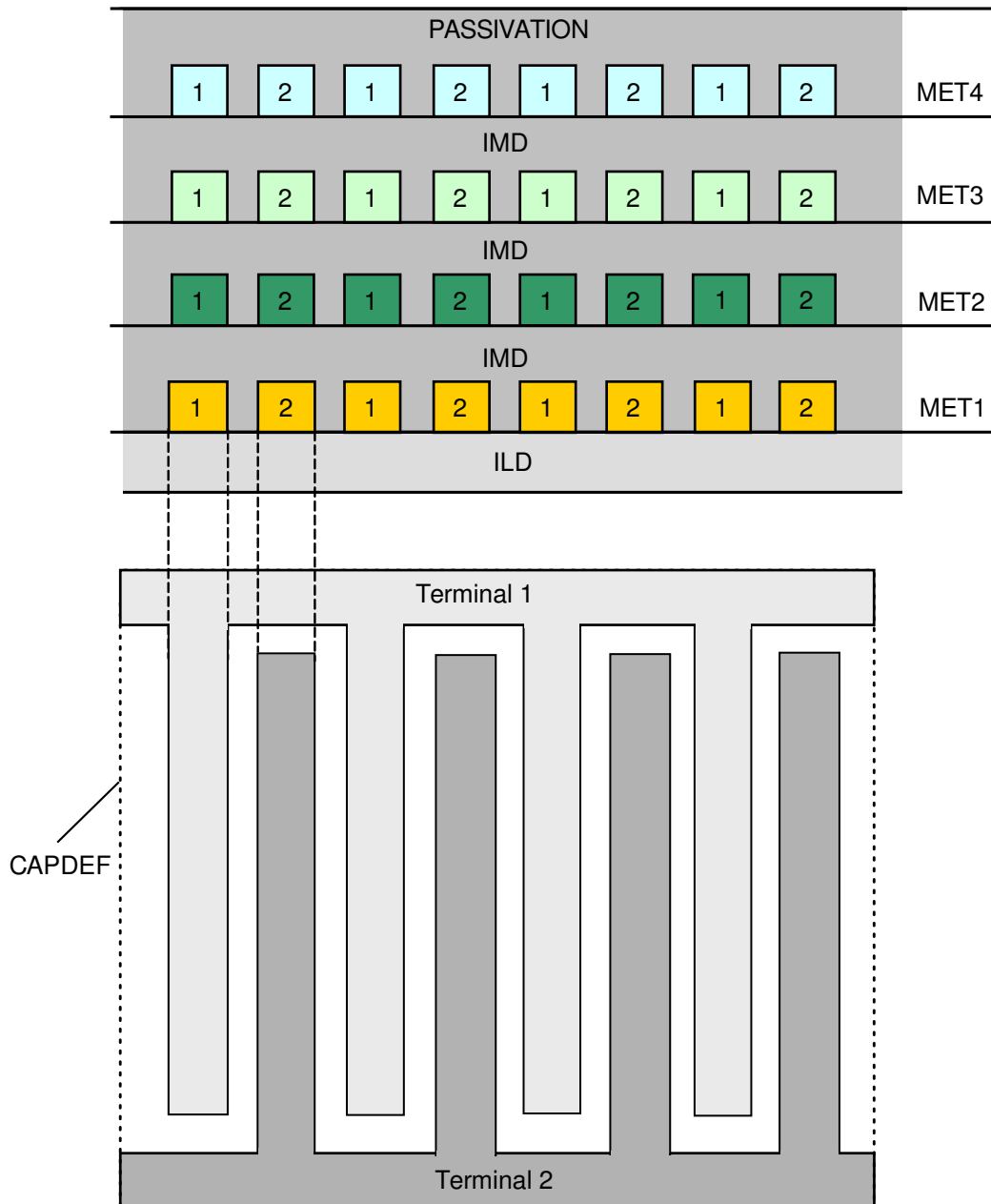


Figure 3.95 csf4, csf4a

3. Layer and Device rules → 3.5 MET5 module

3.5 MET5 module**3.5.1 Layer rules****VIA4**

Name	Description	Value	Unit
B1V4	VIA4 must be covered by MET4 and MET5	-	-
B2V4	VIA4 stripes are only allowed to bend at 135 degrees	-	-
W2V4	Fixed VIA4 stripe width	0.26	μm
	Note: Stripes are only allowed for use with photodiodes or other optical purposes.		
W1V4	Fixed VIA4 size (except dphoc#, dphod#, OPTOVIA4, PHODIO)	0.26 x 0.26	μm x μm
S1V4	Minimum VIA4 spacing	0.26	μm
S5V4	Minimum VIA4 stripe to VIA4 spacing	1.0	μm
E1M4V4	Minimum MET4 enclosure of VIA4	0.01	μm
E1M5V4	Minimum MET5 enclosure of VIA4	0.01	μm
E2M4V4	Minimum MET4 enclosure of VIA4 (in one direction of VIA4 corner)	0.06	μm
E2M5V4	Minimum MET5 enclosure of VIA4 (in one direction of VIA4 corner)	0.06	μm
E3M5V4	Minimum MET5 enclosure of VIA4 stripe	0.1	μm
E5M4V4	Minimum MET4 enclosure of VIA4 stripe	0.1	μm
R1V4	Maximum ratio of VIA4 area to connected GATE area	20.0	-
Q20V4	Check for the right usage of OPTOVIA4	-	-
	Note: VIA4 stripes must only be used for light shielding purpose.		
	Note: DRC reports this check request whenever this layer is present.		

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.5 MET5 module→ 3.5.1 Layer rules→ VIA4

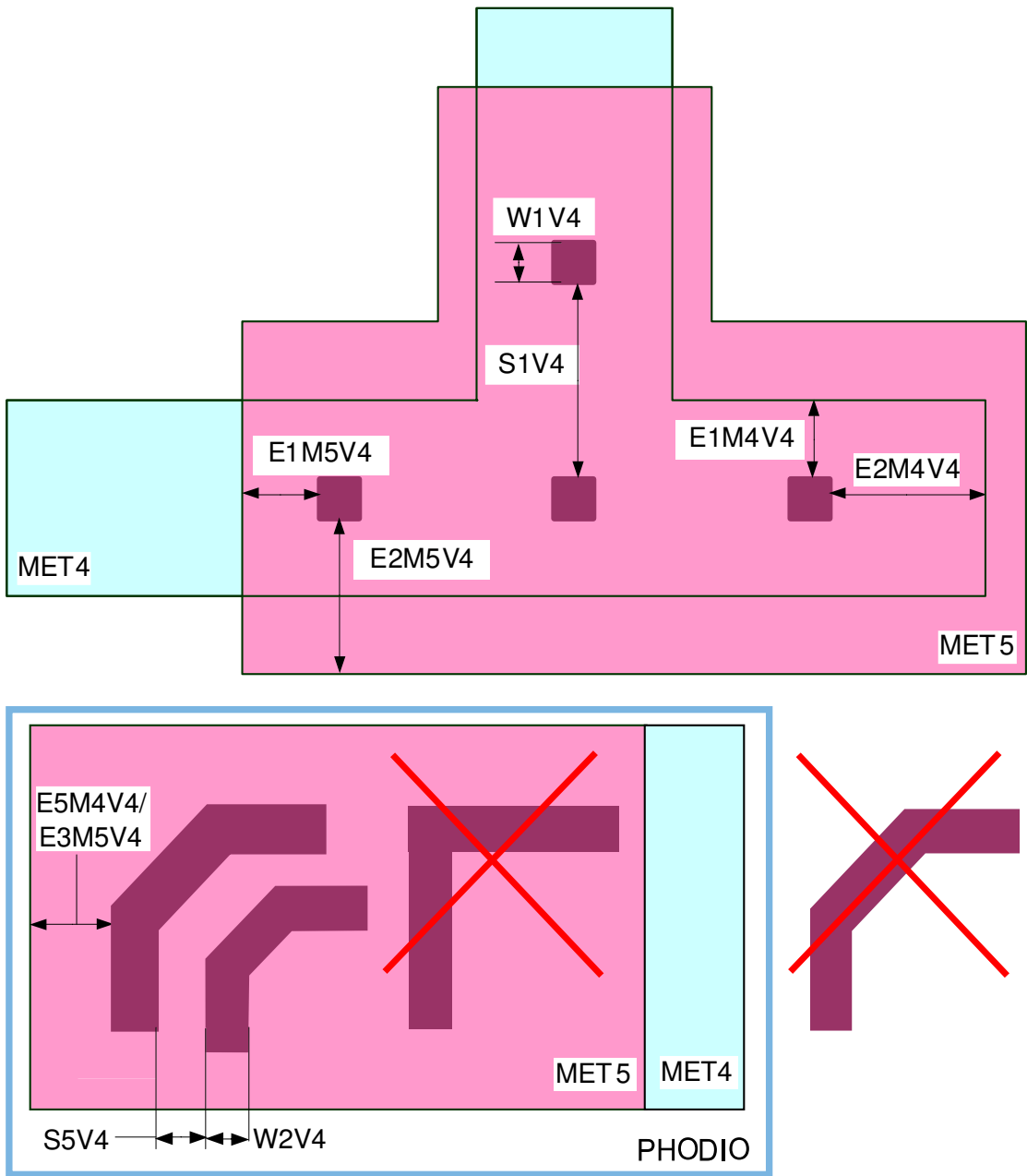
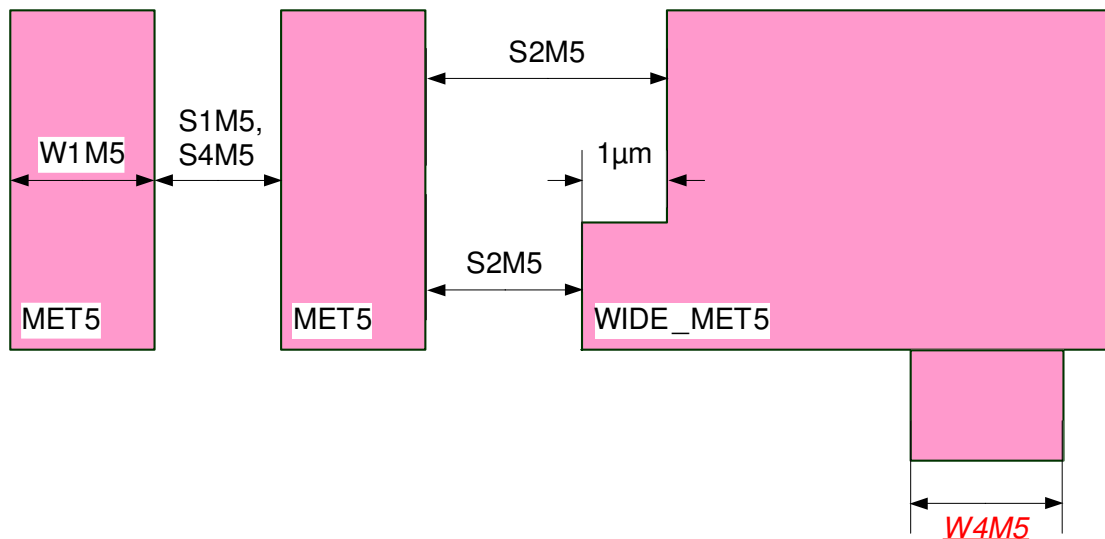


Figure 3.96 VIA4

3. Layer and Device rules → 3.5 MET5 module → 3.5.1 Layer rules → MET5

MET5

Name	Description	Value	Unit
B1M5	All MET5 tracks > 35µm wide to be slotted (except Pads)	-	-
W1M5	Minimum MET5 width	0.28	µm
<u>W4M5</u>	Minimum MET5 width joining wide MET5 track (> 35 µm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 µm from the main track prior to narrowing it.	10.0	µm
<u>W5M5</u>	Maximum MET5 region size Note: MET5 regions are defined as MET5 shapes (single MET5 shapes or a bundle of MET5 shapes, with width > 2.0µm, inclusive of the spacing if the spacing is ≤ 1.0µm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on "my X-FAB". Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	µm x µm
S1M5	Minimum MET5 spacing/notch	0.28	µm
S4M5	Minimum MET5 spacing (different net, tag_60v) Note: Valid for spacing of all MET5 shapes with label tag_60v to all MET5 shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.4	µm
S2M5	Minimum MET5 spacing to WIDE_MET5	0.6	µm
A1M5	Minimum MET5 area	0.202	µm²
<u>R1M5</u>	Minimum ratio of MET5 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2M5</u>	Maximum ratio of MET5 area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1M5P1	Maximum ratio of MET5 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
R2M5P1	Maximum ratio of MET5 area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
Q1M5	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

**Figure 3.97** MET5

3. Layer and Device rules → 3.5 MET5 module → 3.5.1 Layer rules → M5SLOT

M5SLOT

Name	Description	Value	Unit
W2M5	Minimum M5SLOT width	0.6	μm
W3M5	Minimum M5SLOT length	20.0	μm
	Note: If this rule cannot be adhered to, it is suggested that the track in question is drawn as two narrow parallel tracks.		
S3M5	Minimum M5SLOT spacing/notch	10.0	μm
S1M5M4	Minimum M5SLOT spacing to M4SLOT	2.0	μm
	Note: M5SLOT is not allowed over M4SLOT.		
E1M5M5	Minimum MET5 enclosure of M5SLOT	10.0	μm
	Note: M5SLOT without MET5 is not allowed.		

Note: Insert M5SLOTs in direction of current flow.

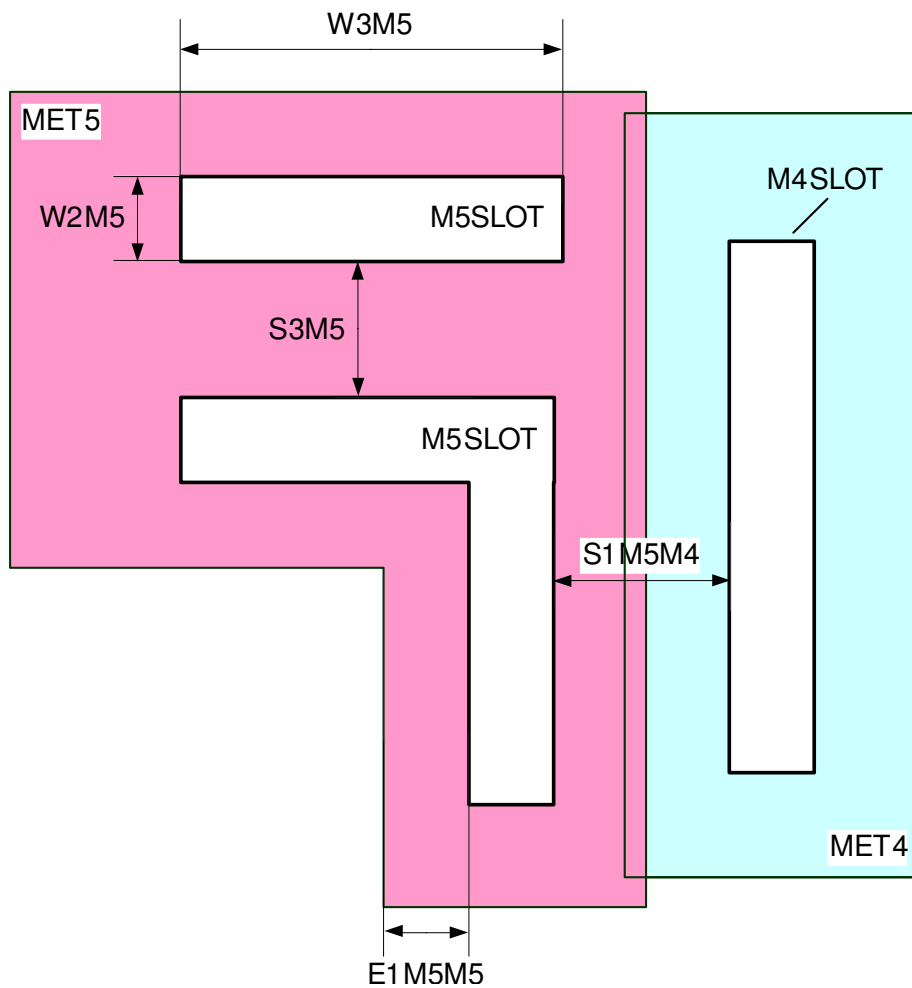


Figure 3.98 M5SLOT

3. Layer and Device rules → 3.5 MET5 module → 3.5.1 Layer rules → M5DUMMY

M5DUMMY

It is recommended to use X-FAB's dummy pattern generation option (DUMMY_FILL included in the DRC for preview) to create dummy pattern. If the generated dummy pattern is not sufficient to meet the minimum requirements for pattern density, customers can draw additional dummy pattern following the dummy design rules (DIFFDUMMY, P1DUMMY, M1DUMMY, M2DUMMY, M3DUMMY, M4DUMMY, M5DUMMY, MTPDUMMY and MTPLDUMMY). For further information, refer to the design related guideline "Dummy Pattern Generation".

Name	Description	Value	Unit
B1Y5	Only rectangular M5DUMMY is allowed	-	-
B2Y5	M5DUMMY overlap of MET5 is not allowed	-	-
B3Y5V4	M5DUMMY overlap of VIA4 is not allowed	-	-
B3Y5VT	M5DUMMY overlap of VIATP is not allowed	-	-
W1Y5	Minimum M5DUMMY width	2.0	μm
W2Y5	Maximum M5DUMMY edge length	20.0	μm
S1Y5	Minimum M5DUMMY spacing	2.0	μm
S10Y5	Minimum M5DUMMY spacing to LOCKED	4.0	μm
S11Y5	Minimum M5DUMMY spacing to LOCKED1	4.0	μm
S1Y5M5	Minimum M5DUMMY spacing to MET5	4.0	μm

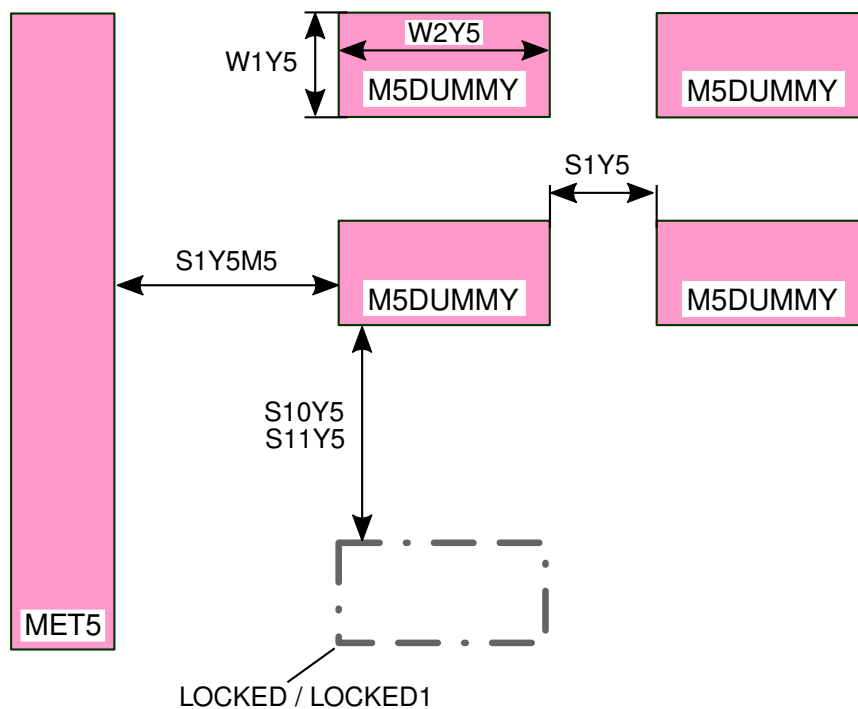


Figure 3.99 M5DUMMY

3.5.2 Device rules**rm5**

Name	Description	Value	Unit
B2M5	VIA4 is not allowed within rm5	-	-
B3M5	VIATP is not allowed within rm5	-	-

Note: rm5 resistor definition: MET5 and M5VERIFY.

3. Layer and Device rules → 3.5 MET5 module → 3.5.2 Device rules → rm5

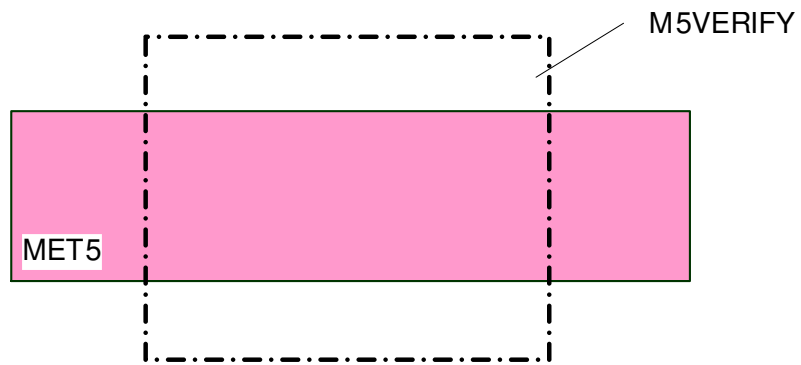


Figure 3.100 rm5

3. Layer and Device rules → 3.5 MET5 module→ 3.5.2 Device rules→ csandwt5

csandwt5

Note: CAPDEF is necessary for csandwt5.

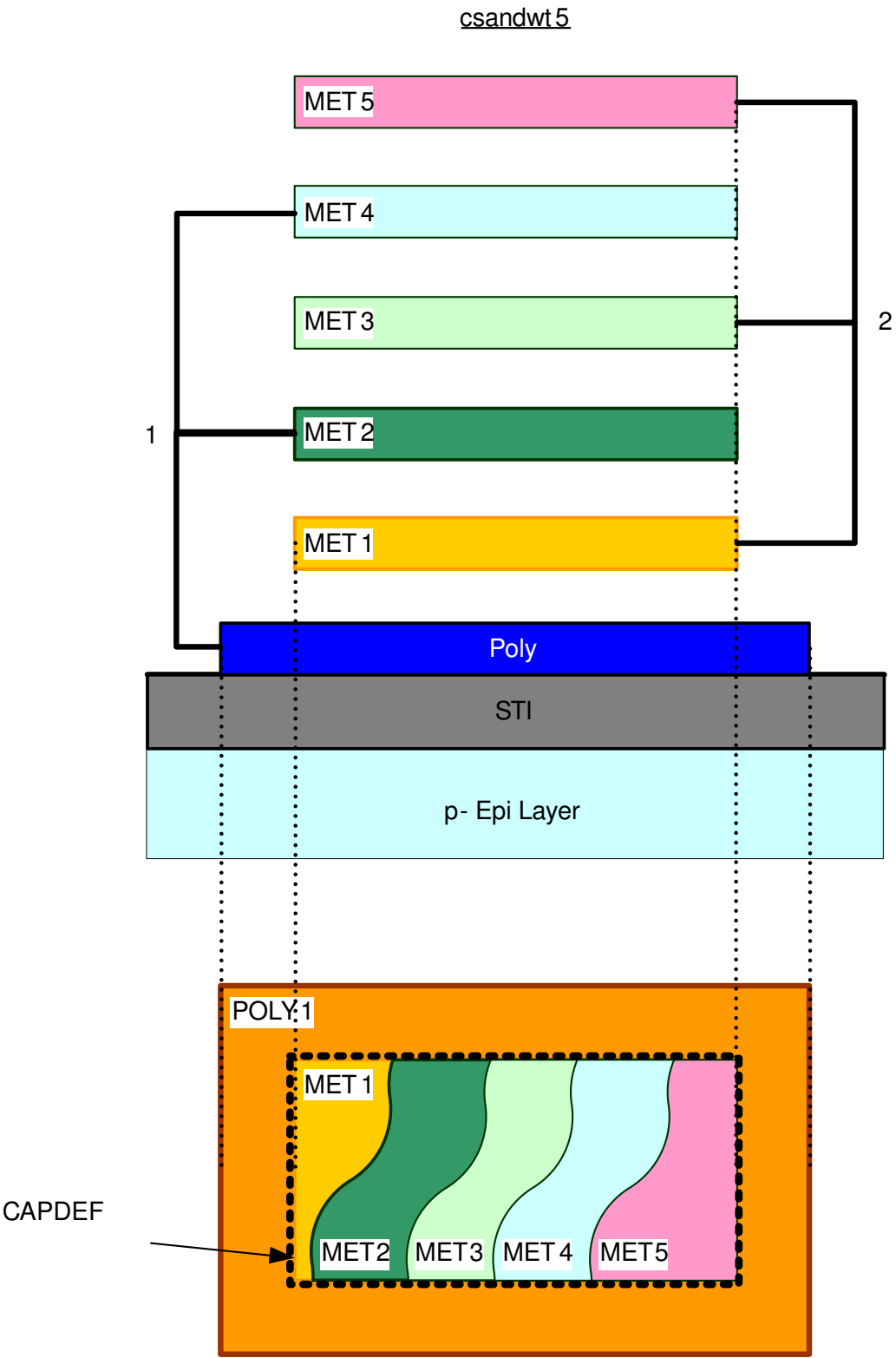


Figure 3.101 csandwt5

3. Layer and Device rules → 3.5 MET5 module→ 3.5.2 Device rules→ cif5a

cif5a

The layout of the device cif5a is fixed and must not be changed. A single cell instance has an area of 4.48 μm x 10.80 μm.

Note: Higher values may be achieved by the formation of arrays.

Note: CAPDEF is necessary for cif5a.

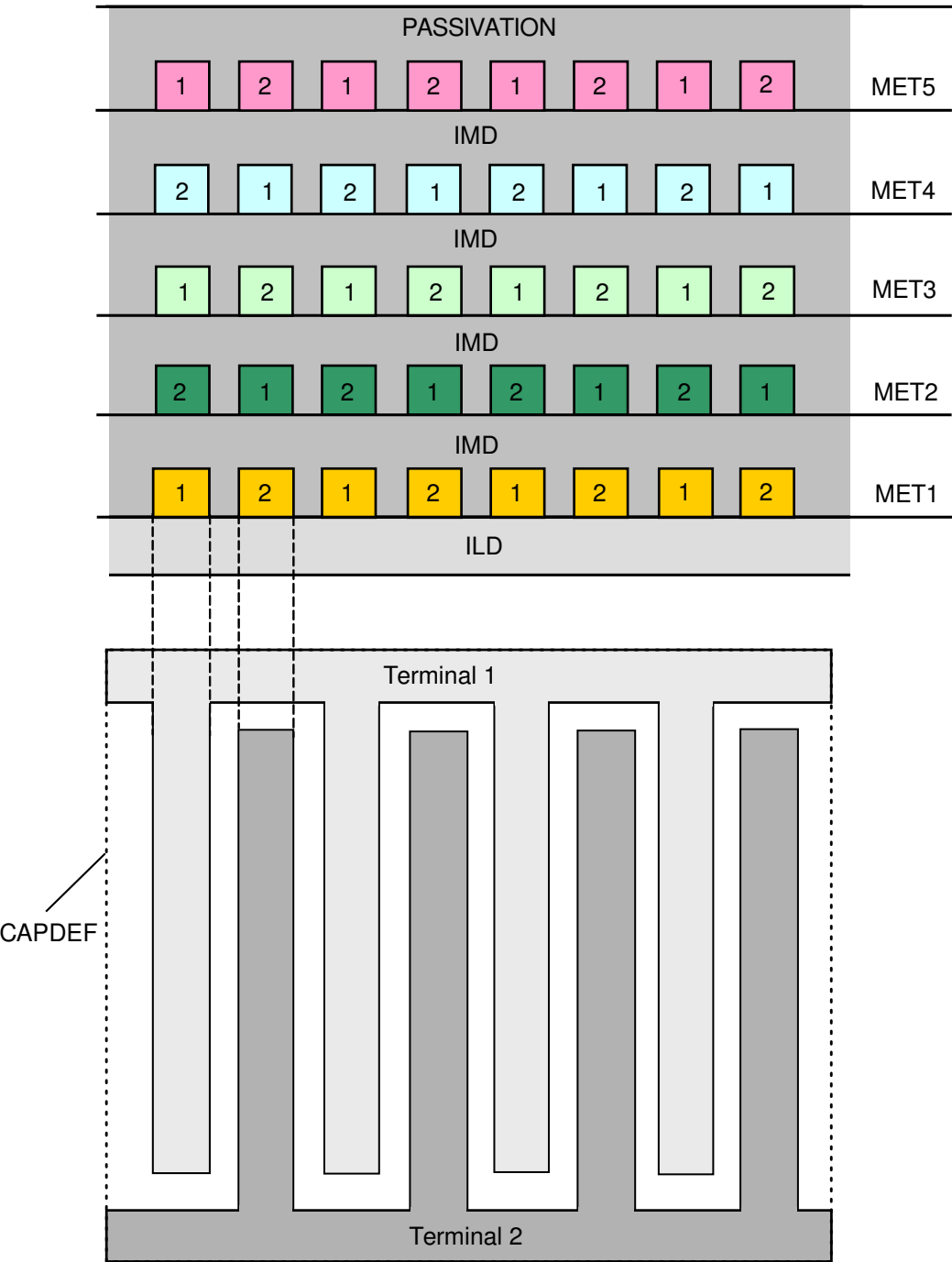


Figure 3.102 cif5a

3. Layer and Device rules → 3.5 MET5 module → 3.5.2 Device rules → csf5, csf5a

csf5, csf5a

The layout of the devices csf5 and csf5a are fixed and must not be changed. For csf5, a single cell instance has an area of $4.48\ \mu\text{m} \times 10.80\ \mu\text{m}$. For csf5a, a single cell instance has an area of $5.76\ \mu\text{m} \times 11.10\ \mu\text{m}$.

Note: CAPDEF is necessary for csf5 and csf5a.

Note: Higher values may be achieved by the formation of arrays.

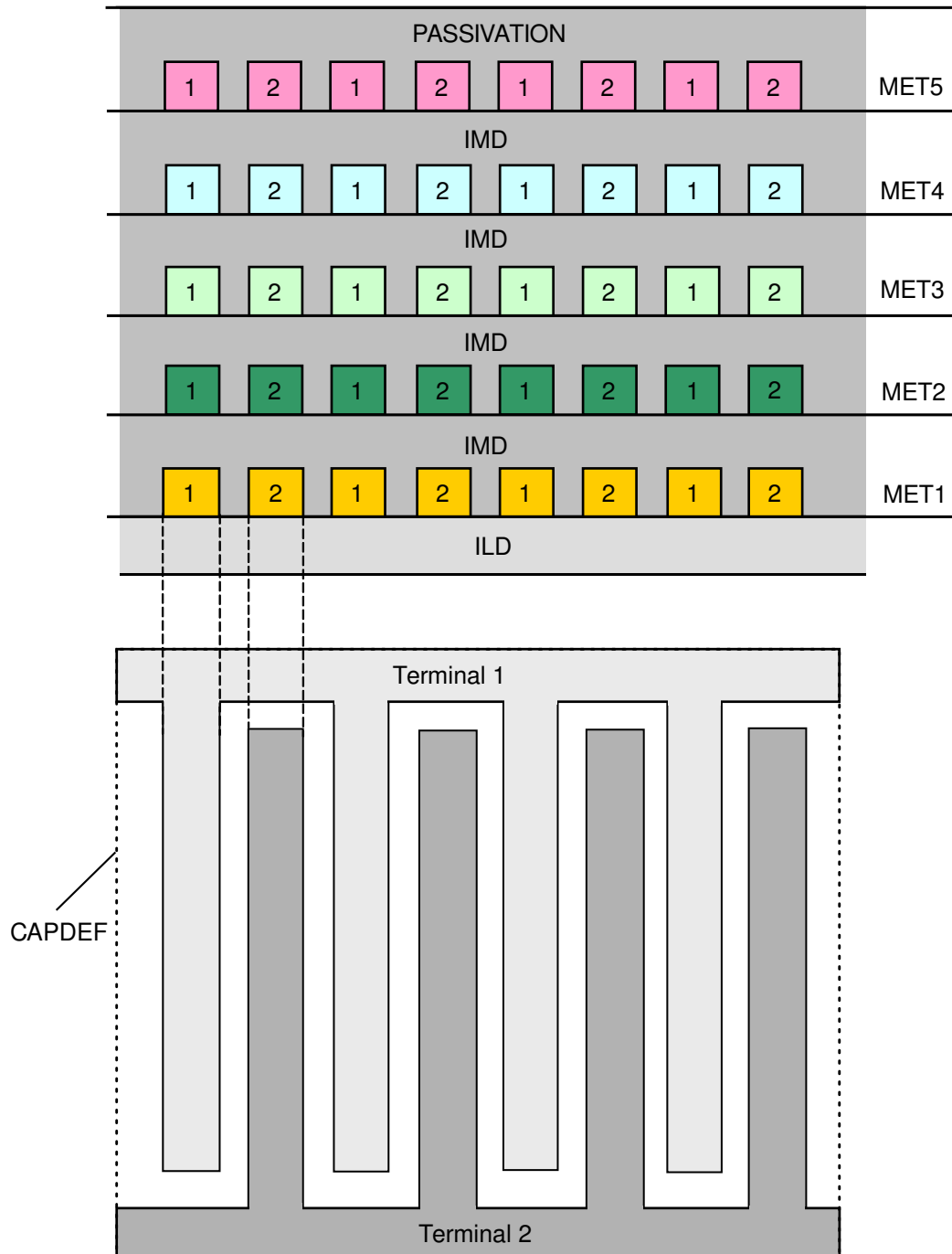


Figure 3.103 csf5, csf5a

3. Layer and Device rules → 3.6 METMID module

3.6 METMID module**3.6.1 Layer rules****VIATP**

Name	Description	Value	Unit
B1VT	VIATP must be covered by METTP	-	-
B1VTM3	VIATP must be covered by MET3 Note: Valid if MET4 module is not selected	-	-
B1VTM4	VIATP must be covered by MET4 Note: Valid if MET4 module is selected and MET5 module is not selected.	-	-
B1VTM5	VIATP must be covered by MET5 Note: Valid if MET5 module is selected	-	-
B2VT	VIATP stripes are only allowed to bend at 135 degrees	-	-
W2VT	Fixed VIATP stripe width Note: Stripes are only allowed for use with photodiodes or other optical purposes.	0.36	µm
W1VT	Fixed VIATP size (except dphoc#, dphod#, OPTOVIATP, PHODIO)	0.36 x 0.36	µm x µm
S1VT	Minimum VIATP spacing	0.35	µm
S9VT	Minimum VIATP stripe to VIATP spacing	1.0	µm
E1M3VT	Minimum MET3 enclosure of VIATP Note: Valid if MET4 module is not selected	0.01	µm
E1M4VT	Minimum MET4 enclosure of VIATP Note: Valid if MET4 module is selected and MET5 module is not selected.	0.01	µm
E1M5VT	Minimum MET5 enclosure of VIATP Note: Valid if MET5 module is selected	0.01	µm
E1MTVT	Minimum METTP enclosure of VIATP	0.09	µm
E2M3VT	Minimum MET3 enclosure of VIATP (in one direction of VIATP corner) Note: Valid if MET4 module is not selected	0.06	µm
E2M4VT	Minimum MET4 enclosure of VIATP (in one direction of VIATP corner) Note: Valid if MET4 module is selected and MET5 module is not selected.	0.06	µm
E2M5VT	Minimum MET5 enclosure of VIATP (in one direction of VIATP corner) Note: Valid if MET5 module is selected	0.06	µm
E3MTVT	Minimum METTP enclosure of VIATP stripe	0.2	µm
E4M5VT	Minimum MET5 enclosure of VIATP stripe Note: Valid if MET5 module is selected	0.2	µm
E5M3VT	Minimum MET3 enclosure of VIATP stripe Note: Valid if MET4 module is not selected	0.2	µm
E5M4VT	Minimum MET4 enclosure of VIATP stripe Note: Valid if MET4 module is selected and MET5 module is not selected.	0.2	µm
R1VT	Maximum ratio of VIATP area to connected GATE area	20.0	-
Q20VT	Check for the right usage of OPTOVIATP Note: VIATP stripes must only be used for light shielding purpose. Note: DRC reports this check request whenever this layer is present.	-	-

Note: Bond pads require interconnecting vias between the metal layers. See section "Bond Pad".

3. Layer and Device rules → 3.6 METMID module→ 3.6.1 Layer rules→ VIATP

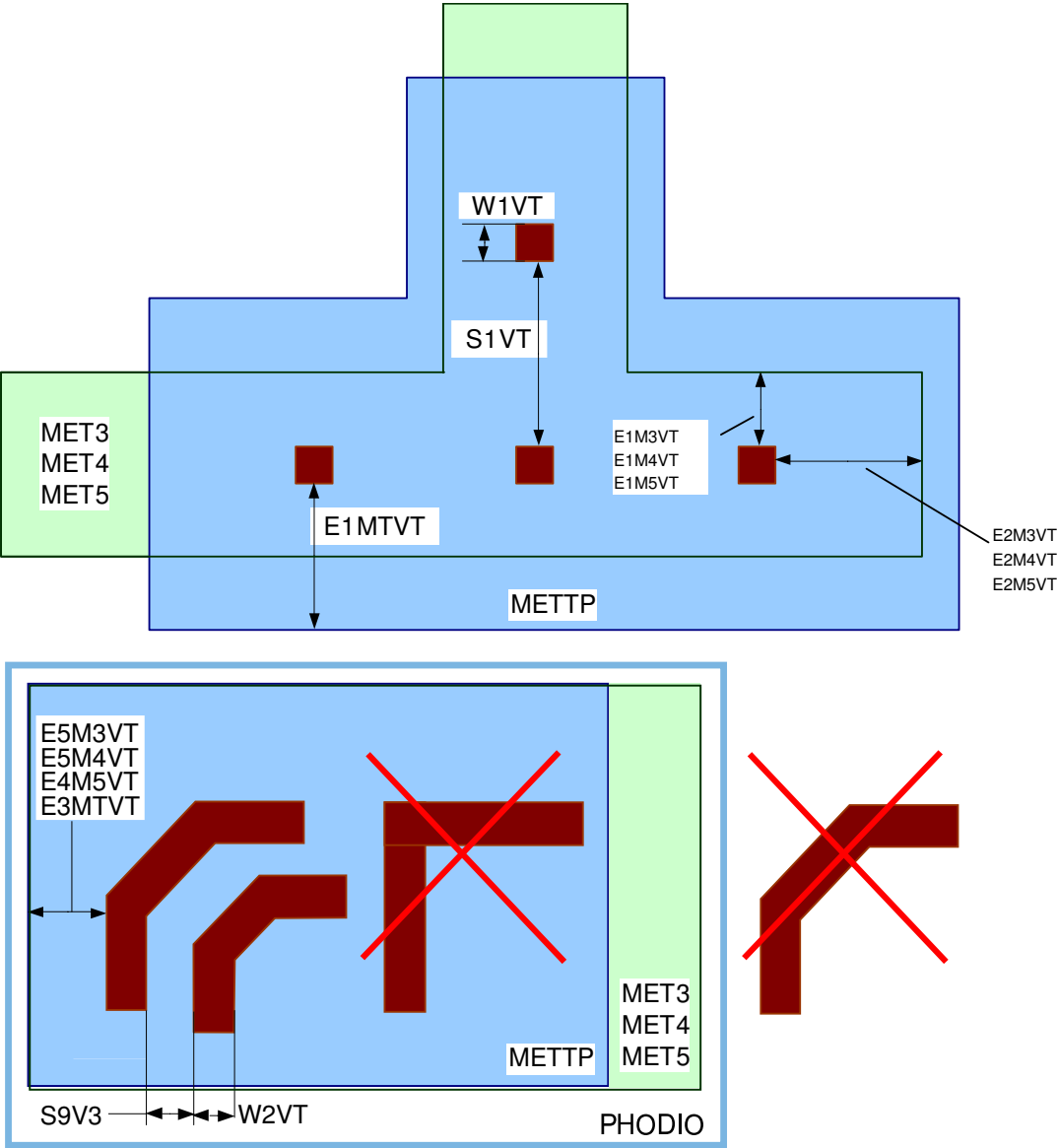
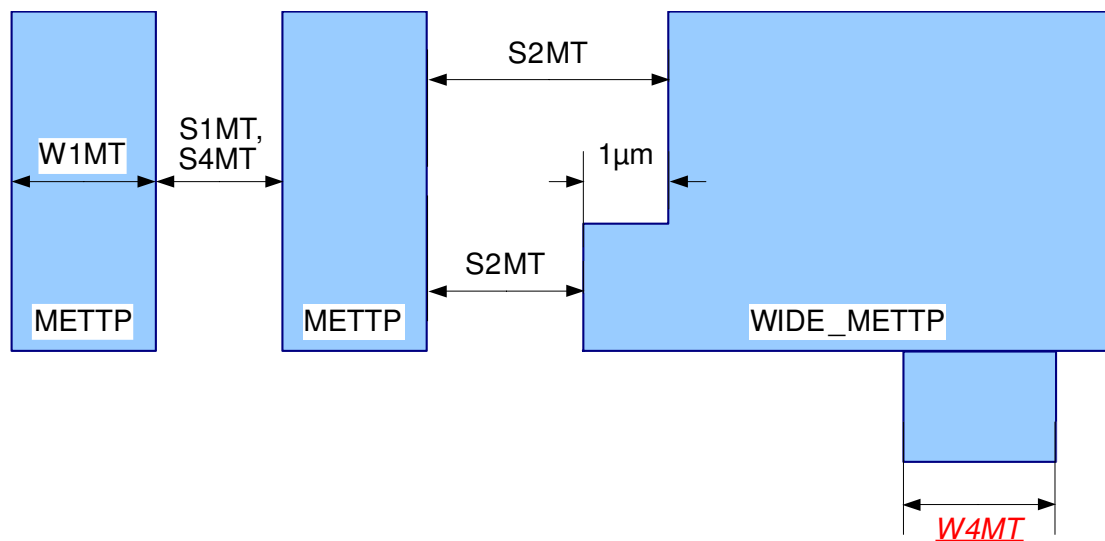


Figure 3.104 VIATP

3. Layer and Device rules → 3.6 METMID module→ 3.6.1 Layer rules→ METTP

METTP

Name	Description	Value	Unit
B1MT	All METTP tracks > 35µm wide to be slotted (except Pads)	-	-
W1MT	Minimum METTP width	0.44	µm
<u>W4MT</u>	Minimum METTP width joining wide METTP track (> 35 µm) Note: No slot is allowed opposite the join. It is recommended to maintain this width for at least 1 µm from the main track prior to narrowing.	10.0	µm
<u>W5MT</u>	Maximum METTP region size Note: METTP regions are defined as METTP shapes (single METTP shapes or a bundle of METTP shapes, with width > 2.0µm inclusive of the spacing if the spacing is <= 1.0µm) without any other metal layer above. For further information and design guidelines, please refer to the application note about IMD popping on "my X-FAB". Note: Not checked with standard DRC, option for check is available.	17.0 x 17.0	µm x µm
S1MT	Minimum METTP spacing/notch	0.46	µm
S4MT	Minimum METTP spacing (different net, tag_60v) Note: Valid for spacing of all METTP shapes with label tag_60v to all METTP shapes having lower voltage classes. Refer to the design related guideline "Voltage class definitions".	0.5	µm
S2MT	Minimum METTP spacing to WIDE_METTP	0.6	µm
A1MT	Minimum METTP area	0.562	µm ²
<u>R1MT</u>	Minimum ratio of METTP area to EXTENT area Note: Not checked with standard DRC, option for check is available.	30.0	%
<u>R2MT</u>	Maximum ratio of METTP area to EXTENT area Note: Not checked with standard DRC, option for check is available.	65.0	%
R1MTP1	Maximum ratio of METTP area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
R2MTP1	Maximum ratio of METTP area to connected GATE area Note: Refer to section "Antenna Rule definitions" as well.	400.0	-
Q1MT	Resistor terminal net without VLABEL Note: Resistor having VLABEL at one terminal only is not allowed. Refer to the design related guideline "Voltage class definitions".	-	-

**Figure 3.105** METTP